

AI 500

Foundations of

Artificial Intelligence

L13: Convolution Neural Networks

By Murtaza Taj



Agenda

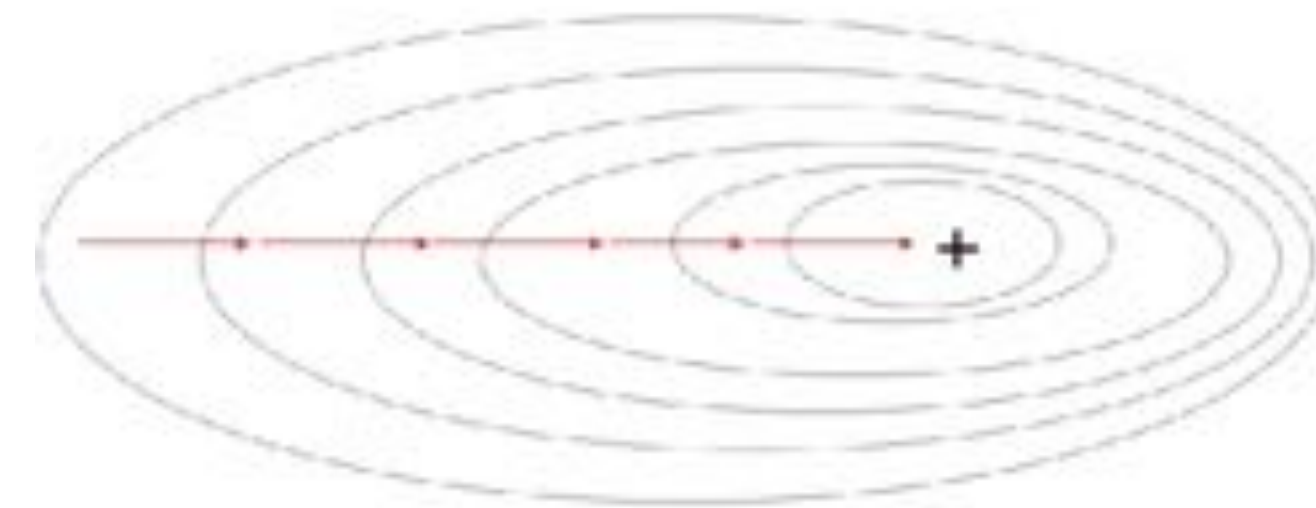
- Optimizers
- Preliminaries
 - Convolution (2D, Padding, Stride)
- Forward and Back Propagation using Convolution
- Transforming Multilayer Perception



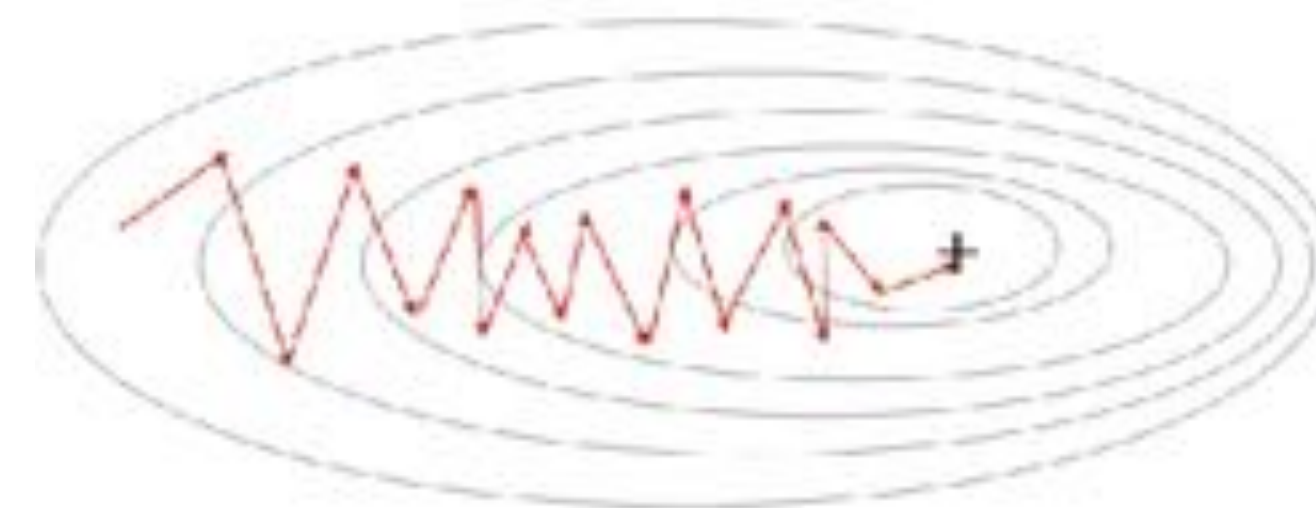
Gradient Descent Variants

- **Batch Gradient descent:**
 - Uses all samples per update
- **Stochastic Gradient Descent:**
 - Uses only one sample per update
- **Mini-batch Gradient Descent:**
 - Uses batch of samples

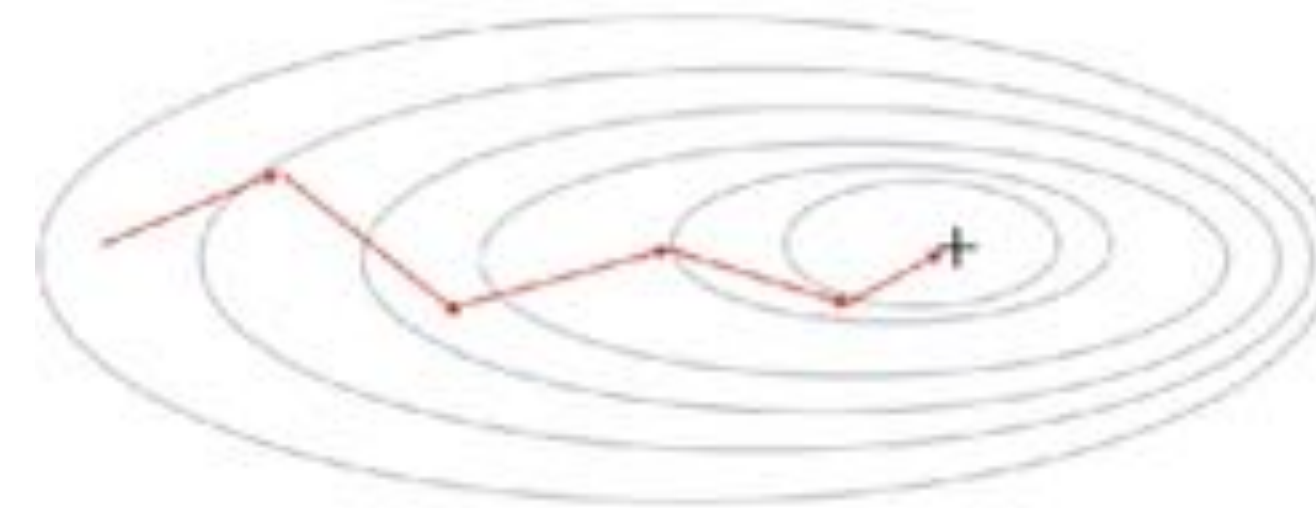
Gradient Descent



Stochastic Gradient Descent



Mini-Batch Gradient Descent



Gradient Descent Variants

- **Batch Gradient descent:**
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$$\Delta w_j = \eta \sum_i x_j^{(i)} (t^{(i)} - y^{(i)})$$

$$w_j \leftarrow w_j + \Delta w_j$$

$$\Delta w_j = \eta x_j^{(i)} (t^{(i)} - y^{(i)})$$

$$w_j \leftarrow w_j + \Delta w_j$$

$$\Delta w_j = \eta \sum_{k \in \text{mini-batch}} x_j^{(k)} (t^{(k)} - y^{(k)})$$

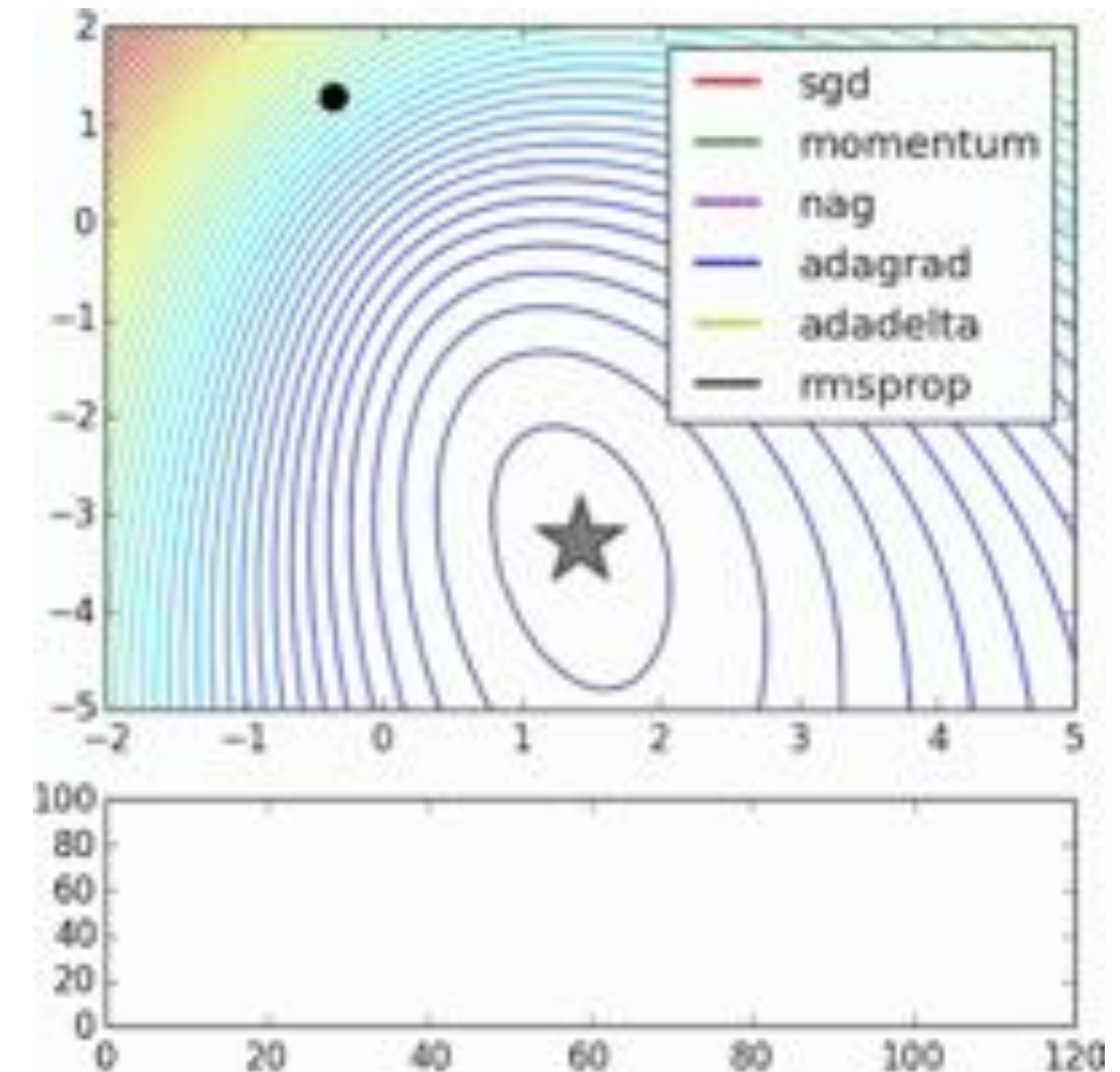
$$w_j \leftarrow w_j + \Delta w_j$$

Optimisers

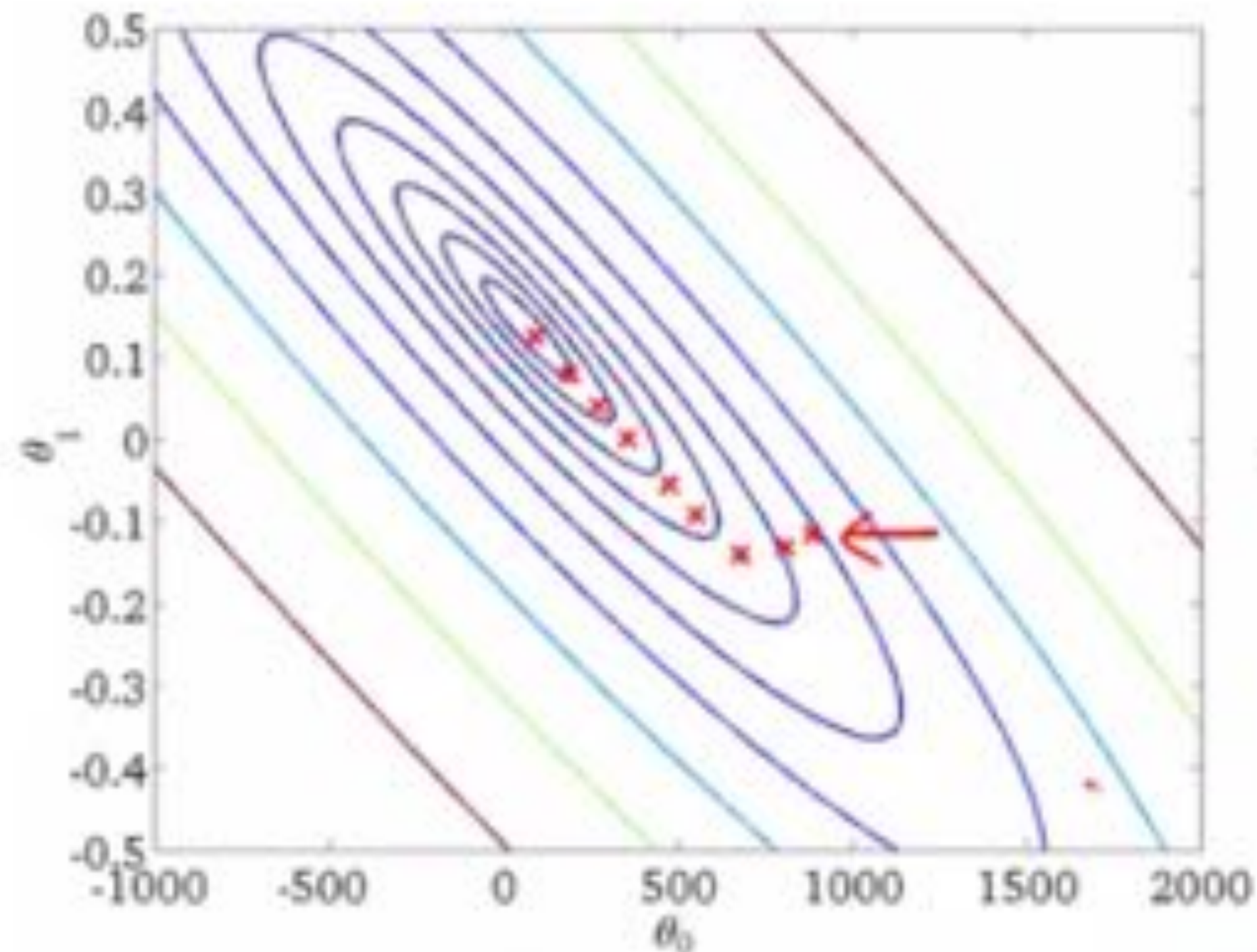
- Most critical challenge to optimizing deep networks is finding the correct trajectory to move in

► More on Optimizers

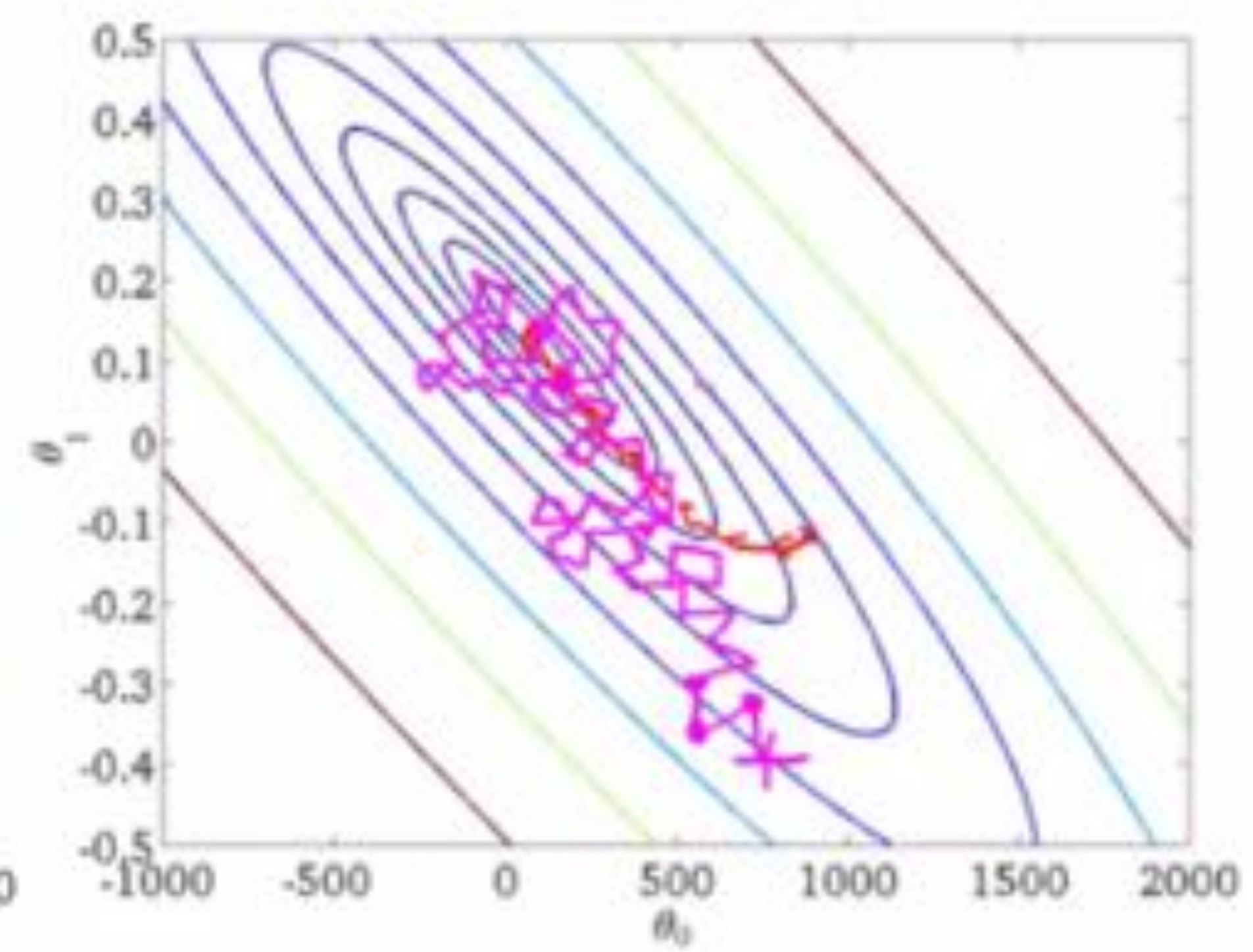
- Nesterov accelerated gradient
- Adagrad
- Adadelta
- RMSprop
- Adam
- AdaMax
- NAdam
- AMSGrad



Gradient Descent Variants



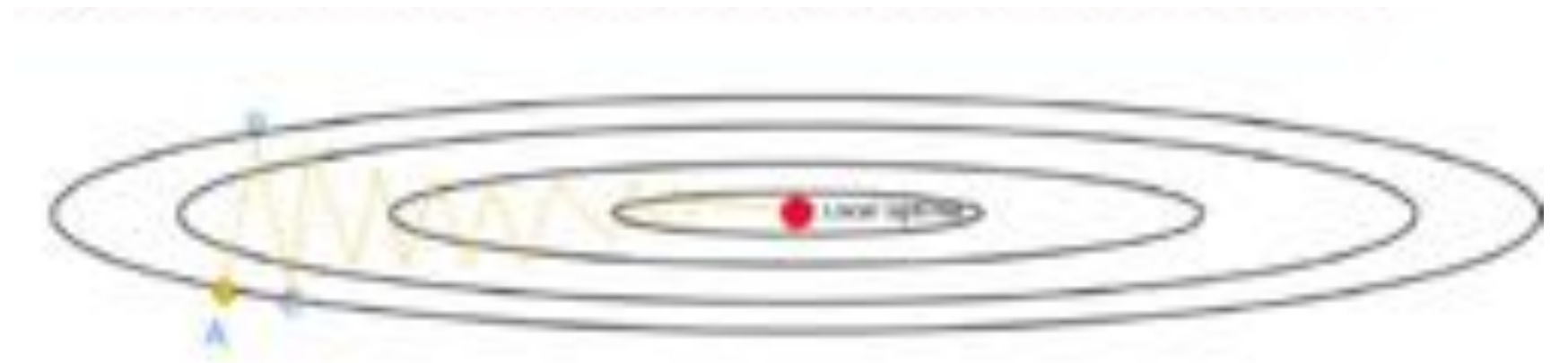
Vanilla Gradient Descent



Stochastic Gradient Descent

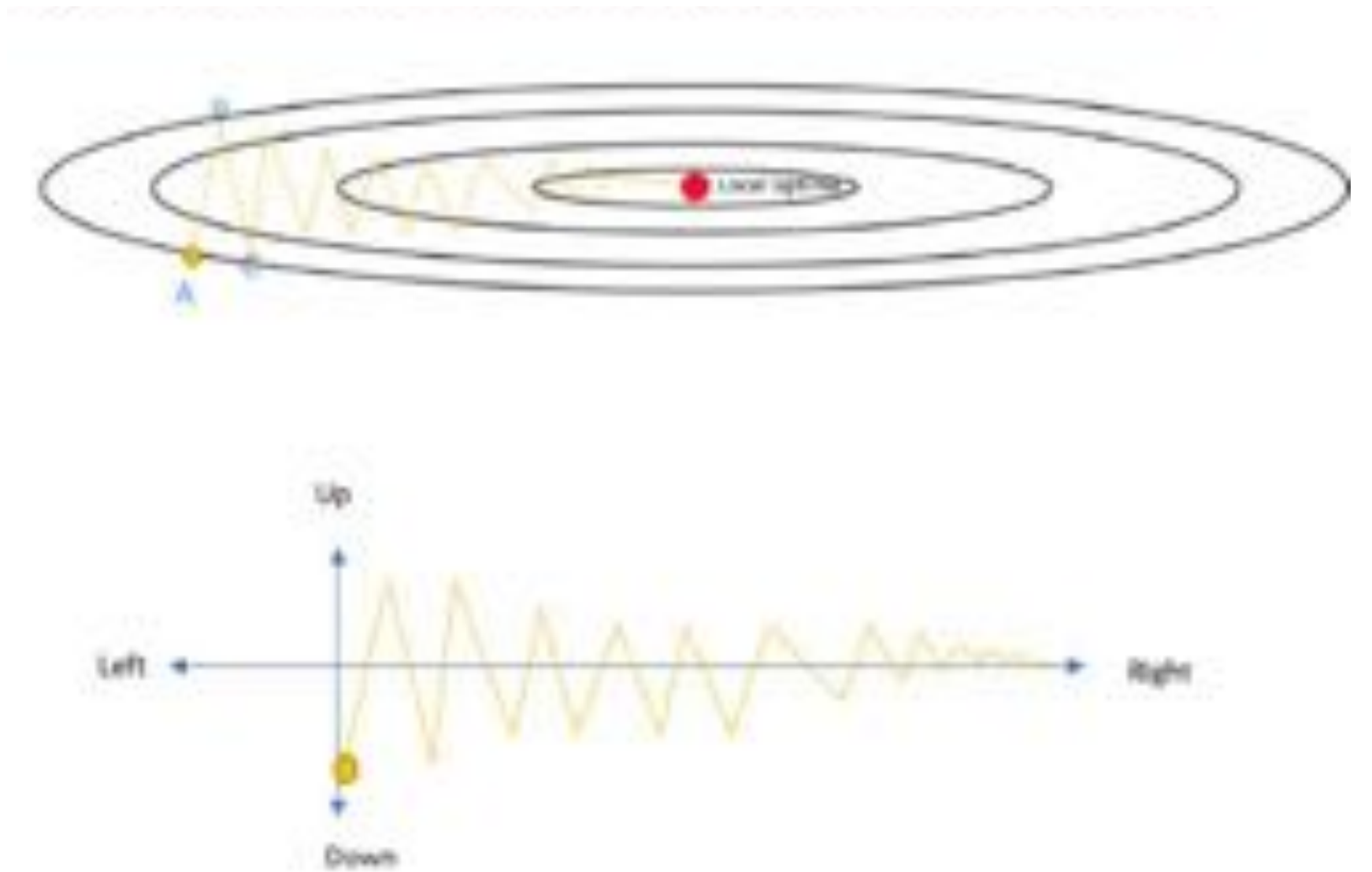
Gradient Descent with Momentum

- GD has trouble navigating ravines, i.e. areas where the surface curves much more steeply in one dimension than in another



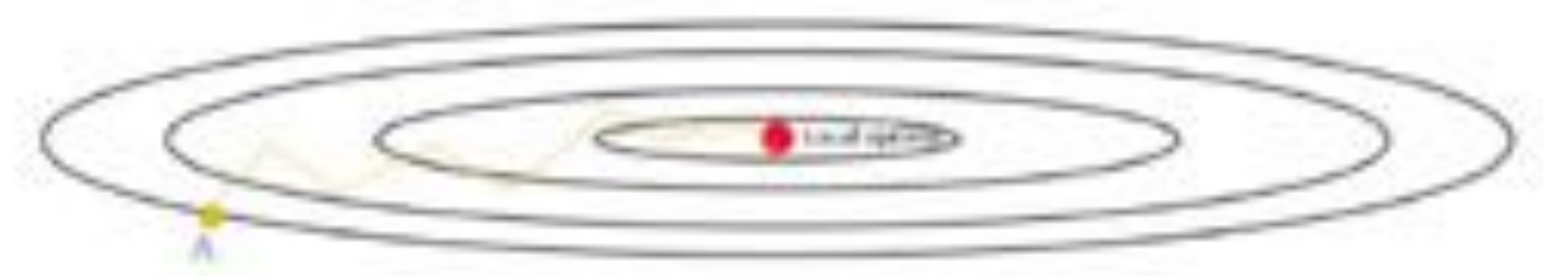
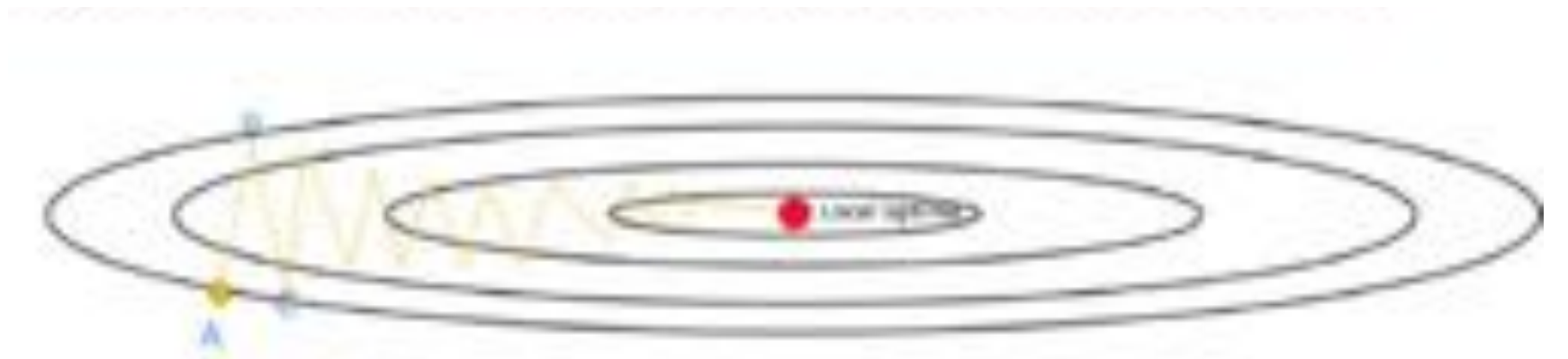
Gradient Descent with Momentum

- GD has trouble navigating ravines, i.e. areas where the surface curves much more steeply in one dimension than in another
- In these scenarios, SGD oscillates across the slopes of the ravine while only making hesitant progress along the bottom toward the local optimum



Gradient Descent with Momentum

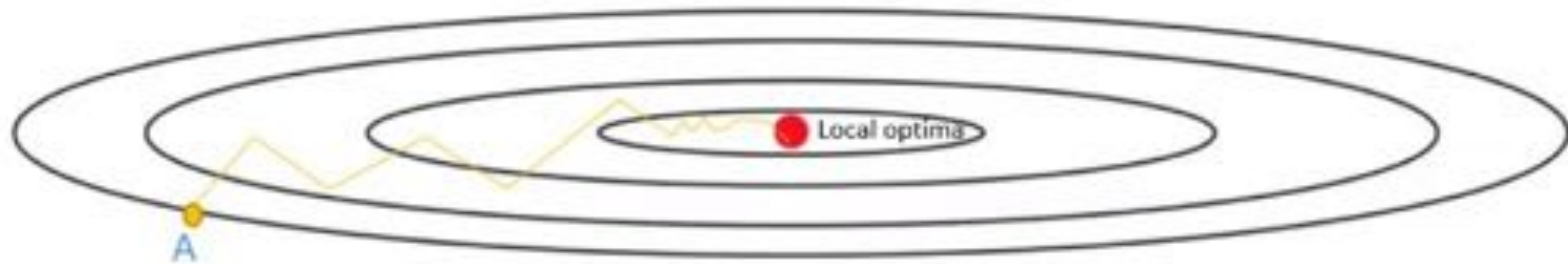
- GD has trouble navigating ravines, i.e. areas where the surface curves much more steeply in one dimension than in another
- In these scenarios, SGD oscillates across the slopes of the ravine while only making hesitant progress along the bottom toward the local optimum
- Momentum helps accelerate SGD in the relevant direction by adding a fraction γ of the update vector of the past time step to the current update vector



$$\mathbf{v}_t = \gamma \mathbf{v}_{t-1} + \eta \nabla_{\mathbf{w}} J(\theta)$$

$$\mathbf{w} = \mathbf{w} - \mathbf{v}_t$$

Gradient Descent with Momentum



- ▶ Gradient Descent

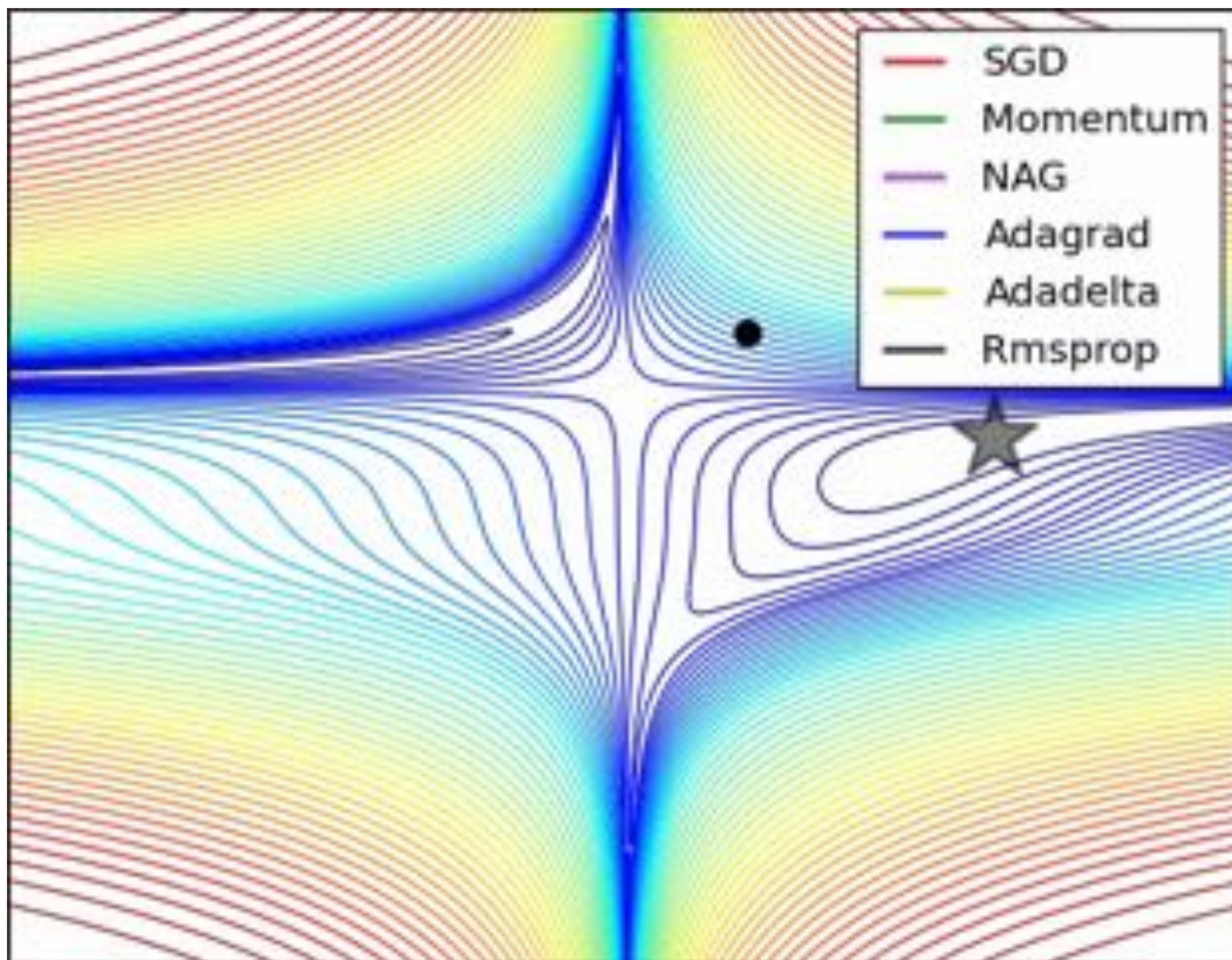
$$\mathbf{w} = \mathbf{w} - \eta \cdot \nabla_{\mathbf{w}} J(\mathbf{w})$$

$$\mathbf{w} = \mathbf{w} - \eta \cdot \nabla_{\mathbf{w}} J(\mathbf{w}; x^{(i)}; y^{(i)})$$

- ▶ Gradient Descent with momentum

$$\mathbf{v}_t = \gamma \mathbf{v}_{t-1} + \eta \nabla_{\mathbf{w}} J(\mathbf{w})$$

$$\mathbf{w} = \mathbf{w} - \mathbf{v}_t$$



Adagrad

- Adapts the learning rate
- smaller updates(i.e. low learning rates) for parameters associated with frequently occurring features,
- larger updates (i.e. high learning rates) for parameters associated with infrequent features. For this reason, it is well-suited for dealing with sparse data.
- Uses a different learning rate for every w_i at every time step t

$$g_{t,j} = \nabla_w J(w_{t,j})$$

$$w_{t+1,j} = w_{t,j} - \frac{\eta}{\sqrt{G_{t,jj} + \epsilon}} \cdot g_{t,j}$$

- $G_t \in \mathbb{R}^{d \times d}$ is a diagonal matrix where each diagonal element j,j is the sum of the squares of the gradients w.r.t. w_j up to time step t

Adadelta & RMS Prop

- Adagrad's main weakness is its accumulation of the squared gradients in the denominator

$$t \rightarrow \infty \quad \eta \rightarrow 0$$

- Instead of inefficiently storing previous squared gradients, the sum of gradients is recursively defined as a decaying average of all past squared gradients.

$$E[g^2]_t = \gamma E[g^2]_{t-1} + (1 - \gamma)g_t^2$$

$$w_{t+1,j} = w_{t,j} - \frac{\eta}{\sqrt{E[g^2]_t + \epsilon}} \cdot g_t$$

$$\Delta w_t = -\frac{\eta}{RMS[g]_t} g_t$$

- RMS Prop

$$E[g^2]_t = 0.9E[g^2]_{t-1} + 0.1g_t^2$$

Summary

► Variants

- Vanilla / Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent

► Optimization Algos

- Momentum
- Nesterov accelerated gradient
- Adagrad
- Adadelata
- RMSprop

$$\mathbf{w} = \mathbf{w} - \eta \cdot \nabla_{\mathbf{w}} J(\mathbf{w})$$

$$\mathbf{w} = \mathbf{w} - \eta \cdot \nabla_{\mathbf{w}} J(\mathbf{w}; x^{(i)}; y^{(i)})$$

$$\mathbf{w} = \mathbf{w} - \eta \cdot \nabla_{\mathbf{w}} J(\mathbf{w}; x^{(i:i+n)}; y^{(i:i+n)})$$

$$\mathbf{v}_t = \gamma \mathbf{v}_{t-1} + \eta \nabla_{\mathbf{w}} J(\mathbf{w})$$

$$\mathbf{w} = \mathbf{w} - \mathbf{v}_t$$

$$\mathbf{v}_t = \gamma \mathbf{v}_{t-1} + \eta \nabla_{\mathbf{w}} J(\mathbf{w} - \gamma \mathbf{v}_{t-1})$$

$$\mathbf{w} = \mathbf{w} - \mathbf{v}_t$$

$$w_{t+1,i} = w_{t,i} - \frac{\eta}{\sqrt{G_{t,ii} + \epsilon}} \cdot g_{t,i}$$

$$E[g^2]_t = \gamma E[g^2]_{t-1} + (1 - \gamma) g_t^2$$
$$w_{t+1,i} = w_{t,i} - \frac{\eta}{\sqrt{E[g^2]_t + \epsilon}} \cdot g_{t,i}$$

$$E[g^2]_t = 0.9 E[g^2]_{t-1} + 0.1 g_t^2$$

More Optimizers

► Optimization Algos

► Adam

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t & \hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \end{aligned}$$

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t$$

► AdaMax

$$\begin{aligned} u_t &= \beta_2^\infty v_{t-1} + (1 - \beta_2^\infty) |g_t|^\infty & w_{t+1} &= w_t - \frac{\eta}{u_t} \hat{m}_t \\ &= \max(\beta_2 \cdot v_{t-1}, |g_t|) \end{aligned}$$

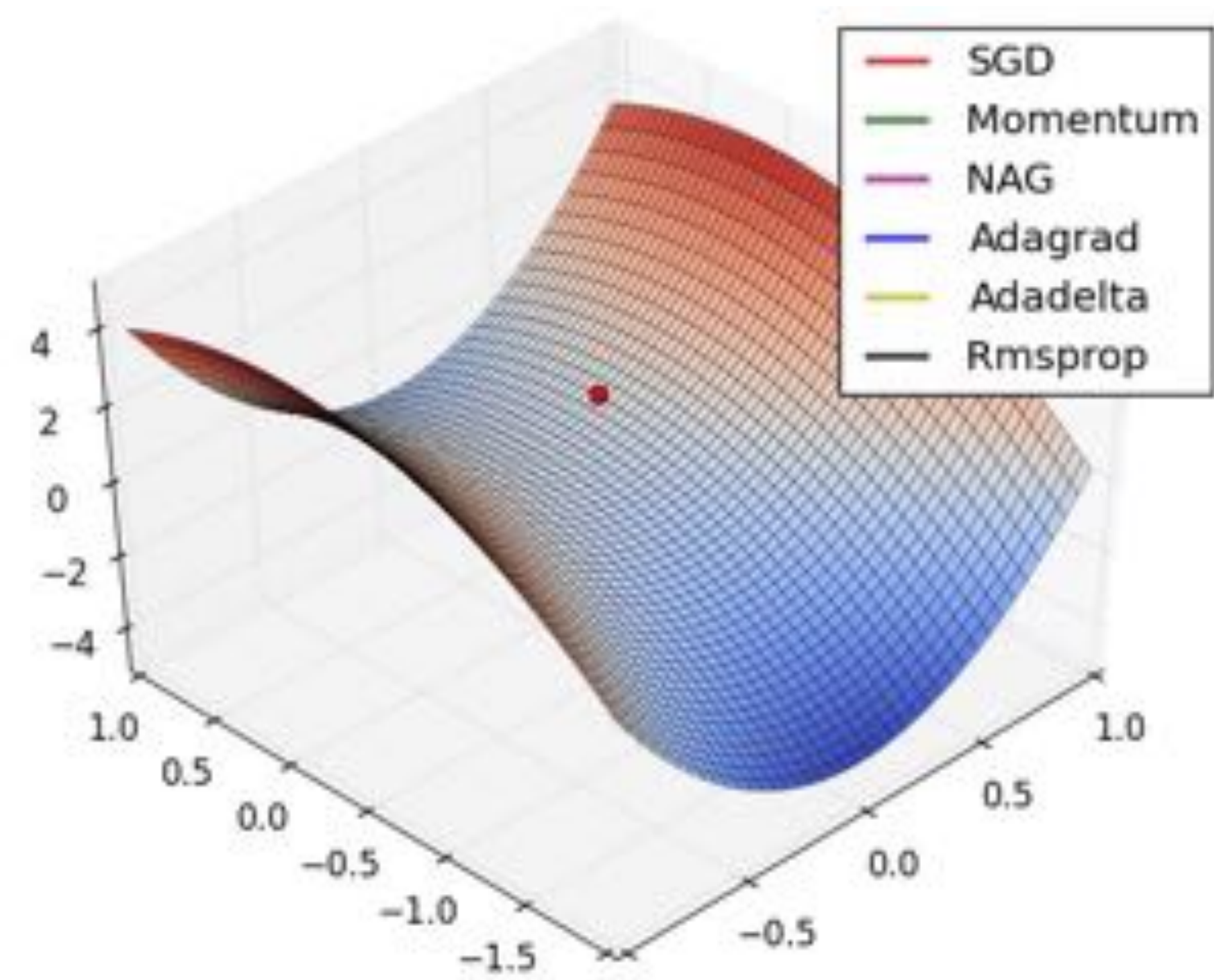
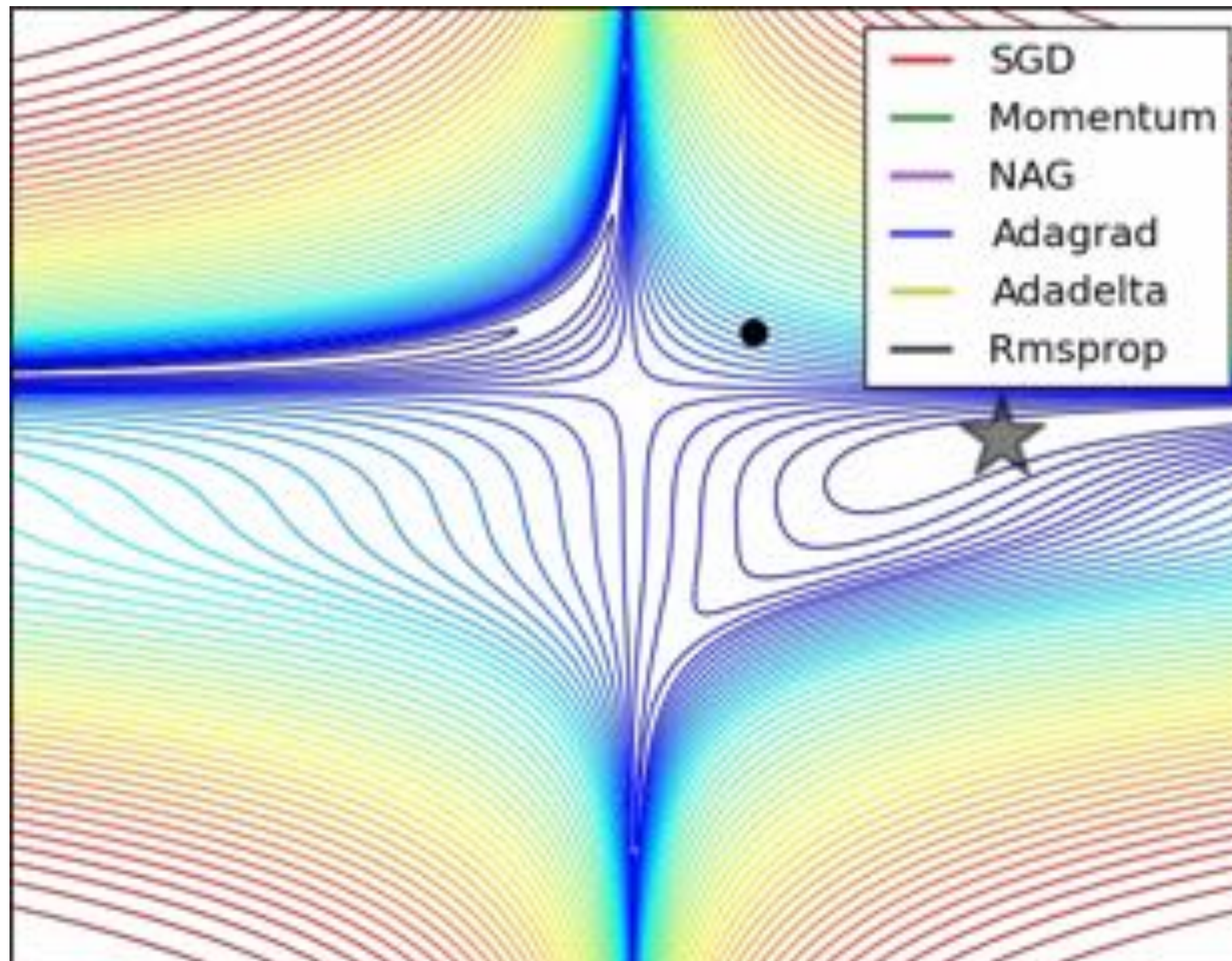
► Nadam

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \left(\beta_1 \hat{m}_t + \frac{(1 - \beta_1) g_t}{1 - \beta_1^t} \right)$$

► AMSGrad

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{v}_t &= \max(\hat{v}_{t-1}, v_t) \\ w_{t+1} &= w_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} m_t \end{aligned}$$

Optimizer Comparison



Reading

Optimising Gradient Descent

<https://runder.io/optimizing-gradient-descent/>

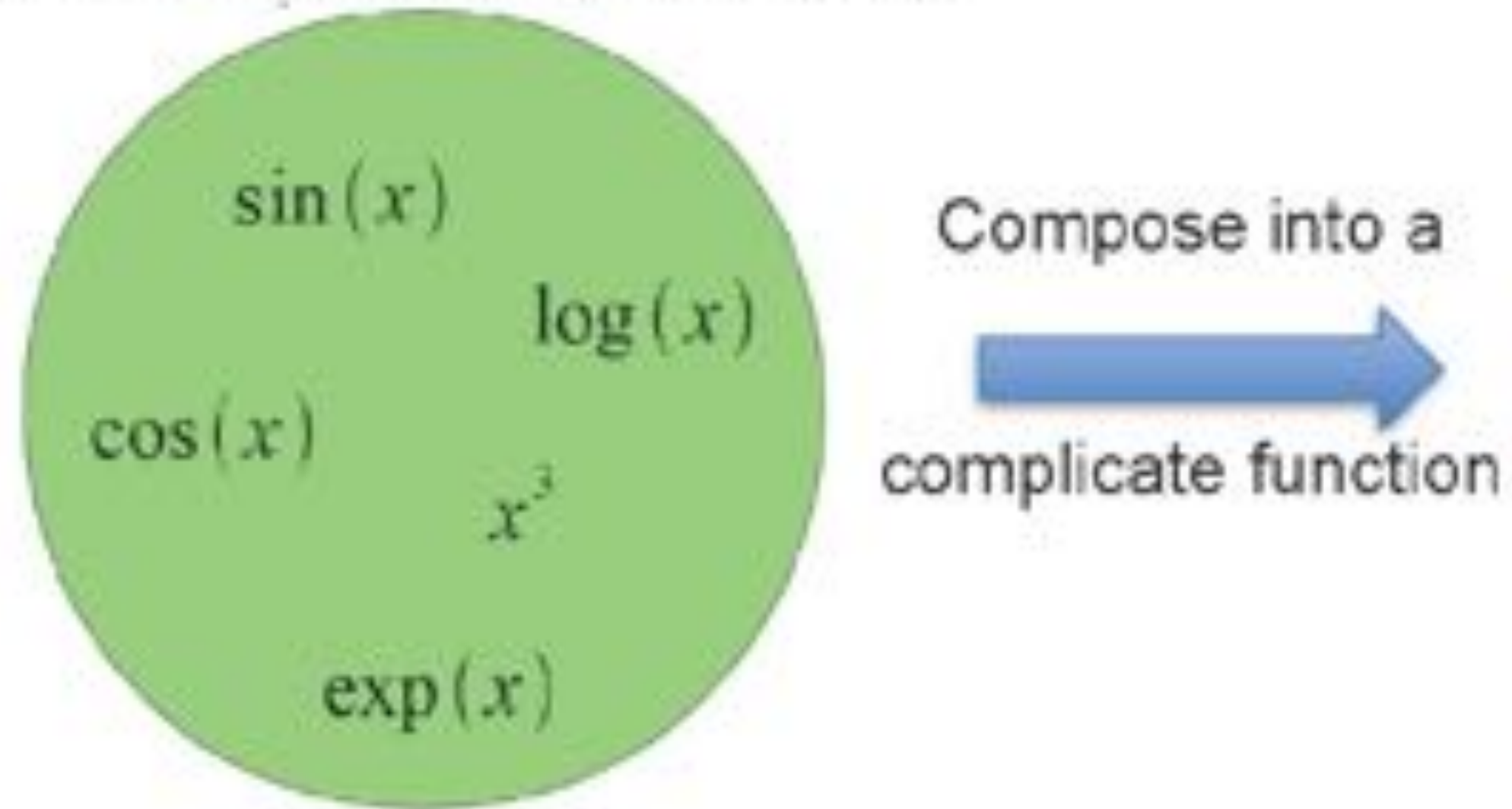
Ch8: Optimization for Training Deep Models

<https://www.deeplearningbook.org/contents/optimization.html>

Deep Learning vs. Shallow Learning

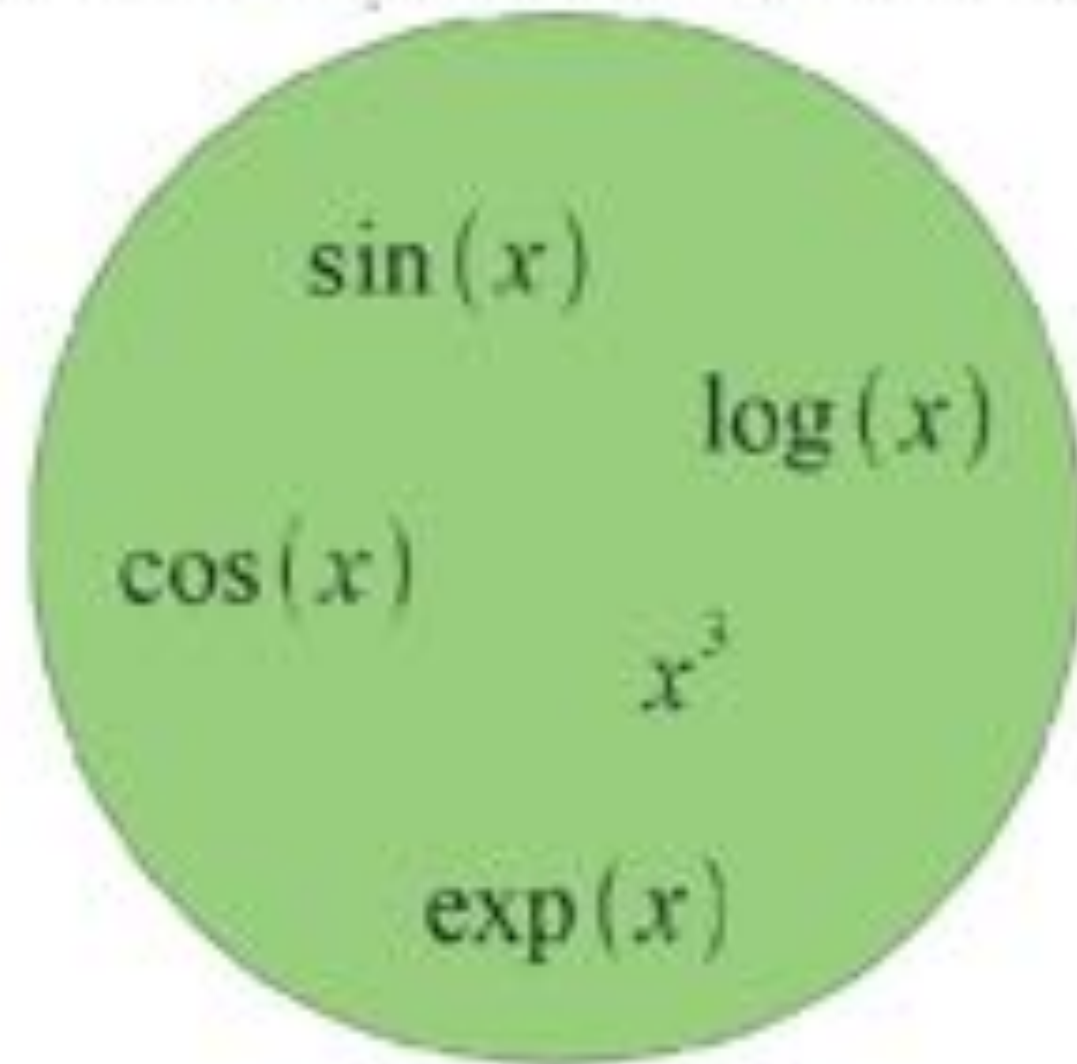
Building a Complicated Function

Given a library of simple functions



Building a Complicated Function

Given a library of simple functions



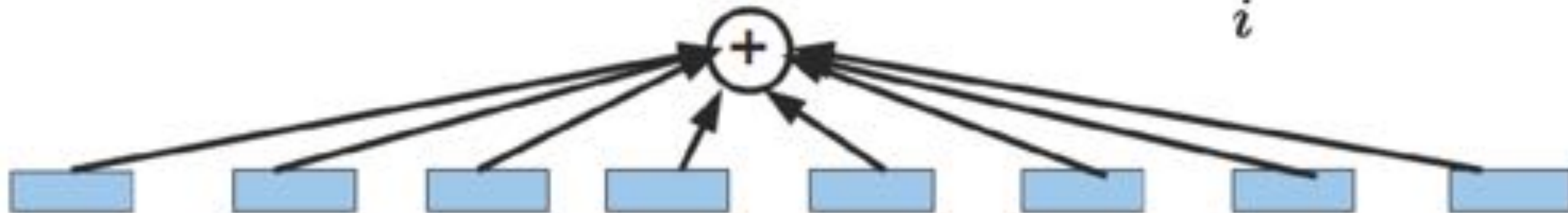
Compose into a
complicate function



Idea 1: Linear Combinations

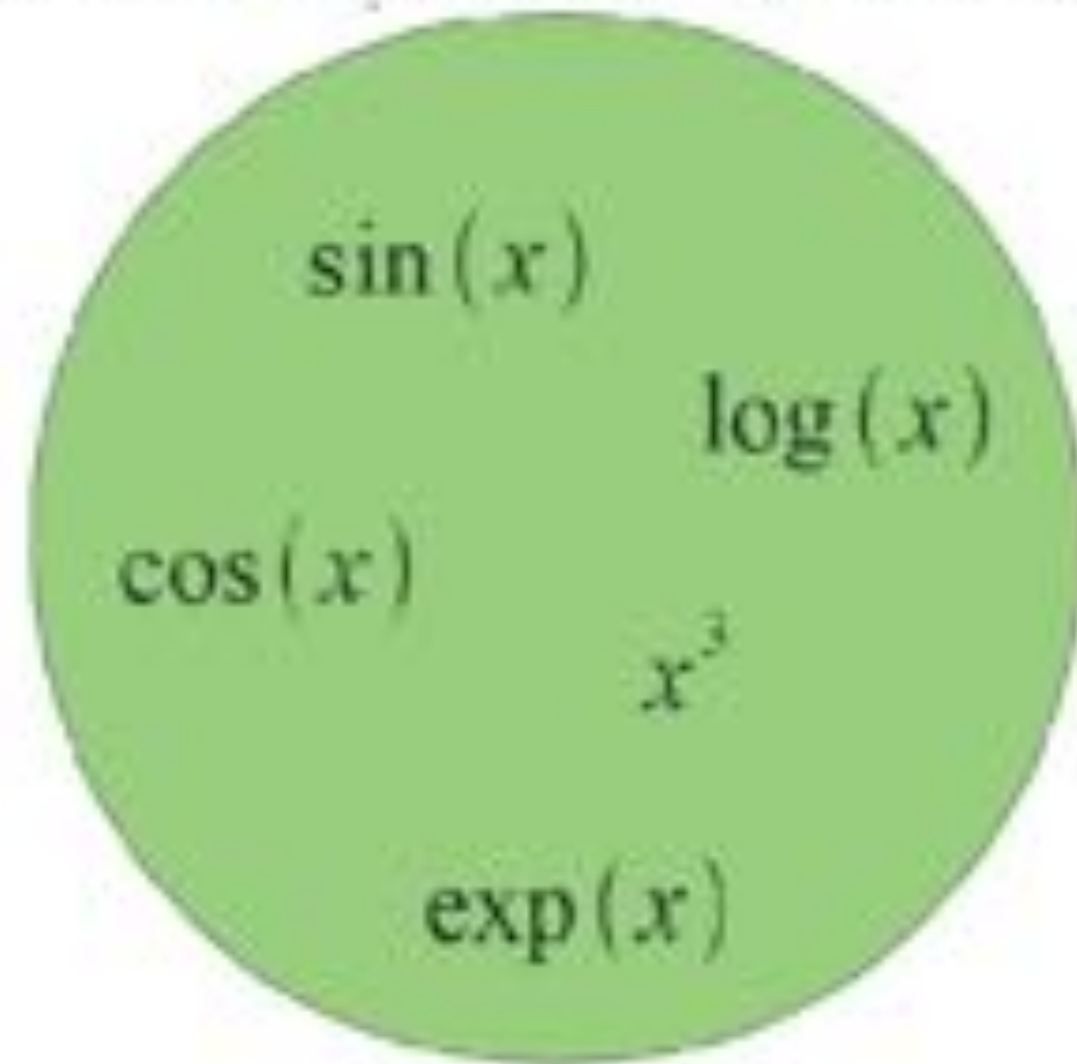
- Boosting
- Kernels
- ...

$$f(x) = \sum_i \alpha_i g_i(x)$$



Building a Complicated Function

Given a library of simple functions



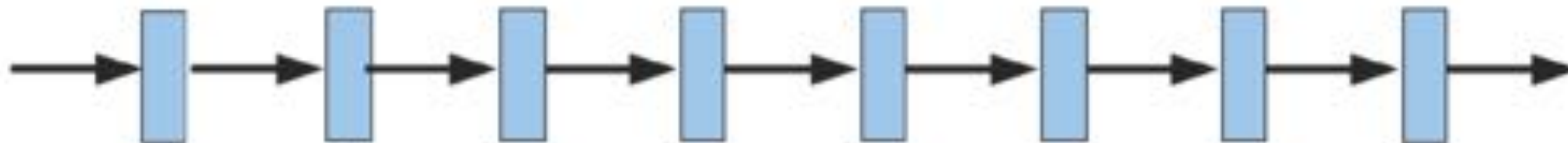
Compose into a

 complicate function

Idea 2: Compositions

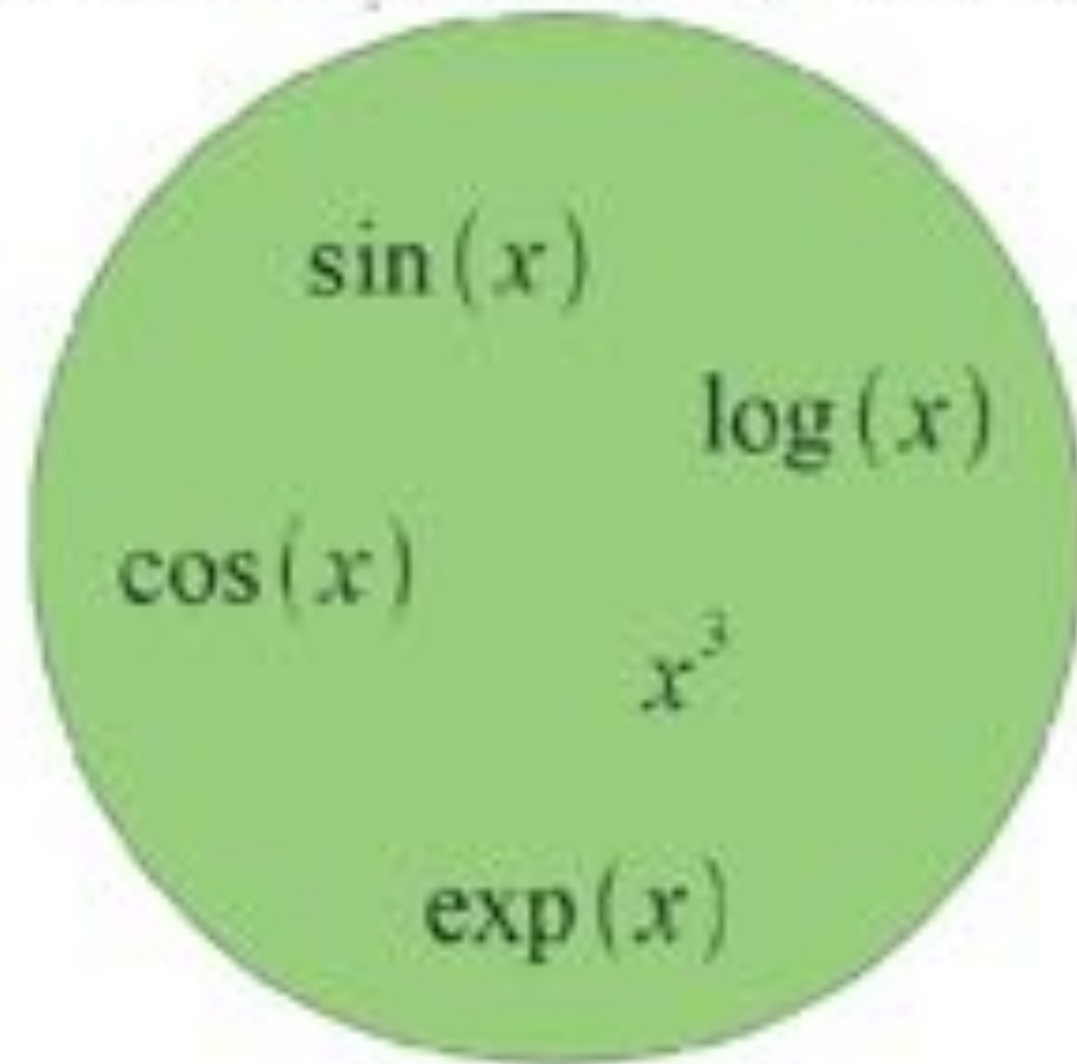
- Deep Learning
- Grammar models
- Scattering transforms...

$$f(x) = g_1(g_2(\dots(g_n(x) \dots)))$$



Building a Complicated Function

Given a library of simple functions



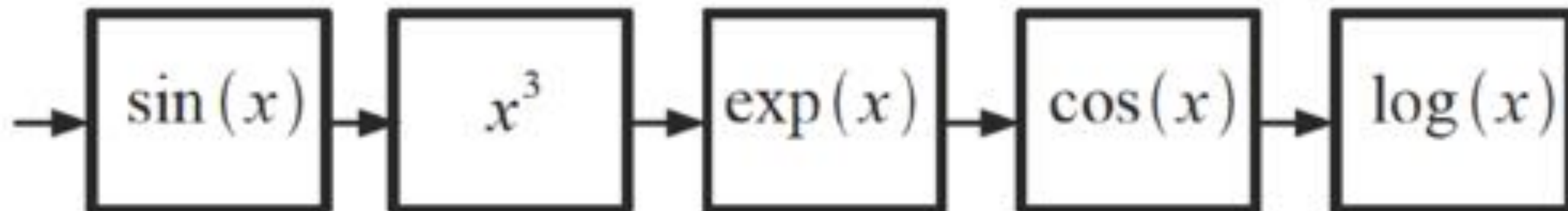
Compose into a

 complicate function

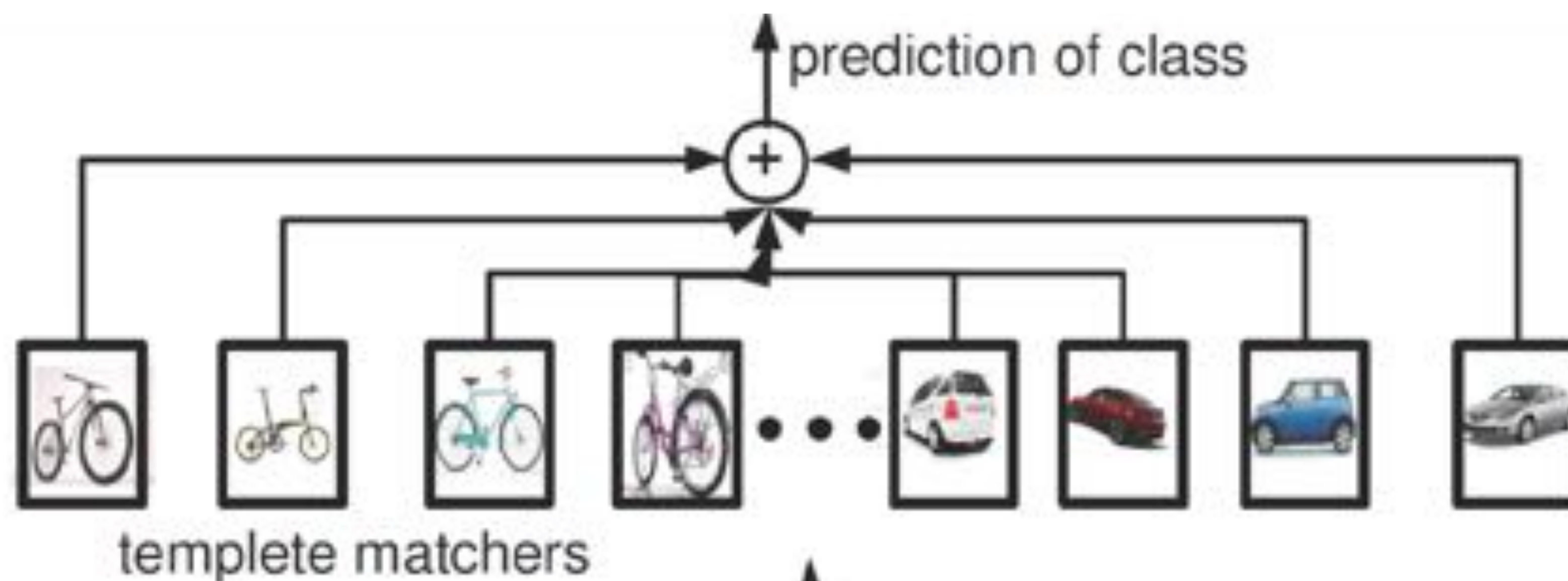
Idea 2: Compositions

- Deep Learning
- Grammar models
- Scattering transforms...

$$f(x) = \log(\cos(\exp(\sin^3(x))))$$



Linear Combination

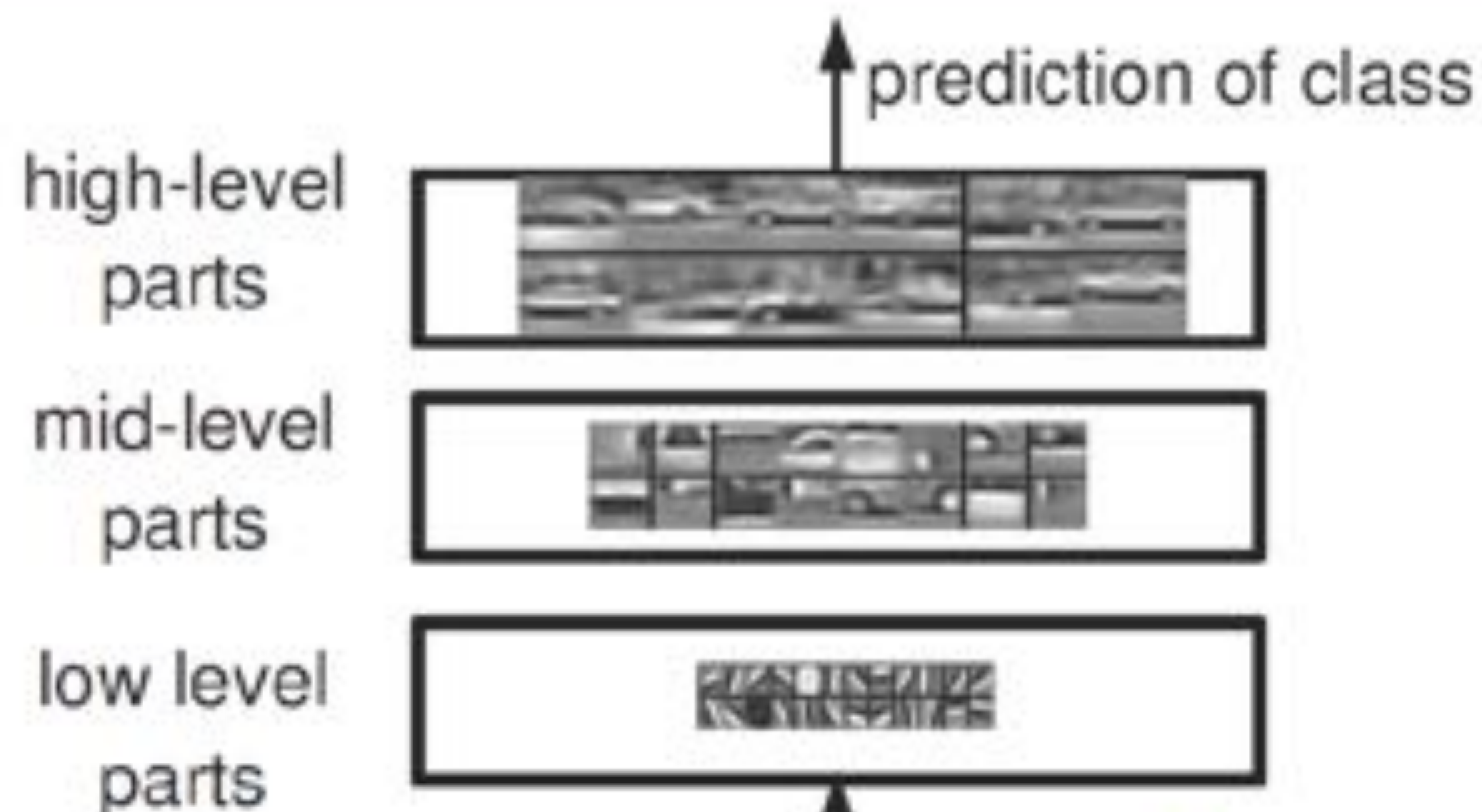


**BAD: it may require
an exponential nr. of
templates!!!**



Input image

Composition

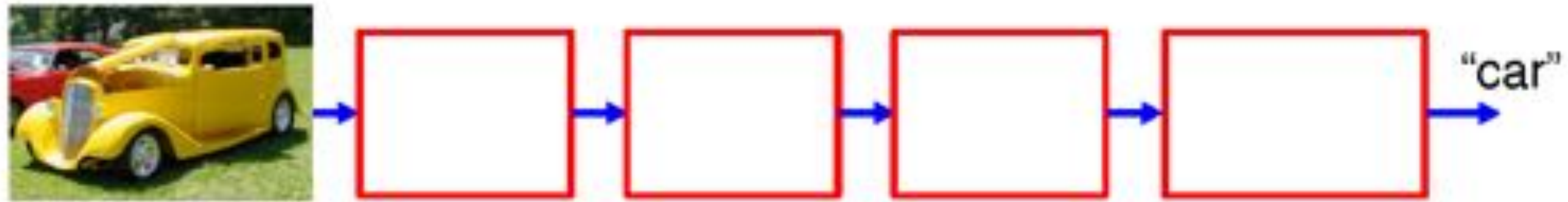


Input image

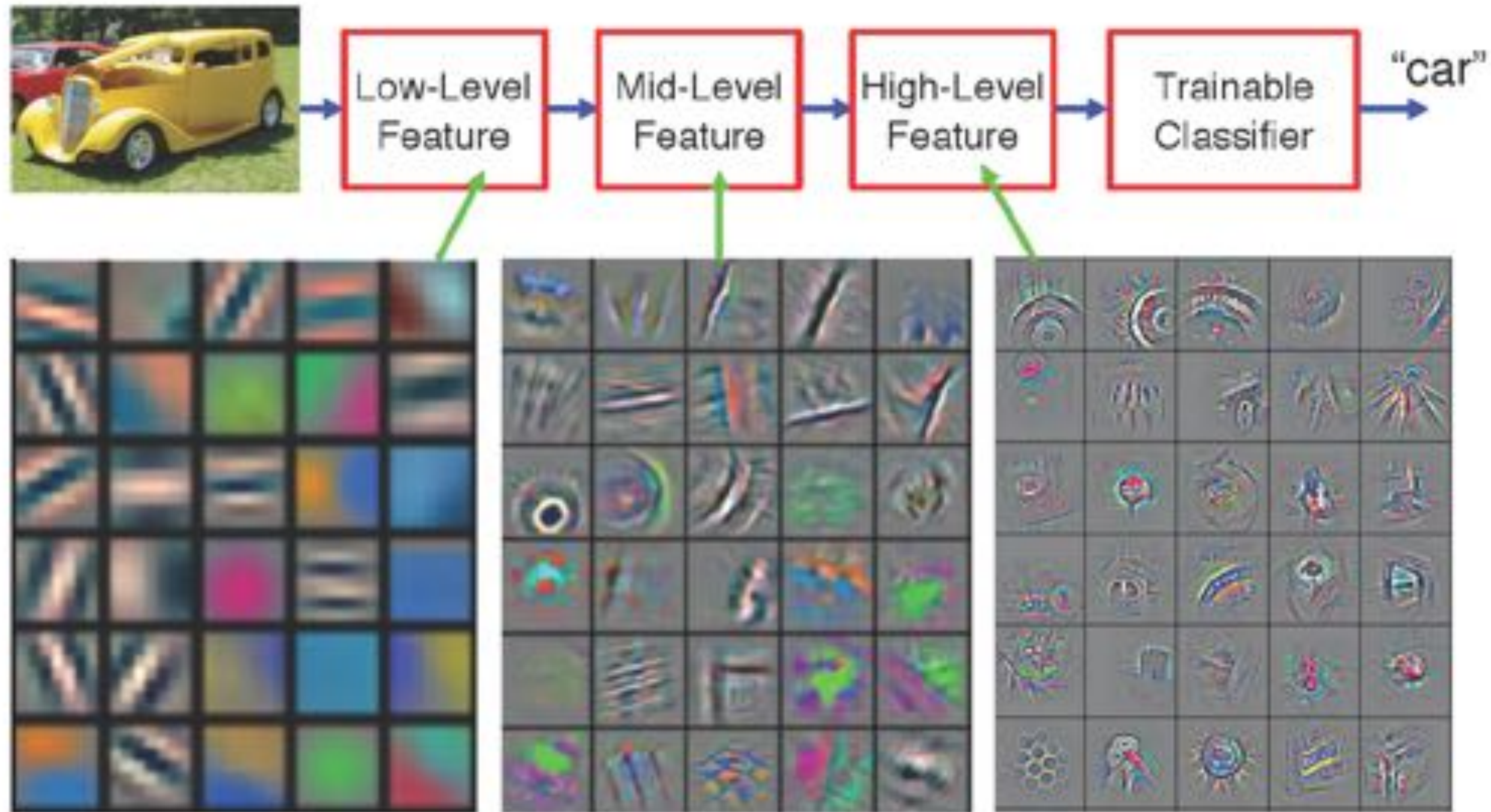
- reuse of intermediate parts
- distributed representations

**GOOD: (exponentially)
more efficient**

Deep Learning = Hierarchical Composition



Deep Learning = Hierarchical Composition



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]



Face detectors



Face parts
(combination
of edges)



edges

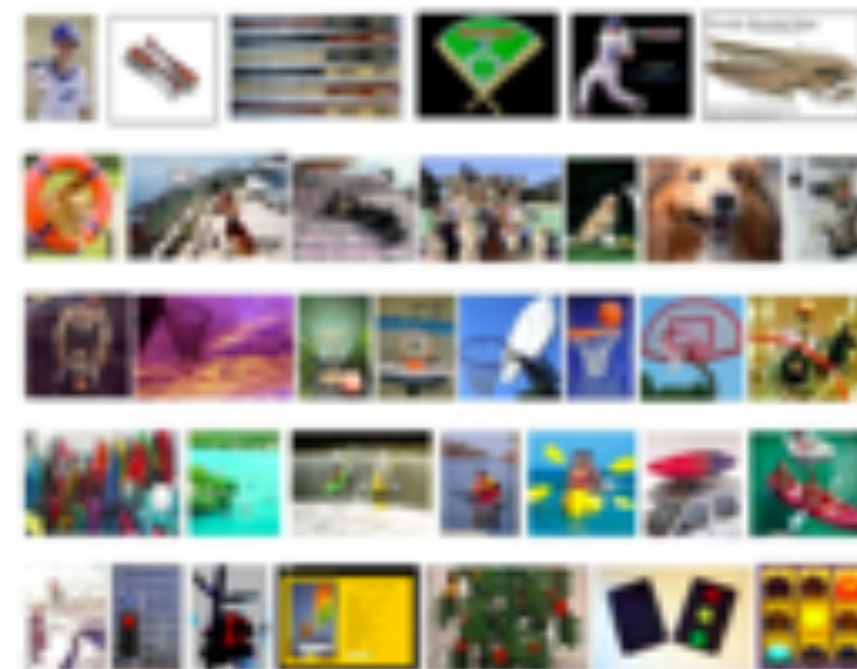


pixels

Convolution Neural Networks

Relentless Research on Visual Recognition

Caltech 101



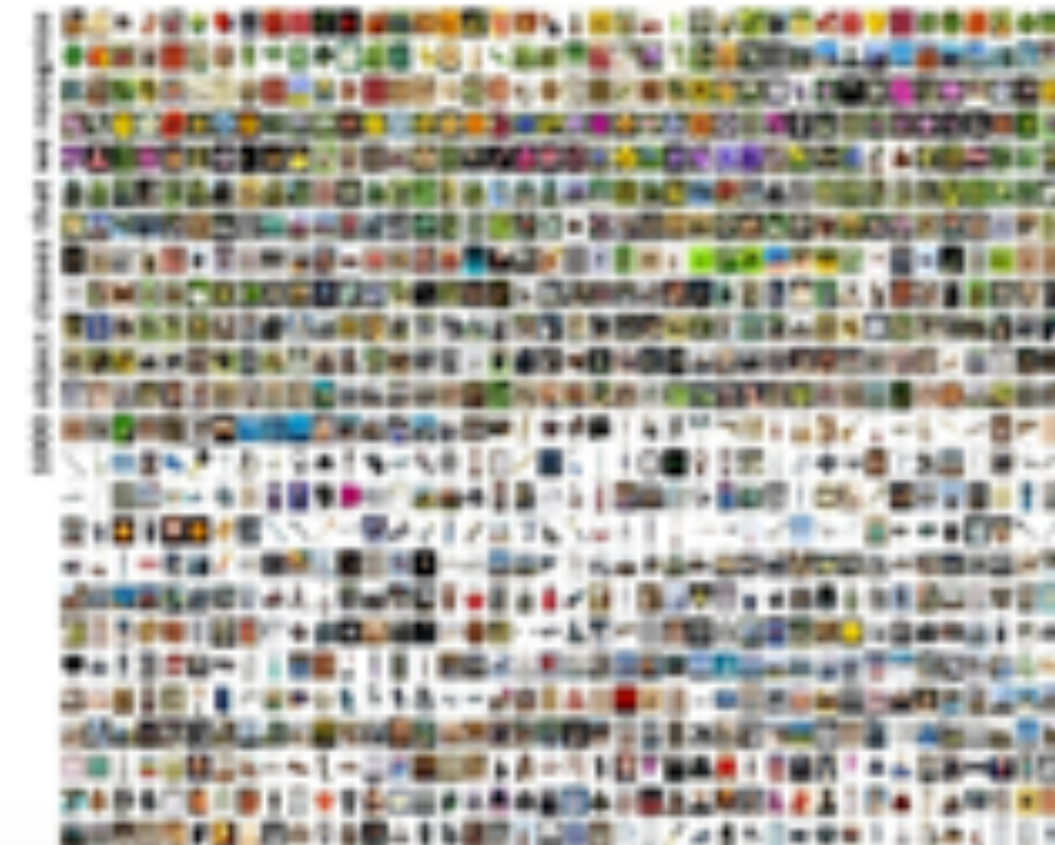
80 Million Tiny Images



PASCAL VOC



ImageNet





IMGENET

www.image-net.org

22K categories and **14M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
- Food
- Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
- Scenes
 - Indoor
 - Geological Formations
- Sport Activities

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes

1,431,167 images



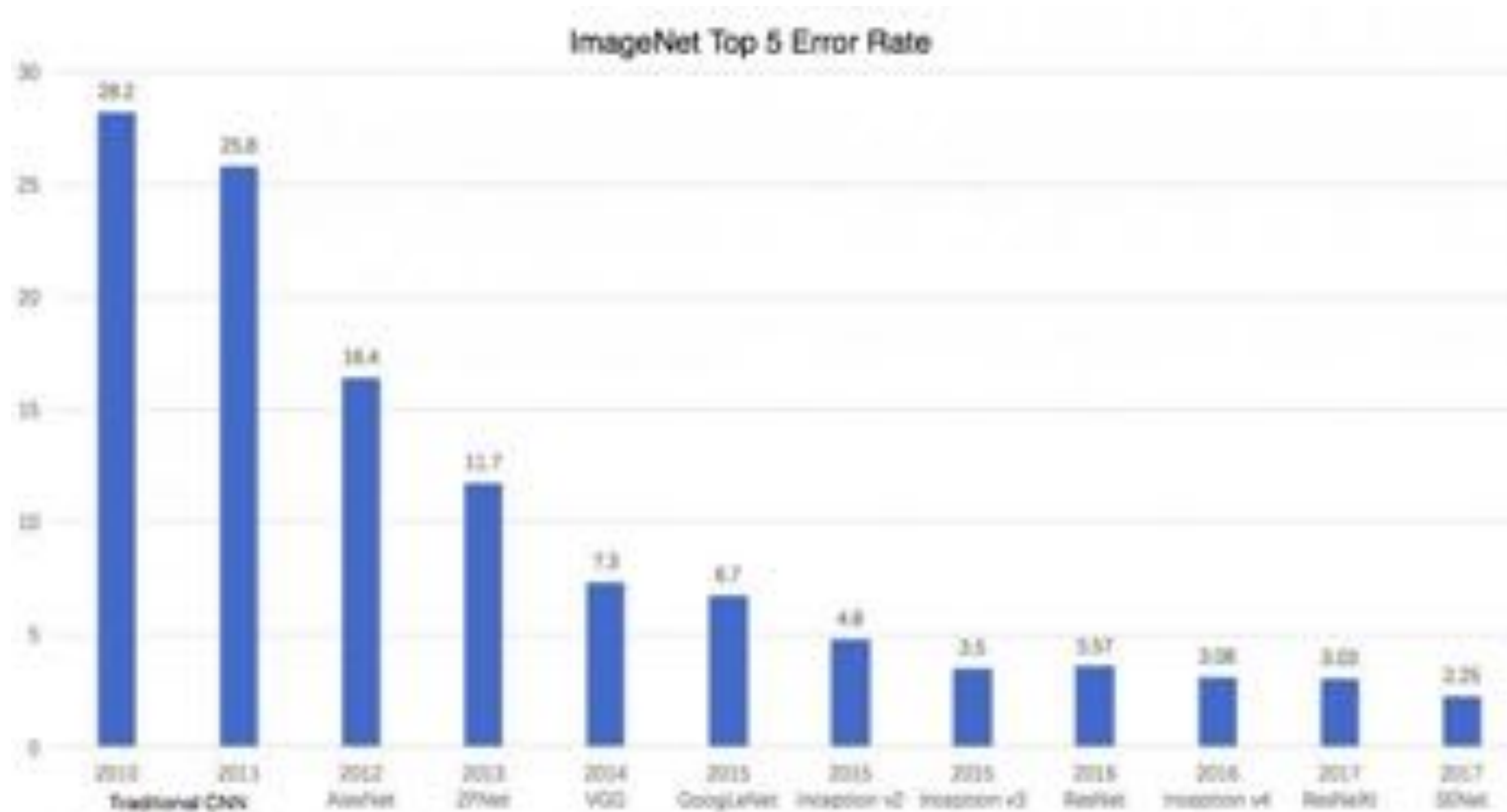
Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle



Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle

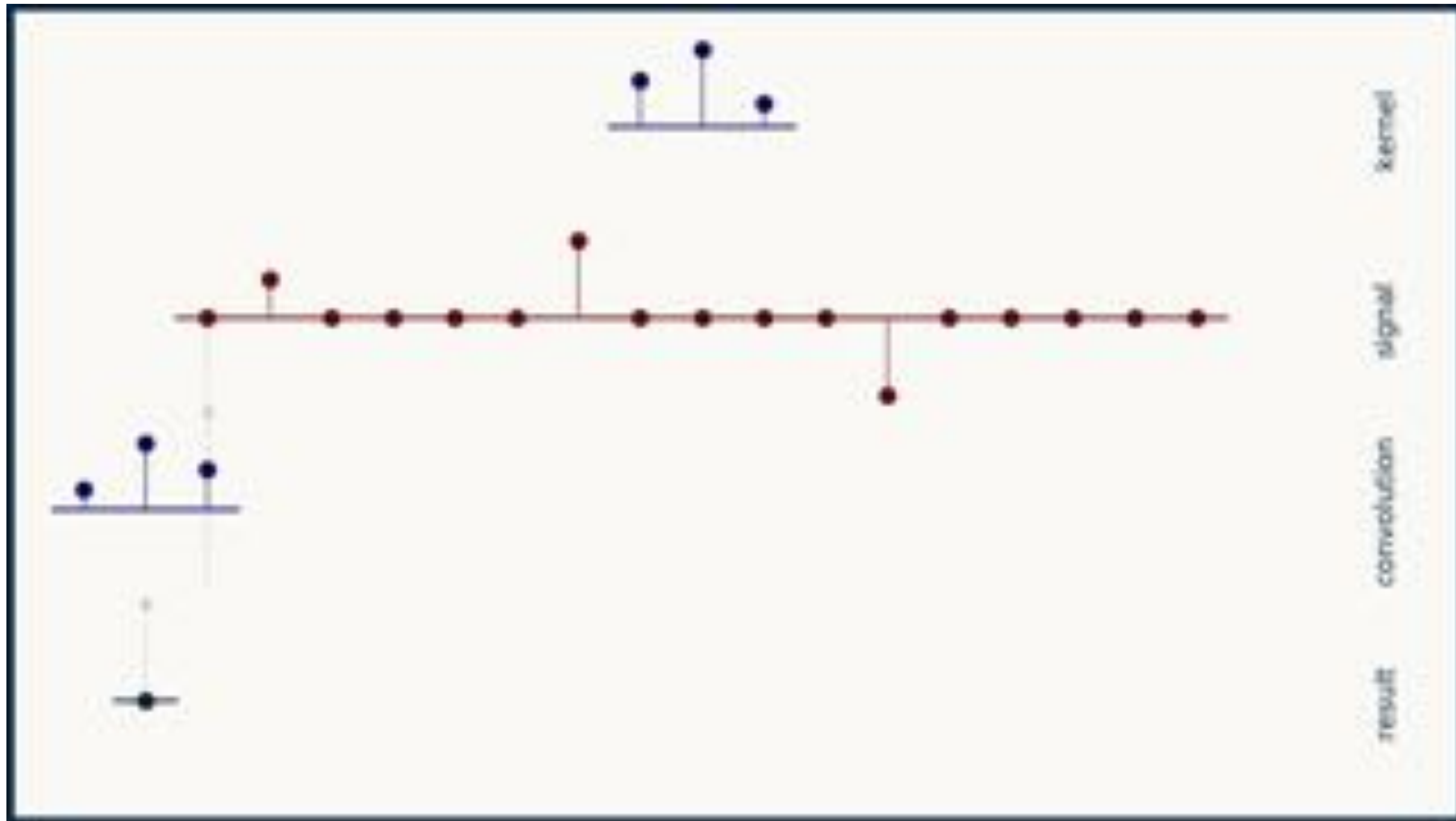


ImageNet Challenge



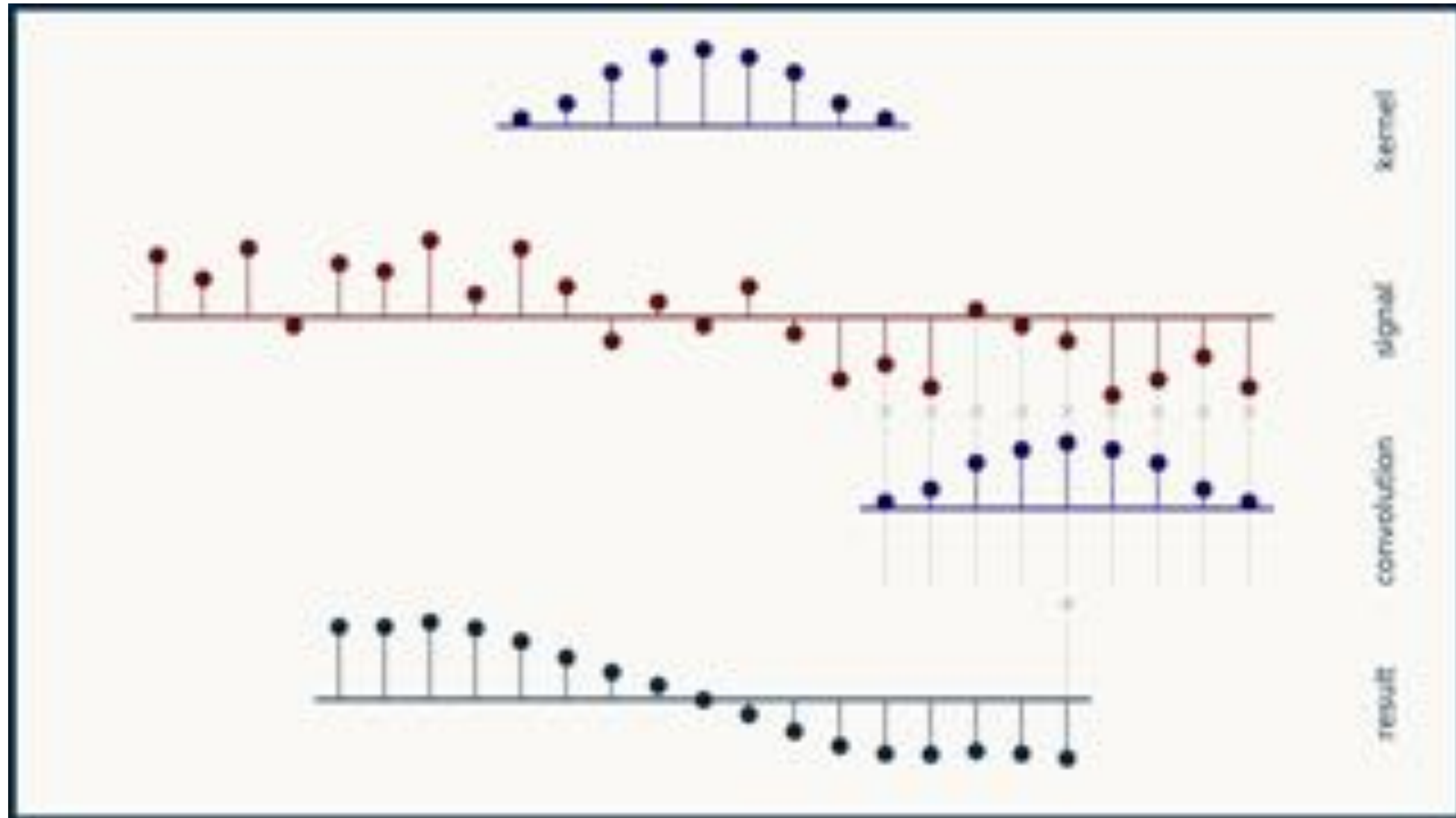
1D Convolution

1D Convolution



https://e2eml.school/images/conv1d/bb_copy.gif

1D Convolution



https://e2eml.school/images/conv1d/bb_copy.gif

Example

M - 2*round down(filter length/2)

[illegible]

2D Convolution

Digital Image

0
Black



255
White

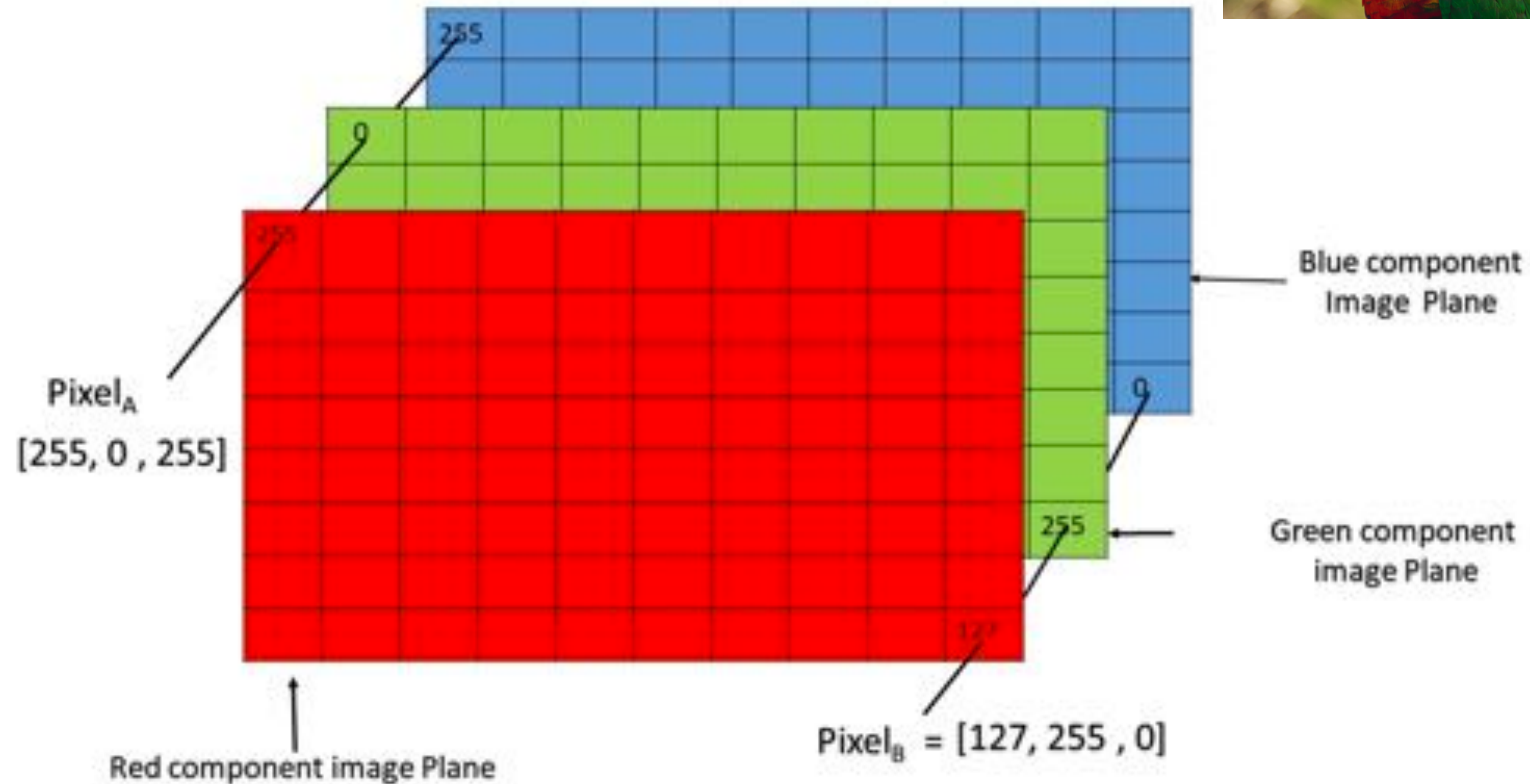


$f(x+1,y+0) - f(x,y)$
 $44-57 = -13$

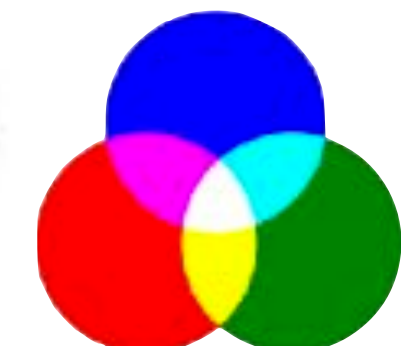
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2	184	193	188	187	191	196	207	209	156	154	179	163	133	119	128	155	178	195	175	198	221	197	169	171
3	168	185	187	178	185	205	220	210	173	161	171	166	168	165	165	174	185	174	166	212	211	203	186	182
4	150	177	182	168	180	211	222	199	184	172	173	181	196	196	189	181	164	143	186	181	187	199	188	183
5	158	179	164	144	170	211	215	180	175	179	149	131	147	178	200	173	127	163	172	164	176	191	191	187
6	165	164	133	129	162	210	214	171	164	124	76	64	70	94	161	154	160	169	157	168	177	175	187	186
7	158	182	117	131	156	201	208	176	153	57	44	71	74	60	100	177	200	171	157	172	171	166	180	183
8	151	117	115	138	153	181	200	188	124	59	45	67	77	63	95	196	209	174	162	168	163	167	171	167
9	150	132	142	154	156	171	201	196	90	54	58	67	68	66	113	185	203	169	153	157	149	155	155	149
10	146	150	173	166	146	175	217	212	136	69	44	61	64	60	124	156	190	172	140	140	133	133	140	144
11	135	140	164	172	167	201	234	226	182	130	36	40	49	52	125	118	190	201	169	145	141	138	134	144
12	116	123	142	182	209	227	239	235	198	94	33	33	35	53	111	78	170	221	203	159	147	141	139	148
13	103	119	141	183	215	223	237	206	144	60	41	38	41	71	78	78	108	201	211	161	136	130	146	156
14	131	146	162	189	199	177	169	165	170	64	53	60	94	153	108	68	66	137	209	179	150	120	127	134
15	146	166	160	137	135	161	197	214	191	66	56	94	140	160	109	61	65	85	188	195	167	117	98	100
16	128	127	115	152	180	198	205	208	177	57	60	113	147	139	89	65	68	70	142	189	172	131	101	90
17	74	108	175	184	167	177	179	189	157	55	61	90	114	72	50	63	69	66	95	166	158	130	103	83
18	87	139	180	179	173	164	146	153	140	58	65	73	83	58	42	53	63	64	72	122	125	113	91	75
19	104	131	132	176	196	157	138	134	118	49	63	47	49	46	41	47	55	59	62	80	107	113	85	68
20	135	139	109	148	158	123	132	149	129	49	42	33	29	33	37	39	44	51	56	63	98	102	73	60

20 x 24 gray scale image
Pixel value: 0 - 255

Digital Image



Pixel of an RGB image are formed from the corresponding pixel of the three component images



Digital Image



R



G



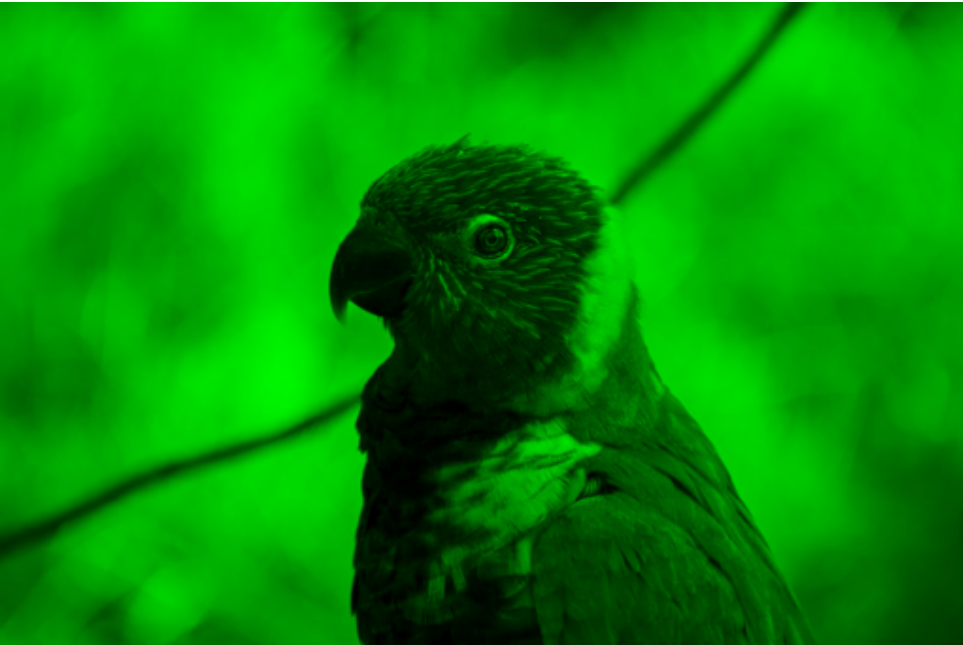
B



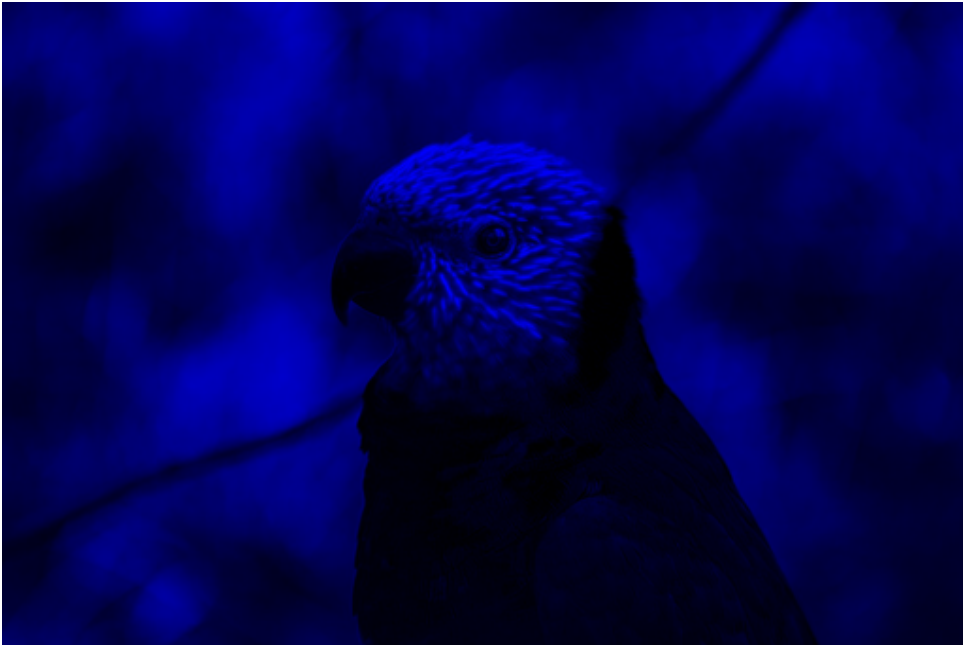
R



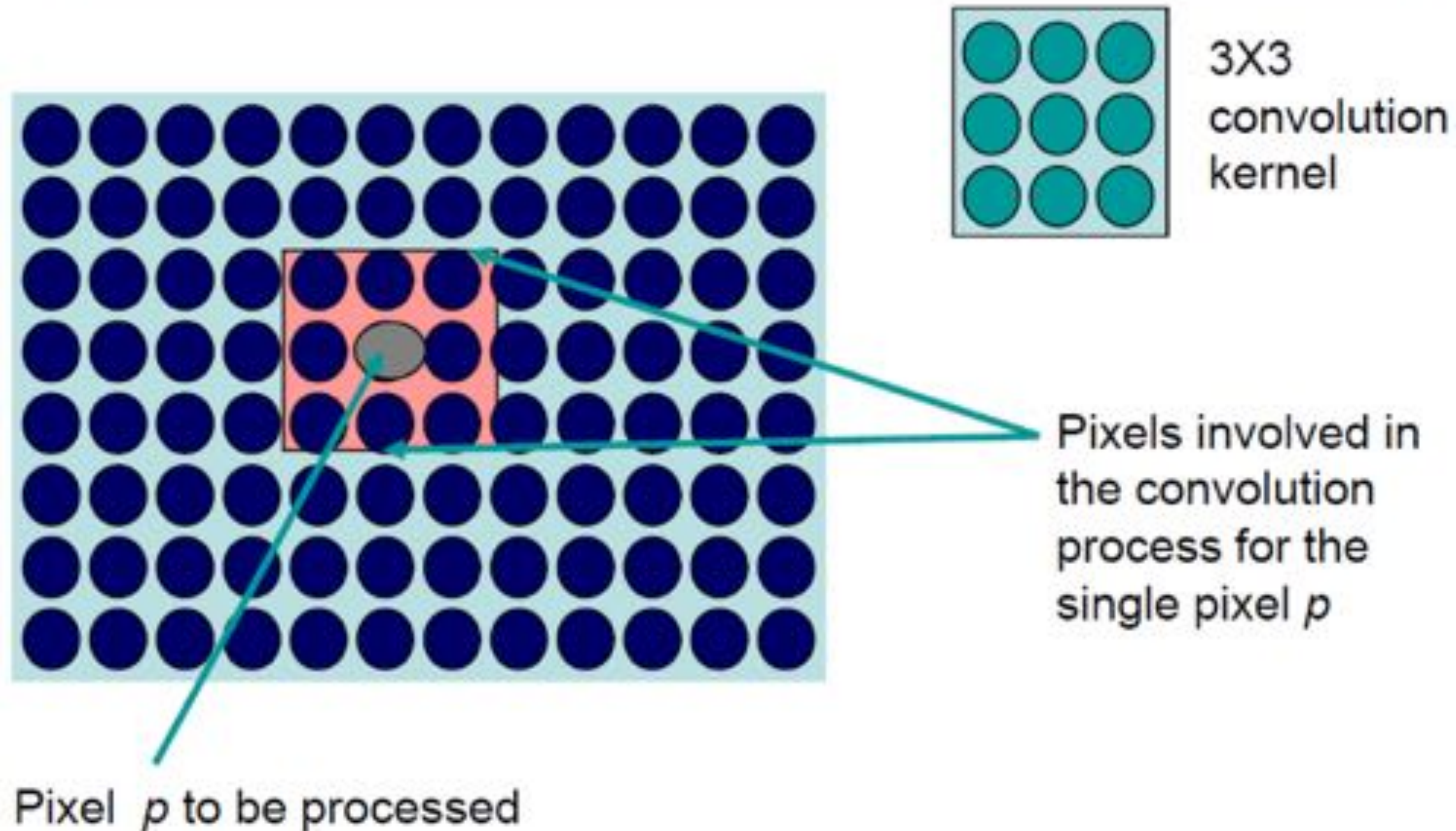
G



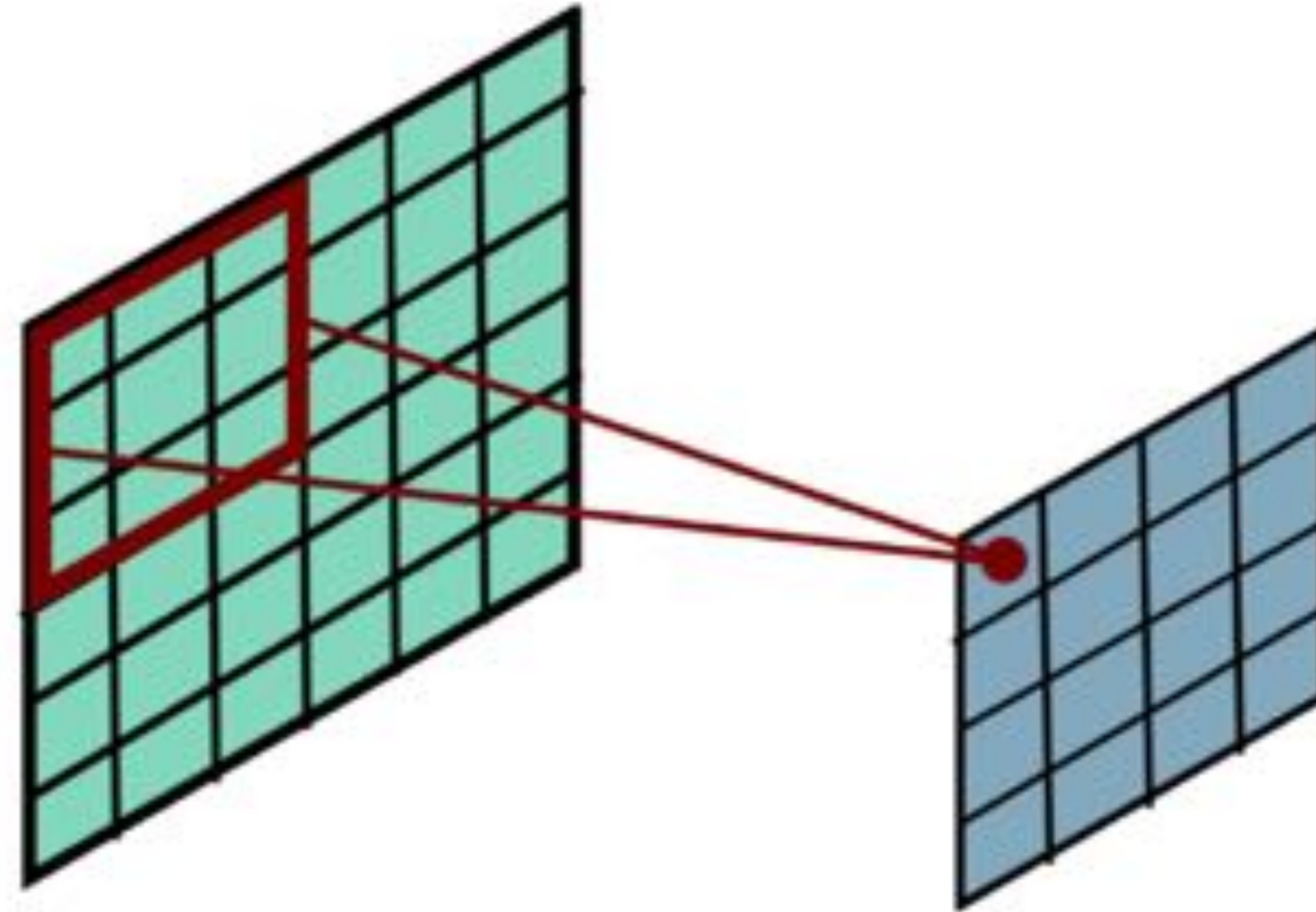
B



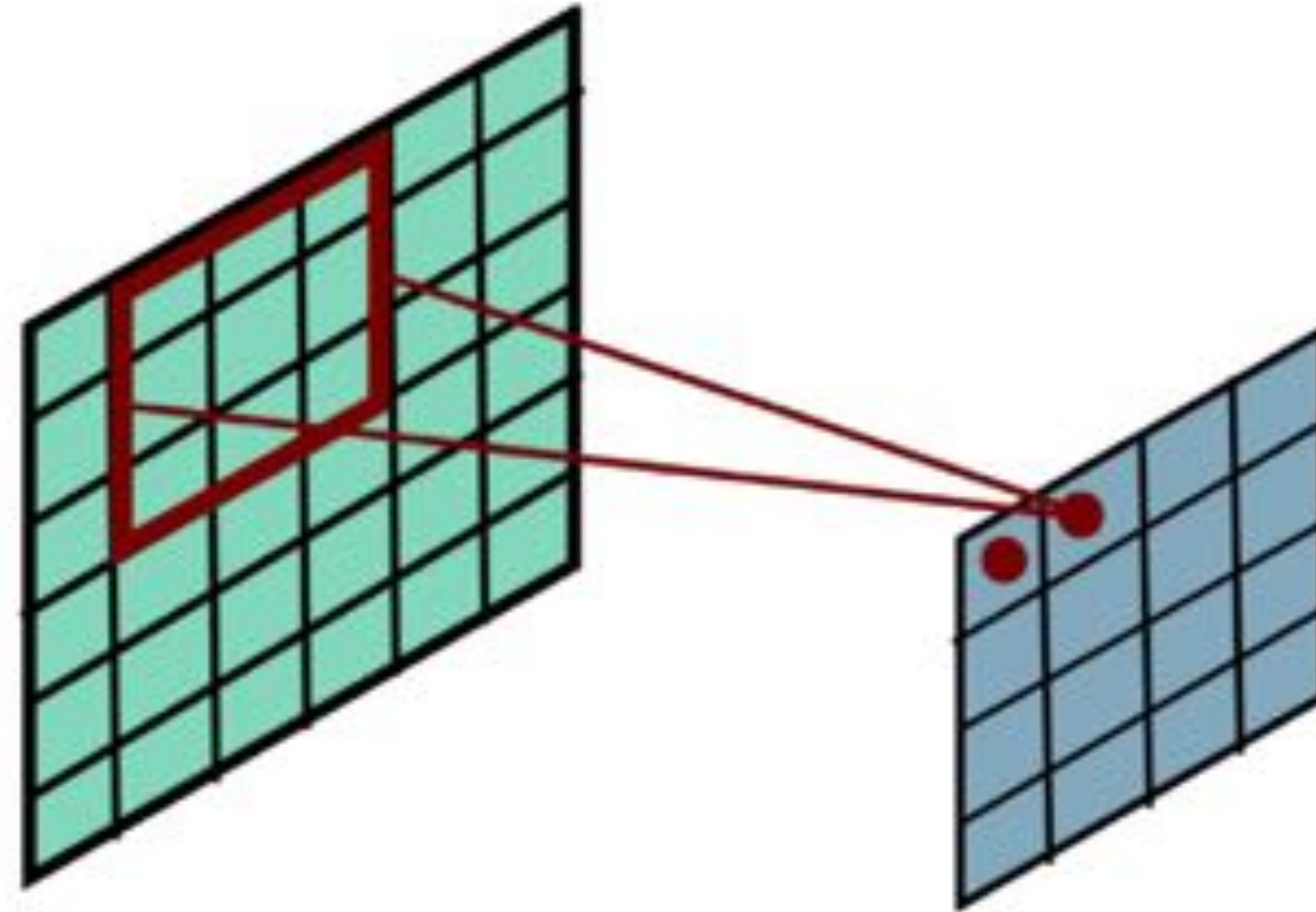
Discrete 2D Convolution



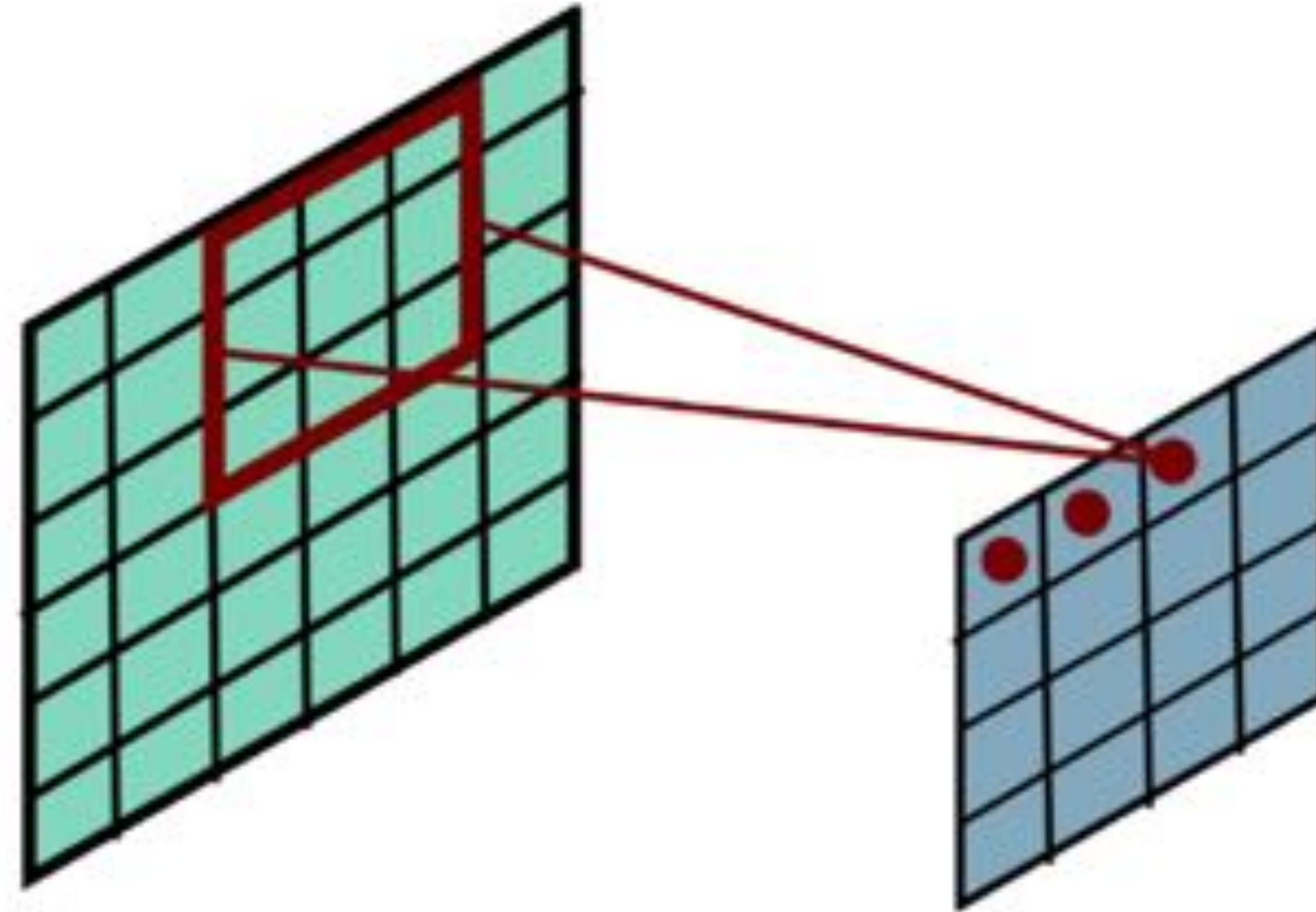
Convolution Procedure



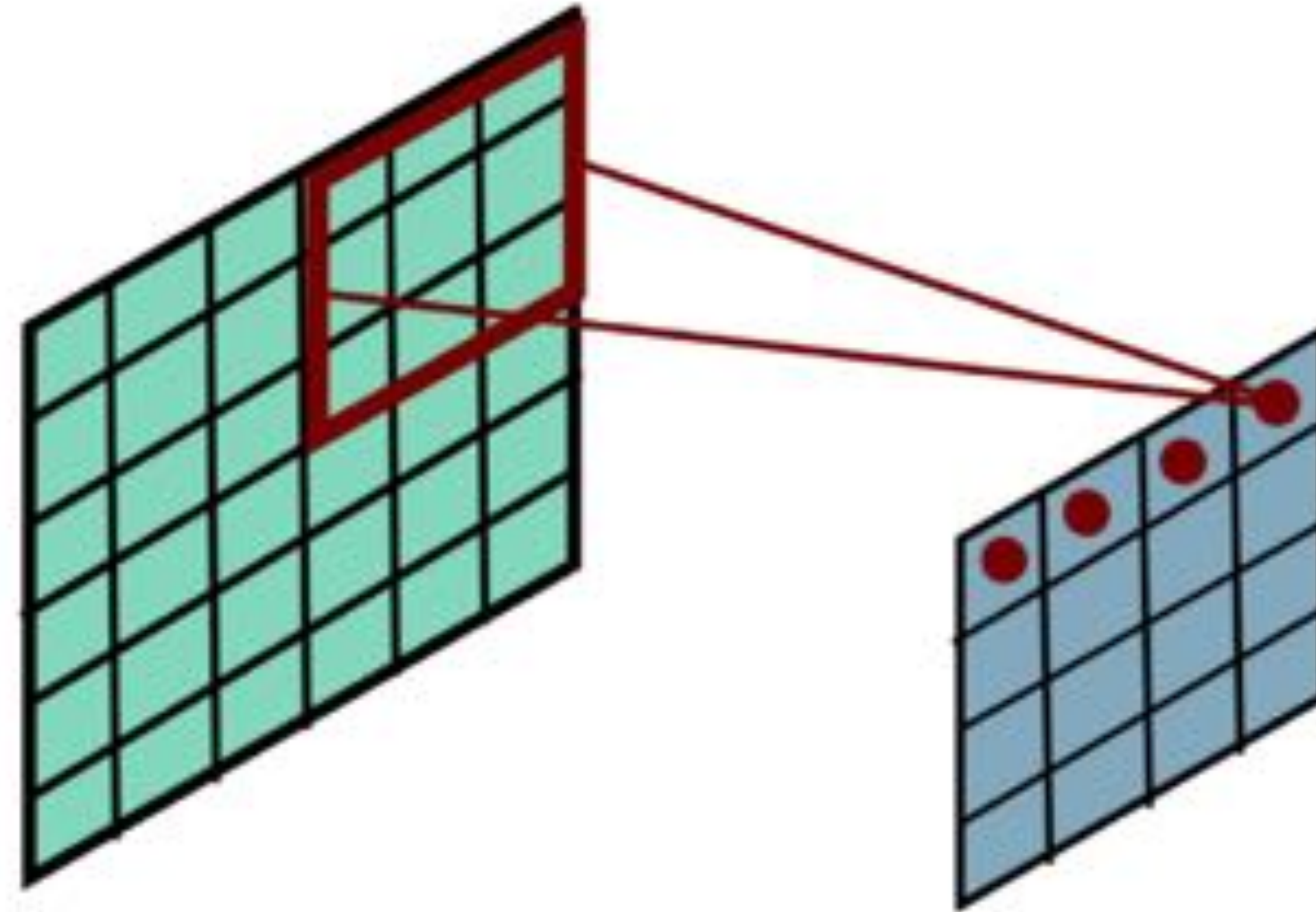
Convolution Procedure



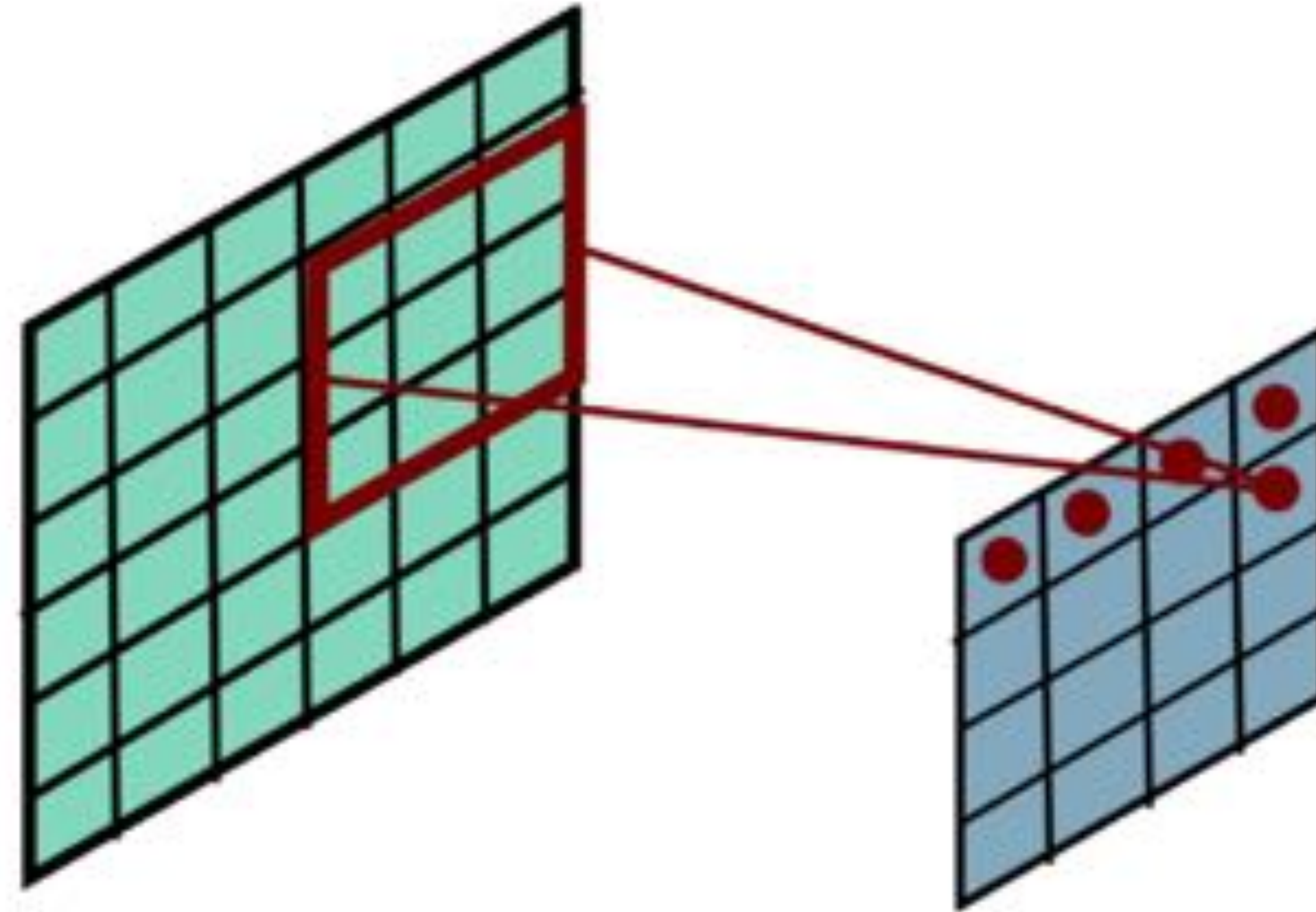
Convolution Procedure



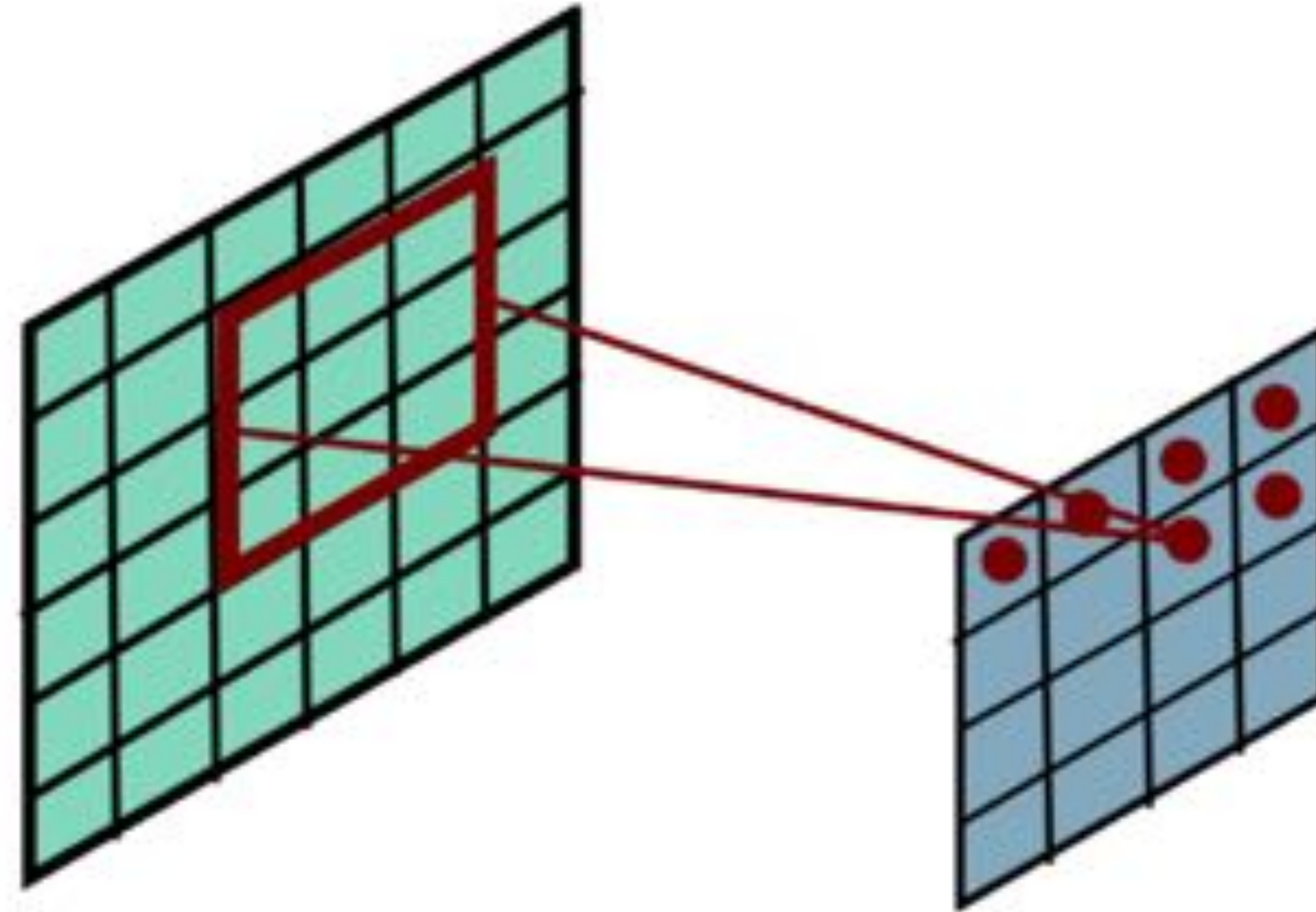
Convolution Procedure



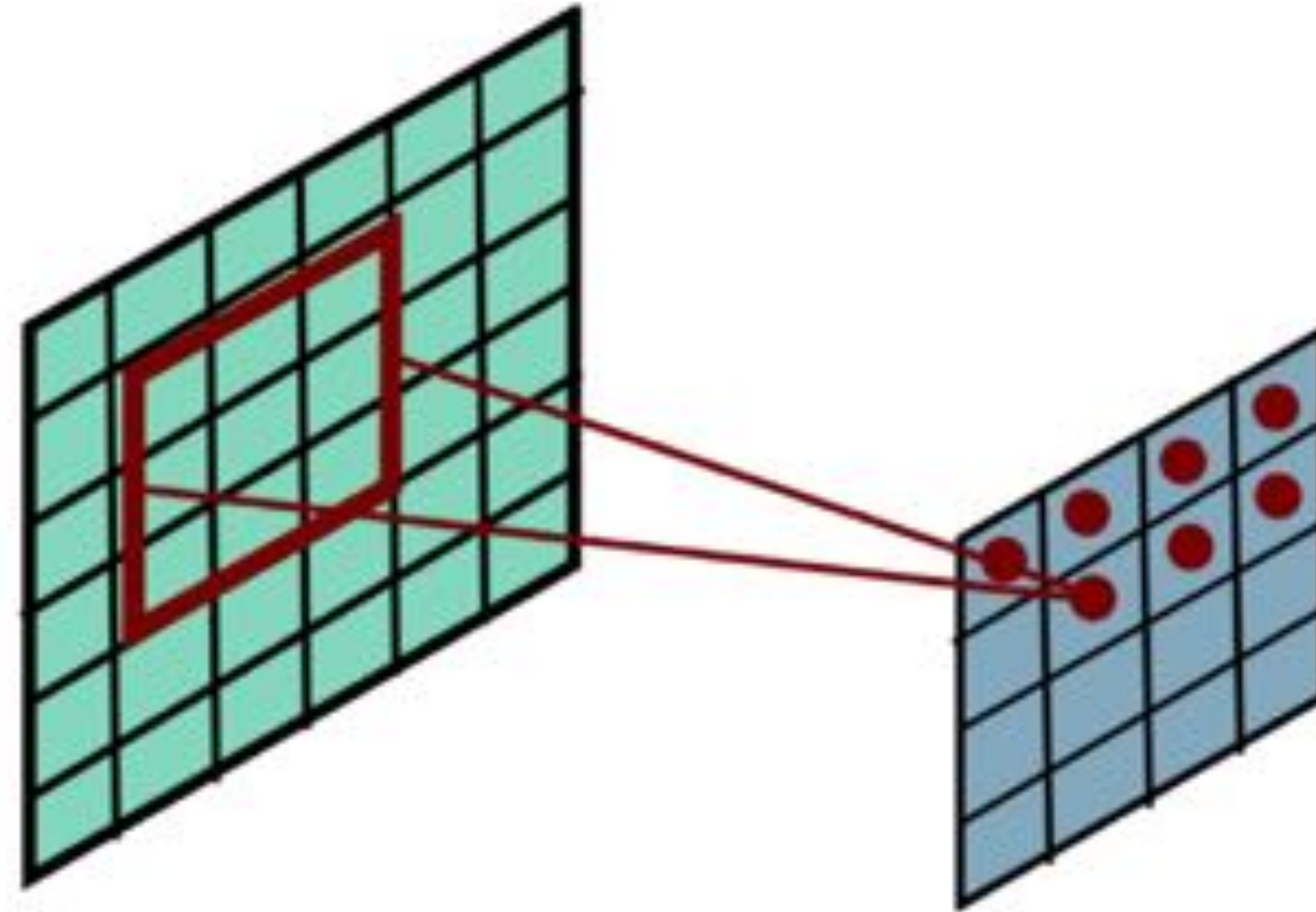
Convolution Procedure



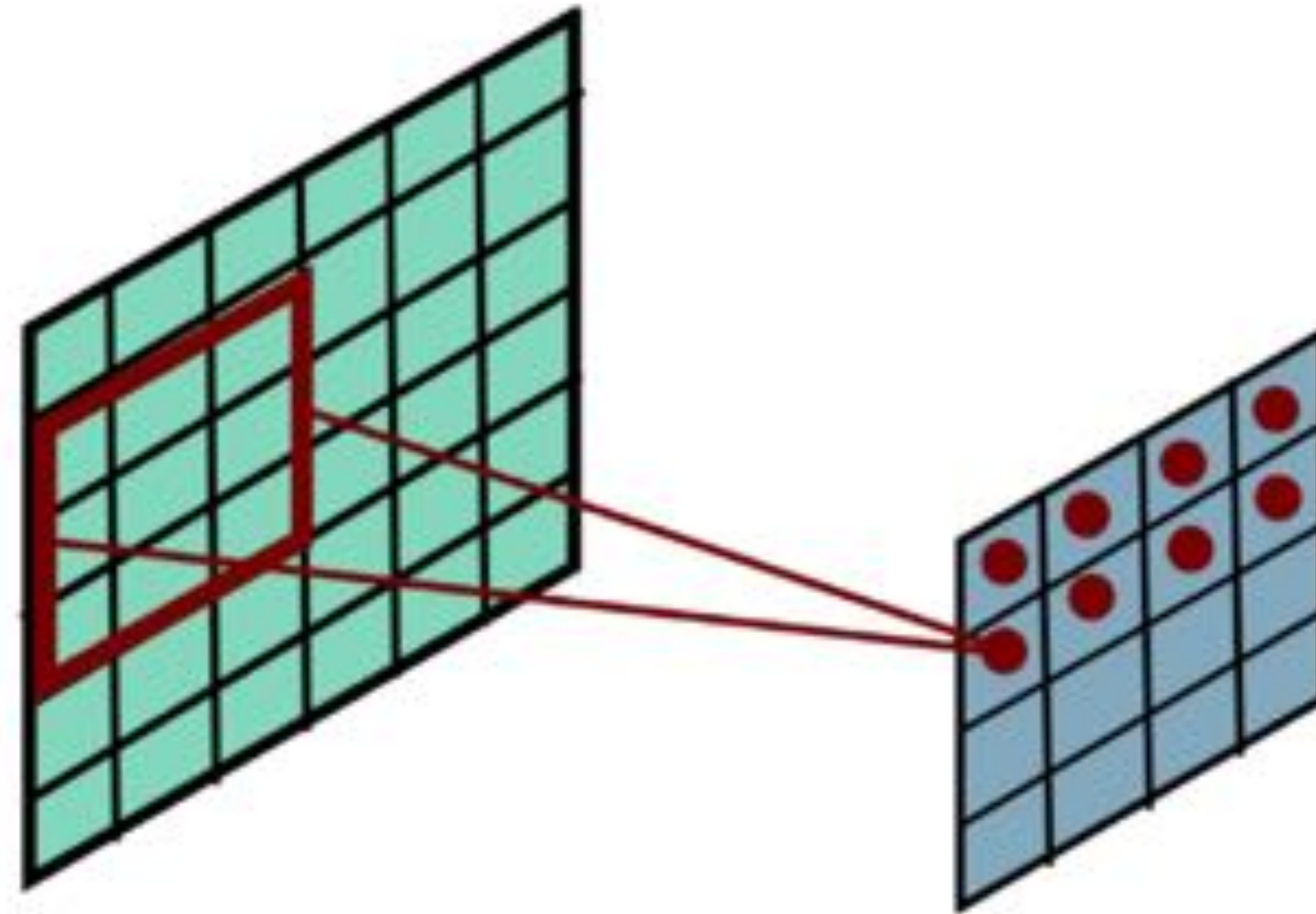
Convolution Procedure



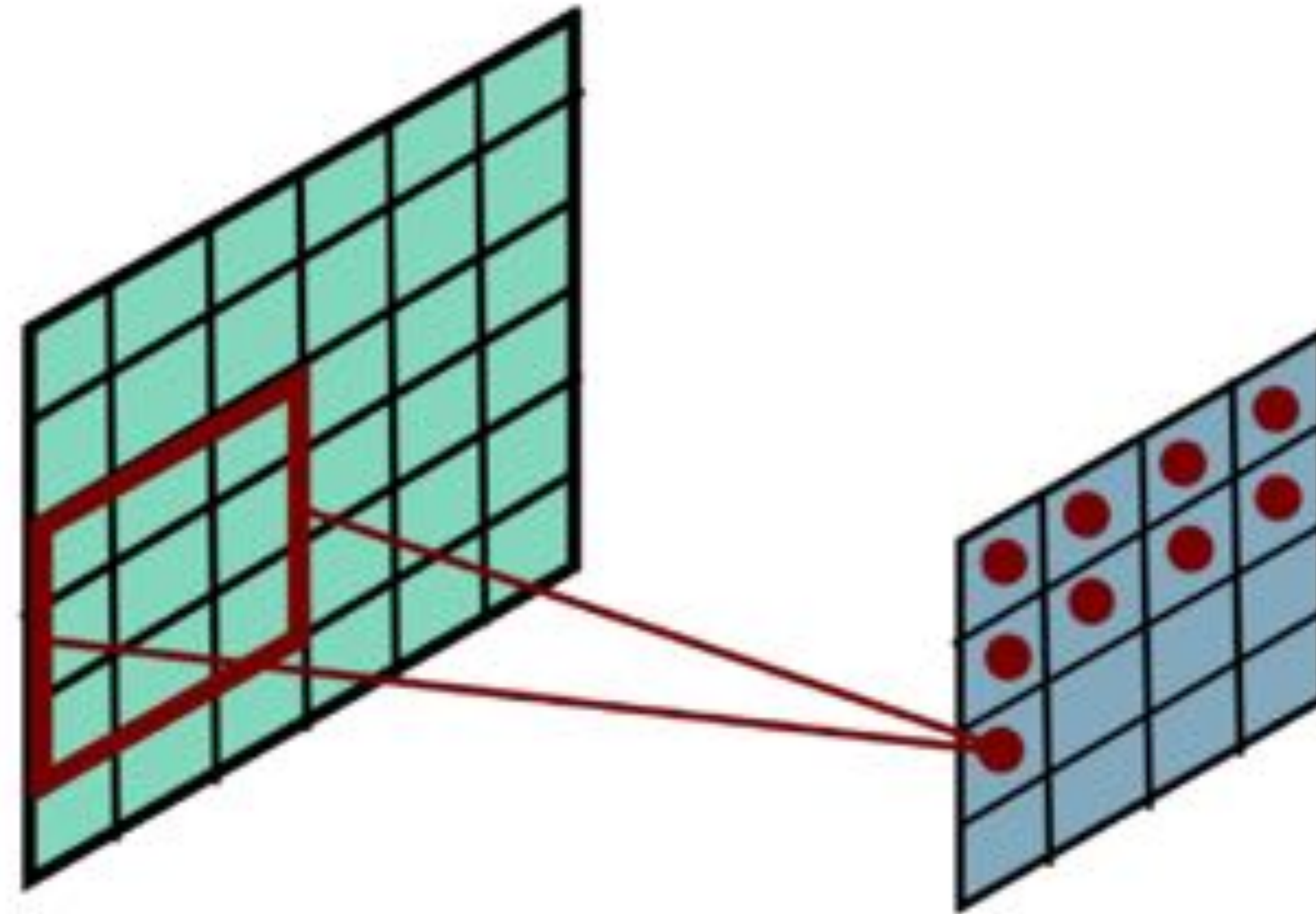
Convolution Procedure



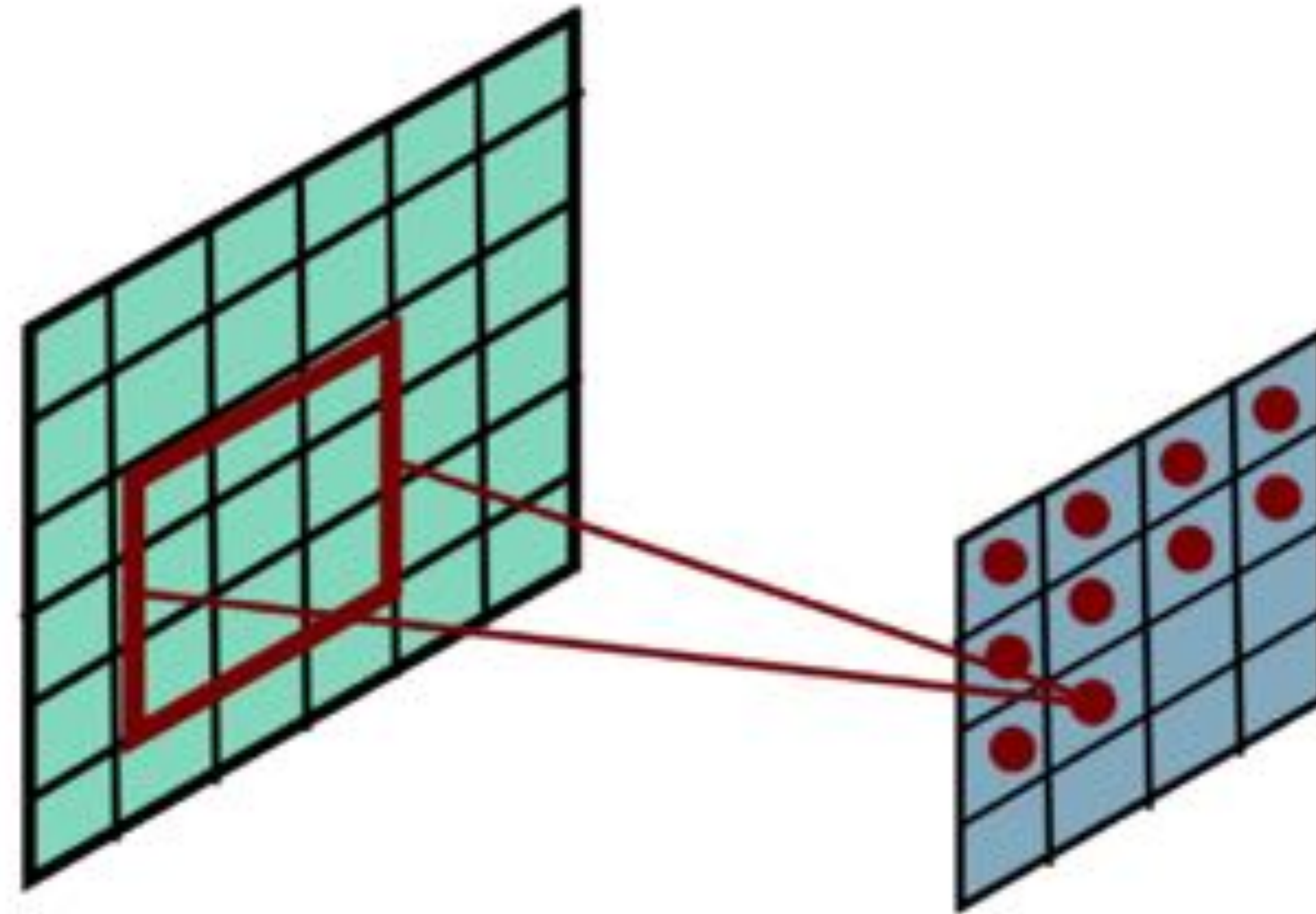
Convolution Procedure



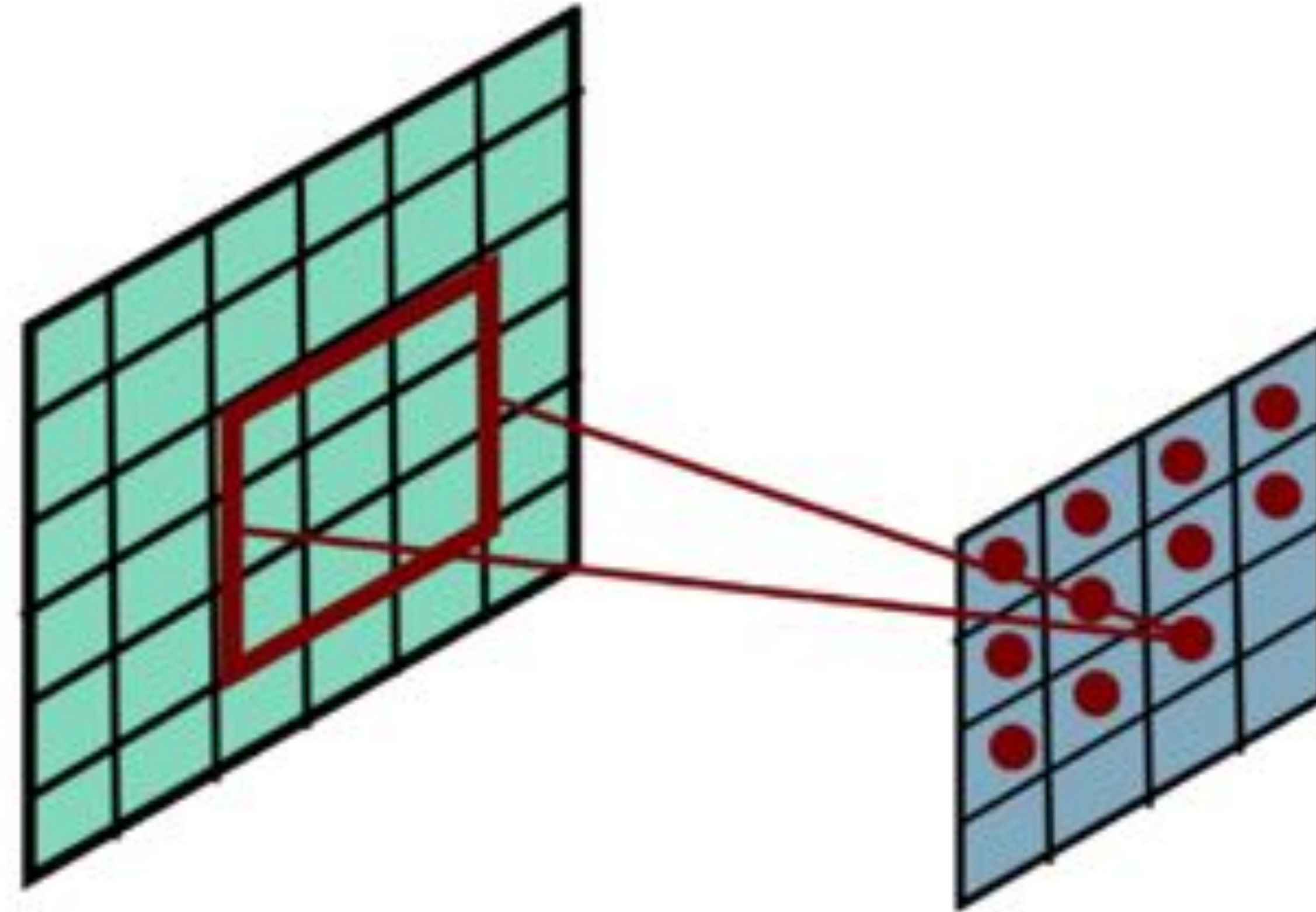
Convolution Procedure



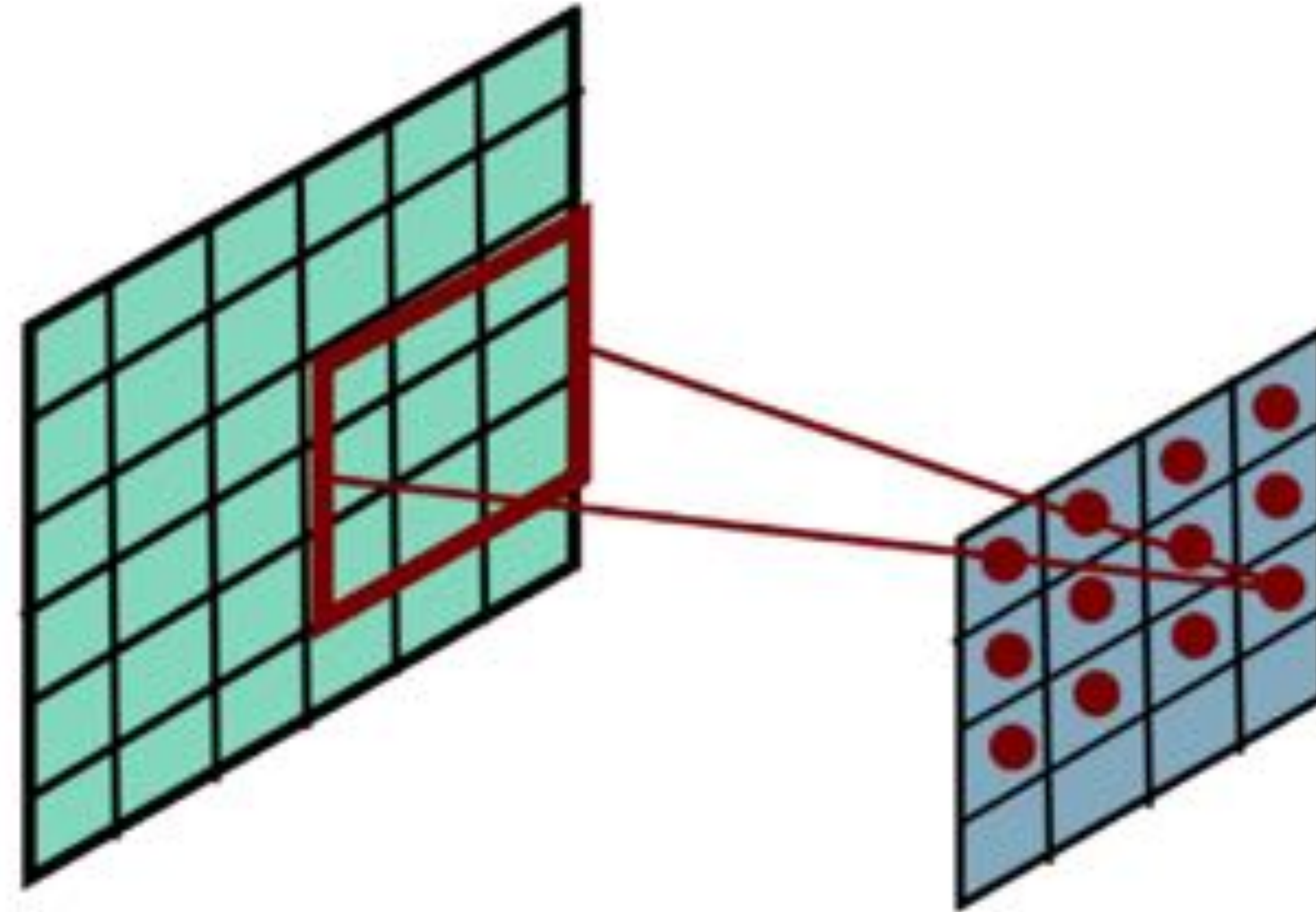
Convolution Procedure



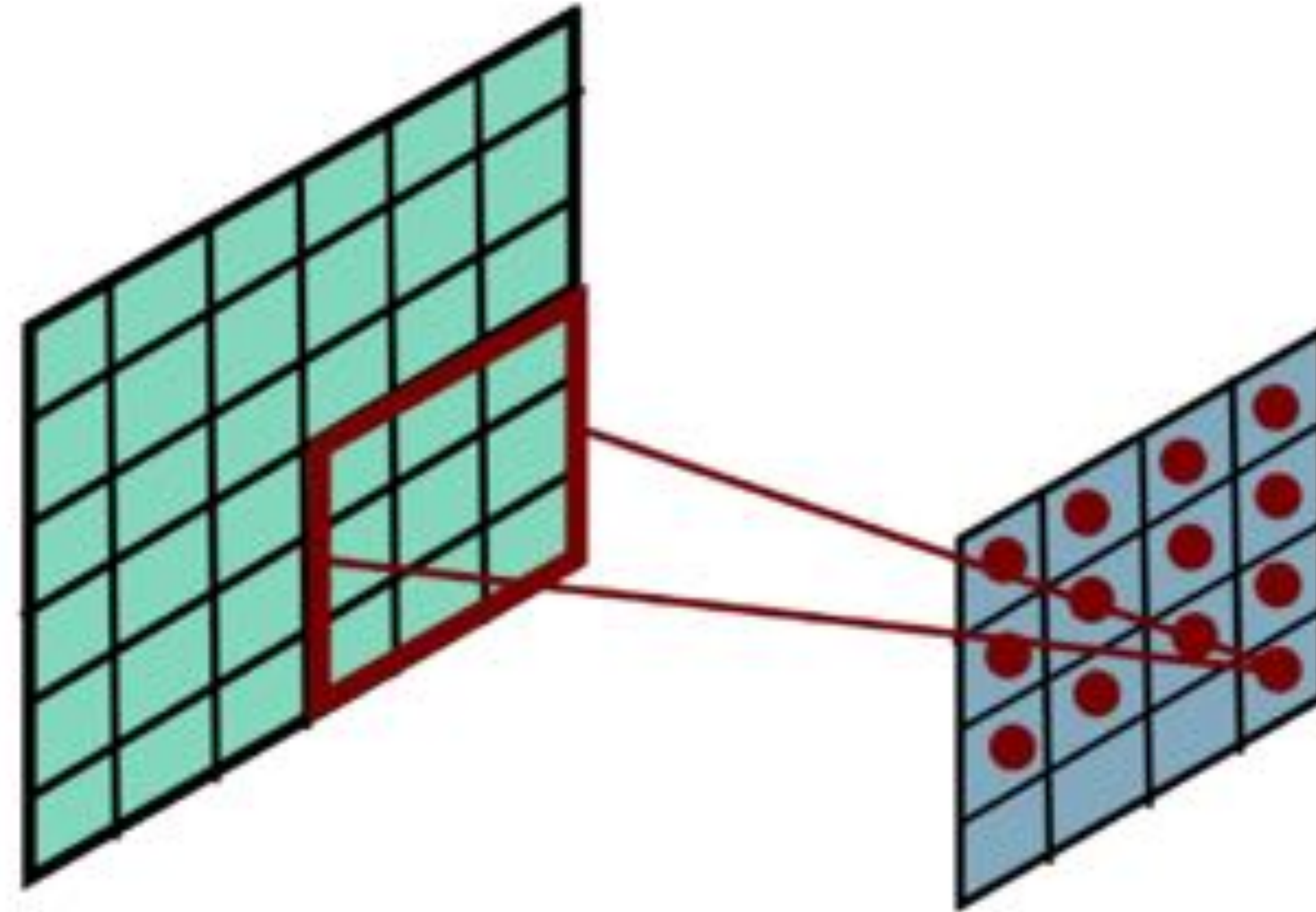
Convolution Procedure



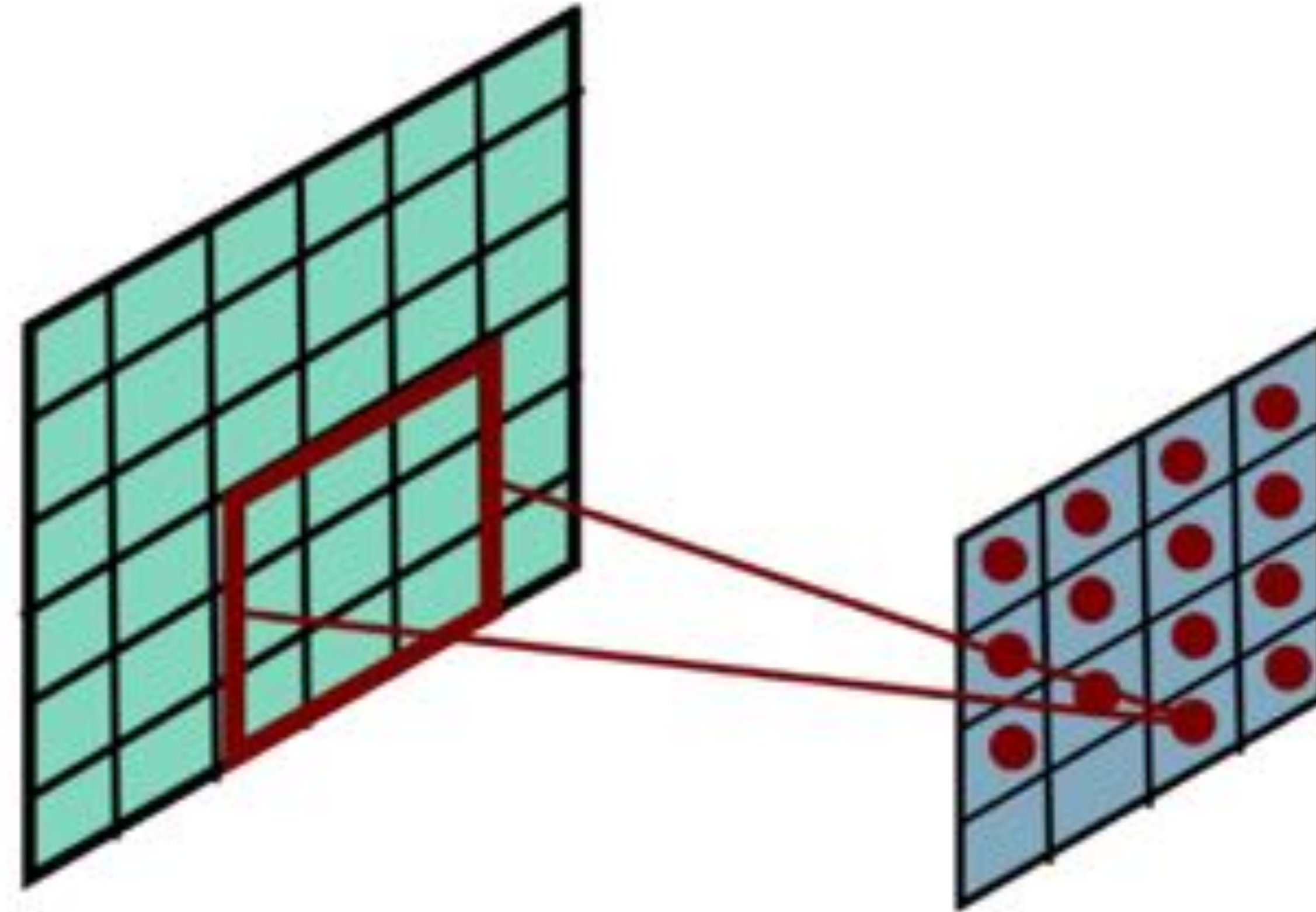
Convolution Procedure



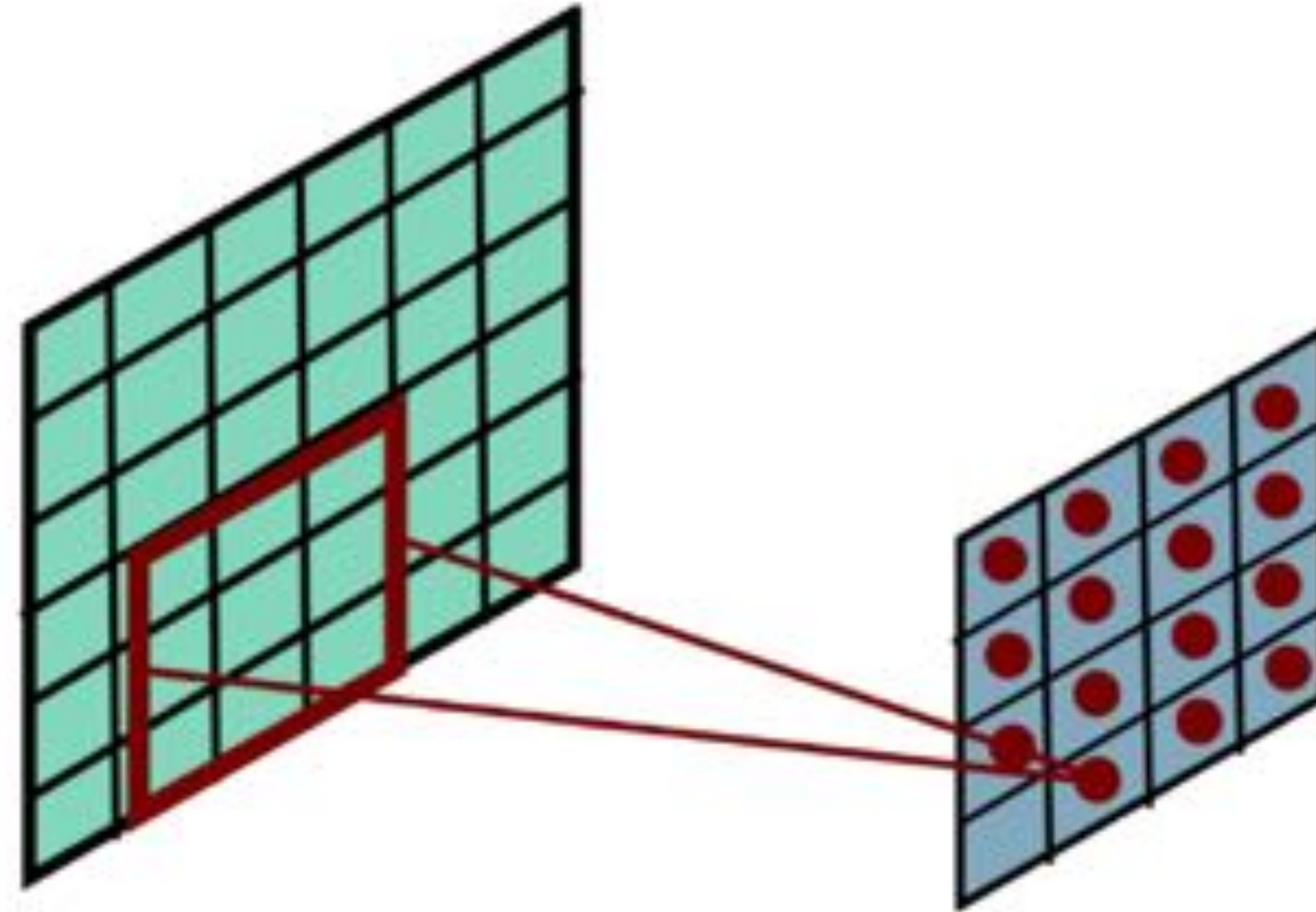
Convolution Procedure



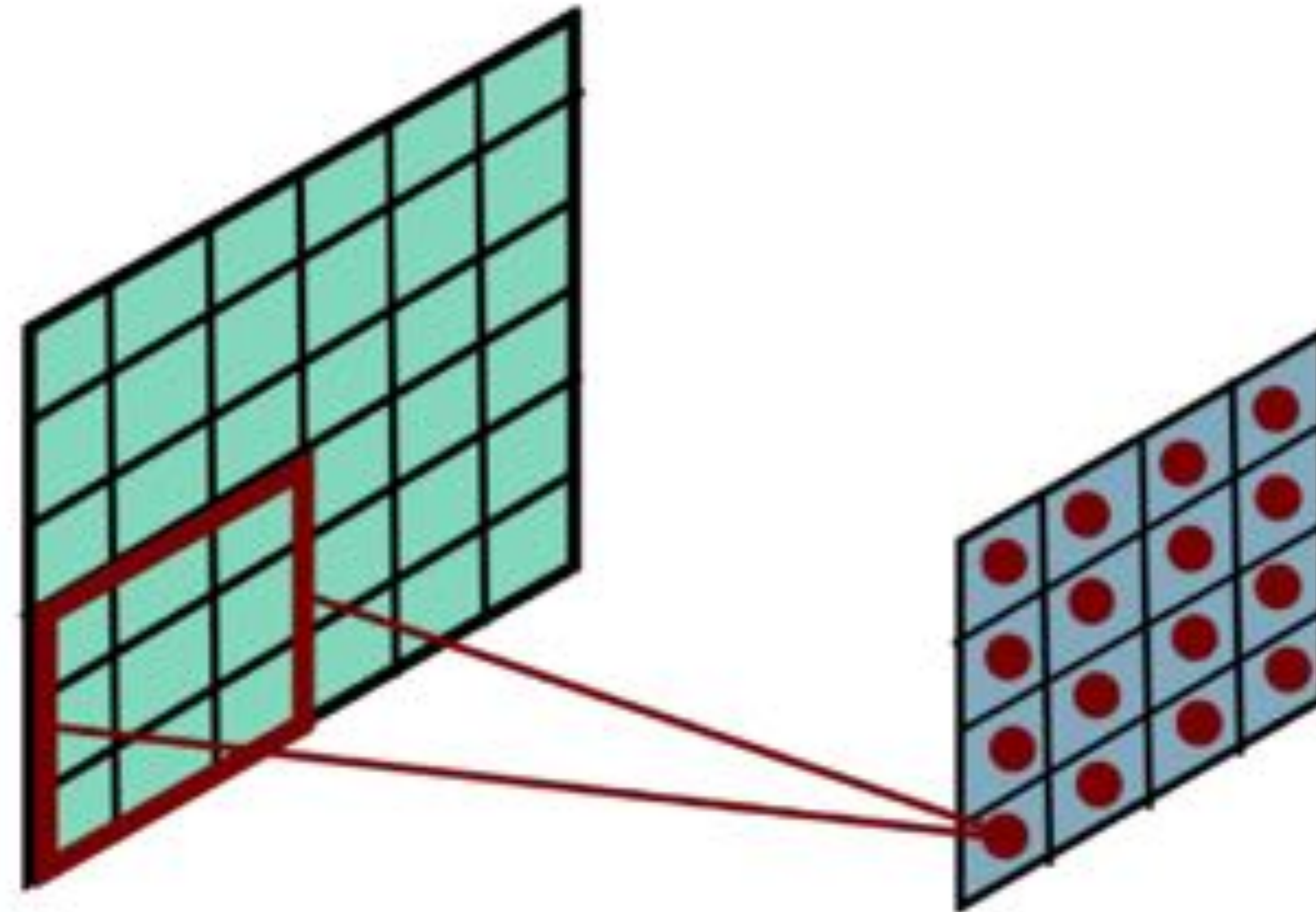
Convolution Procedure



Convolution Procedure

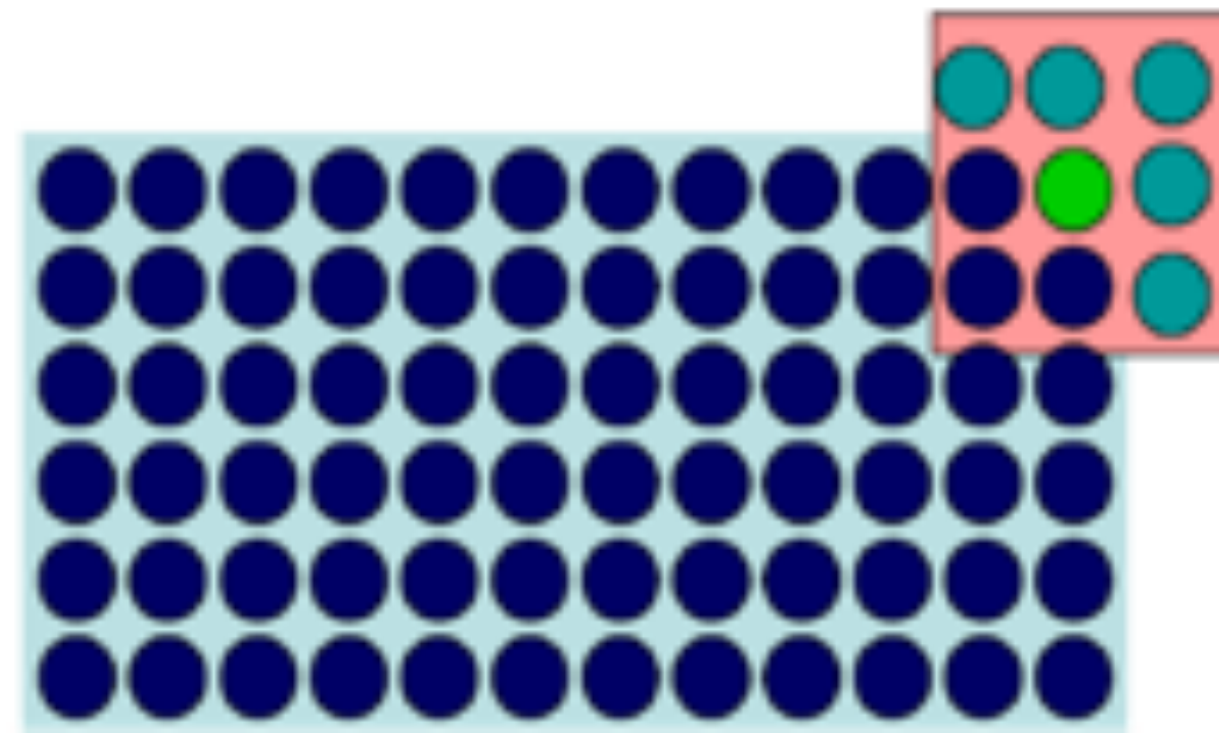


Convolution Procedure



Convolution Procedure

- ▶ Positioning
 - ▶ Place the **centre** of the window over the pixel position to be filtered
- ▶ Multiply
 - ▶ the original pixel values of the image by the filter values corresponding to their location
- ▶ Add
 - ▶ Add together the obtained multiplication results
 - ▶ The result of this addition is the **convolution result**
- ▶ Border
 - ▶ Mirror the border (Padding)



The logo for the AI500 conference. It features a stylized robot head with a face composed of a circuit board. The robot is holding a blue cube. The text 'Foundations of AI' is written vertically on the left, and 'AI500' is written horizontally at the bottom. The background is white with colorful dots and lines.

$$1x_1 + 1x_0 + 5x_{-1} + 4x_1 + 5x_0 + 4x_{-1} + 9x_1 + 1x_0 + 3x_{-1} = 2$$

Numerical Example

Image					Filter			Conv Output		
1	1	5	7	3	1	0	-1			
4	5	4	7	8	1	0	-1	2	-8	-5
9	1	3	1	6	1	0	-1	3	0	-3
2	4	5	2	1				3	3	1
1	3	1	2	1				-5	4	3
1	3	3	2	4						

$1 \times 1 + 5 \times 0 + 7 \times -1 + 5 \times 1 + 4 \times 0 + 7 \times -1 + 1 \times 1 + 3 \times 0 + 1 \times -1 = -8$

Numerical Example

Image	Filter	Conv Output																																																										
<table><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td><td>7</td><td>8</td></tr><tr><td>9</td><td>1</td><td>3</td><td>1</td><td>6</td></tr><tr><td>2</td><td>4</td><td>5</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>3</td><td>2</td><td>4</td></tr></table>	1	1	5	7	3	4	5	4	7	8	9	1	3	1	6	2	4	5	2	1	1	3	1	2	1	1	3	3	2	4	<table><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr></table>	1	0	-1	1	0	-1	1	0	-1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td>2</td><td>-8</td><td>-5</td></tr><tr><td></td><td>3</td><td>0</td><td>-3</td></tr><tr><td></td><td>3</td><td>3</td><td>1</td></tr><tr><td></td><td>-5</td><td>4</td><td>3</td></tr></table>					2	-8	-5		3	0	-3		3	3	1		-5	4	3
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$5 \times 1 + 7 \times 0 + 3 \times -1 + 4 \times 1 + 7 \times 0 + 8 \times -1 + 3 \times 1 + 1 \times 0 + 6 \times -1 = -5$																																																												

Numerical Example

Image	Filter	Conv Output																																																														
<table><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td><td>7</td><td>8</td></tr><tr><td>9</td><td>1</td><td>3</td><td>1</td><td>6</td></tr><tr><td>2</td><td>4</td><td>5</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>3</td><td>2</td><td>4</td></tr></table>	1	1	5	7	3	4	5	4	7	8	9	1	3	1	6	2	4	5	2	1	1	3	1	2	1	1	3	3	2	4	<table><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr></table>	1	0	-1	1	0	-1	1	0	-1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td>2</td><td>-8</td><td>-5</td></tr><tr><td></td><td>3</td><td>0</td><td>-3</td></tr><tr><td></td><td>3</td><td>3</td><td>1</td></tr><tr><td></td><td>-5</td><td>4</td><td>3</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>					2	-8	-5		3	0	-3		3	3	1		-5	4	3				
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$4 \times 1 + 5 \times 0 + 4 \times -1 + 9 \times 1 + 1 \times 0 + 3 \times -1 + 2 \times 1 + 4 \times 0 + 5 \times -1 = 3$																																																																

Numerical Example

Image	Filter	Conv Output																																																			
<table><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td><td>7</td><td>8</td></tr><tr><td>9</td><td>1</td><td>3</td><td>1</td><td>6</td></tr><tr><td>2</td><td>4</td><td>5</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>3</td><td>2</td><td>4</td></tr></table>	1	1	5	7	3	4	5	4	7	8	9	1	3	1	6	2	4	5	2	1	1	3	1	2	1	1	3	3	2	4	<table><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr></table>	1	0	-1	1	0	-1	1	0	-1	<table><tr><td>2</td><td>-8</td><td>-5</td></tr><tr><td>3</td><td>0</td><td>-3</td></tr><tr><td>3</td><td>3</td><td>1</td></tr><tr><td>-5</td><td>4</td><td>3</td></tr></table>	2	-8	-5	3	0	-3	3	3	1	-5	4	3
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$5 \times 1 + 4 \times 0 + 7 \times -1 + 1 \times 1 + 3 \times 0 + 1 \times -1 + 4 \times 1 + 5 \times 0 + 2 \times -1 = 0$																																																					

Numerical Example

Image	Filter	Conv Output																																																														
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$4 \times 1 + 7 \times 0 + 8 \times -1 + 3 \times 1 + 1 \times 0 + 6 \times -1 + 5 \times 1 + 2 \times 0 + 1 \times -1 = -3$																																																																

$$9x_1 + 1x_0 + 3x_{-1} + 2x_1 + 4x_0 + 5x_{-1} + 1x_1 + 3x_0 + 1x_{-1} = 3$$

Numerical Example

Image					Filter			Conv Output			
1	1	5	7	3	1	0	-1				
4	5	4	7	8	1	0	-1		2	-8	-5
9	1	3	1	6	1	0	-1		3	0	-3
2	4	5	2	1					3	3	1
1	3	1	2	1					-5	4	3
1	3	3	2	4							
$1 \times 1 + 3 \times 0 + 1 \times -1 + 4 \times 1 + 5 \times 0 + 2 \times -1 + 3 \times 1 + 1 \times 0 + 2 \times -1 = 3$											

Numerical Example

Image	Filter	Conv Output																																																									
<table><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td><td>7</td><td>8</td></tr><tr><td>9</td><td>1</td><td>3</td><td>1</td><td>6</td></tr><tr><td>2</td><td>4</td><td>5</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>3</td><td>2</td><td>4</td></tr></table>	1	1	5	7	3	4	5	4	7	8	9	1	3	1	6	2	4	5	2	1	1	3	1	2	1	1	3	3	2	4	<table><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr></table>	1	0	-1	1	0	-1	1	0	-1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																		
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3	3	1																																																									
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$3 \times 1 + 1 \times 0 + 6 \times -1 + 5 \times 1 + 2 \times 0 + 1 \times -1 + 1 \times 1 + 2 \times 0 + 1 \times -1 = 1$																																																											

Numerical Example

Image	Filter	Conv Output																																																														
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$2 \times 1 + 4 \times 0 + 5 \times -1 + 1 \times 1 + 3 \times 0 + 1 \times -1 + 1 \times 1 + 3 \times 0 + 3 \times -1 = -5$																																																																

Numerical Example

Image					Filter			Conv Output			
1	1	5	7	3	1	0	-1				
4	5	4	7	8	1	0	-1		2	-8	-5
9	1	3	1	6	1	0	-1		3	0	-3
2	4	5	2	1					3	3	1
1	3	1	2	1					-5	4	3
1	3	3	2	4							
$4 \times 1 + 5 \times 0 + 2 \times -1 + 3 \times 1 + 1 \times 0 + 2 \times -1 + 3 \times 1 + 3 \times 0 + 2 \times -1 = 4$											

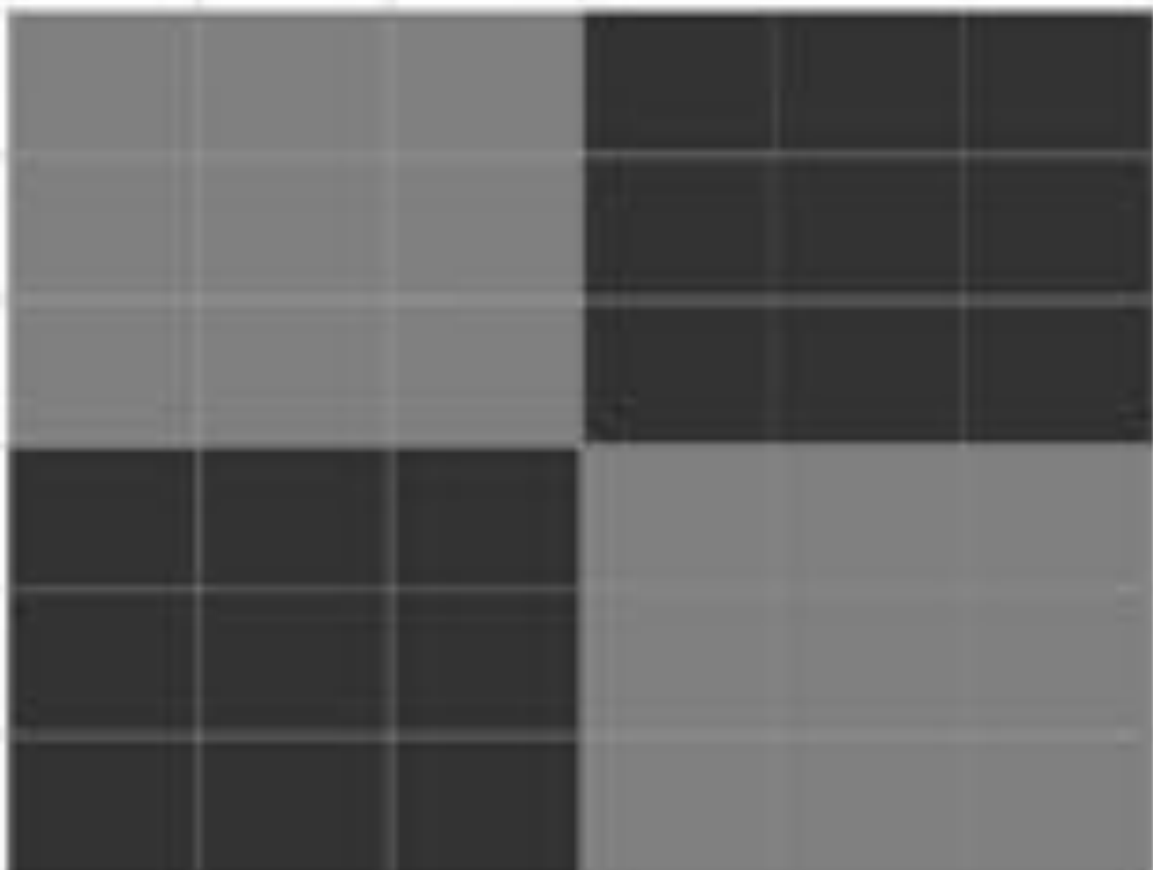
Numerical Example

Image	Filter	Conv Output																																																														
<table><tr><td>1</td><td>1</td><td>5</td><td>7</td><td>3</td></tr><tr><td>4</td><td>5</td><td>4</td><td>7</td><td>8</td></tr><tr><td>9</td><td>1</td><td>3</td><td>1</td><td>6</td></tr><tr><td>2</td><td>4</td><td>5</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td></tr><tr><td>1</td><td>3</td><td>3</td><td>2</td><td>4</td></tr></table>	1	1	5	7	3	4	5	4	7	8	9	1	3	1	6	2	4	5	2	1	1	3	1	2	1	1	3	3	2	4	<table><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td><td>-1</td></tr></table>	1	0	-1	1	0	-1	1	0	-1	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td>2</td><td>-8</td><td>-5</td></tr><tr><td></td><td>3</td><td>0</td><td>-3</td></tr><tr><td></td><td>3</td><td>3</td><td>1</td></tr><tr><td></td><td>-5</td><td>4</td><td>3</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>					2	-8	-5		3	0	-3		3	3	1		-5	4	3				
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$5 \times 1 + 2 \times 0 + 1 \times -1 + 1 \times 1 + 2 \times 0 + 1 \times -1 + 3 \times 1 + 2 \times 0 + 4 \times -1 = 3$																																																																

Smoothing Filter

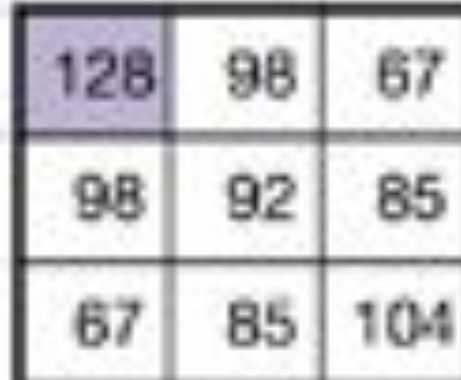
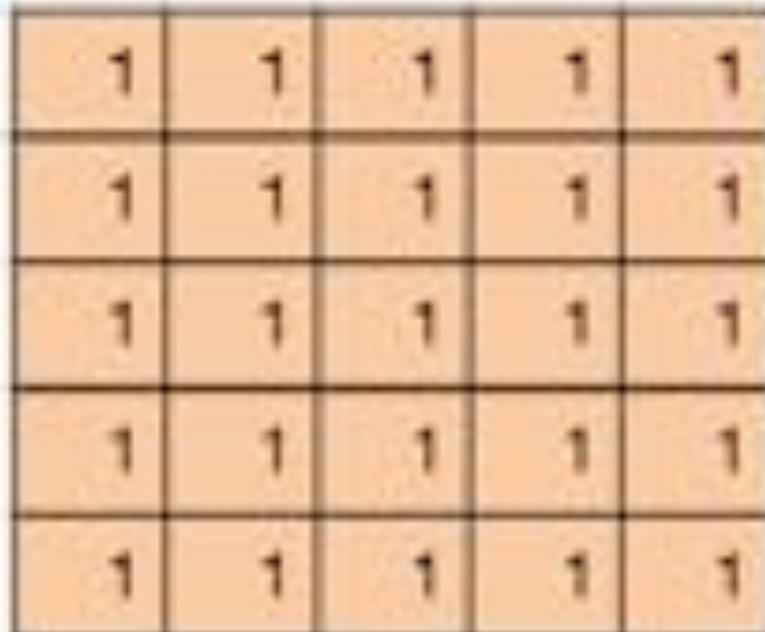
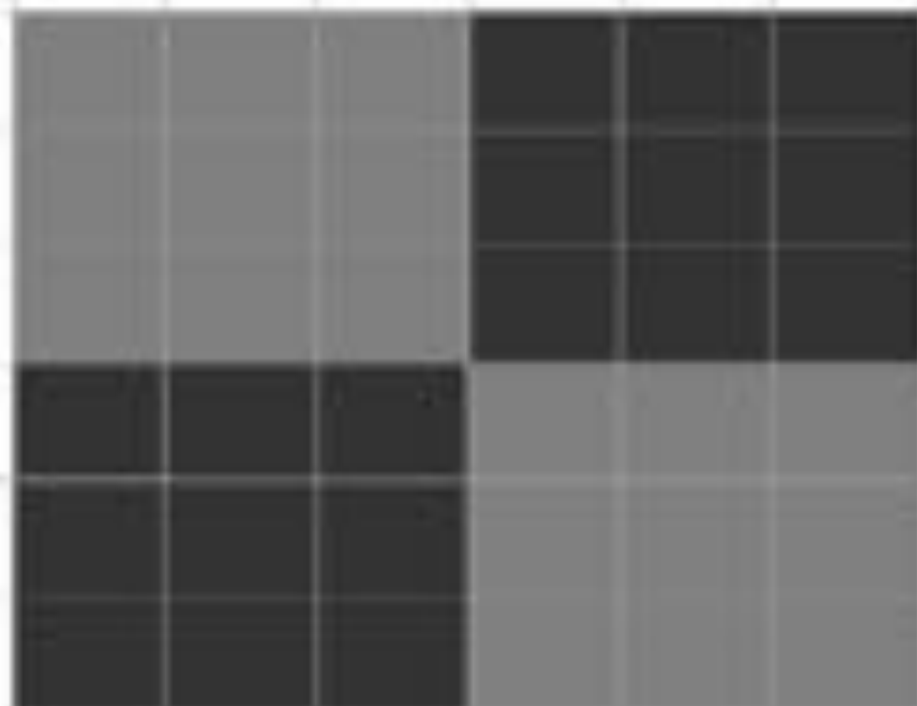
Image						Filter			Conv Output					
128	128	128	52	52	52	1	1	1	x1/9					
128	128	128	52	52	52	1	1	1			128	103	77	52
128	128	128	52	52	52	1	1	1			103	94	86	77
52	52	52	128	128	128						77	86	94	103
52	52	52	128	128	128						52	77	103	128
52	52	52	128	128	128									

Zero Padding - Edge Filter - Horizontal

Image								Filter			Conv Output				
0	0	0	0	0	0	0	0								
0	128	128	128	52	52	52	0	1	1	1	-256	-384	-308	-232	-156
0	128	128	128	52	52	52	0	0	0	0	0	0	0	0	0
0	128	128	128	52	52	52	0	-1	-1	-1	152	228	76	-76	-228
0	52	52	52	128	128	128	0	Ignoring flip			152	228	76	-76	-228
0	52	52	52	128	128	128	0				0	0	0	0	0
0	52	52	52	128	128	128	0				104	156	232	308	384
0	0	0	0	0	0	0	0								
															

Stride 2, Padding 2

Image										Filter					Conv Output				
128	128	128	128	128	52	52	52	52	52	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 x1/25	128	98	67						
128	128	128	128	128	52	52	52	52	52		98	92	85						
128	128	128	128	128	52	52	52	52	52		67	85	104						
52	52	52	52	52	128	128	128	128	128										
52	52	52	52	52	128	128	128	128	128										
52	52	52	52	52	128	128	128	128	128										
52	52	52	52	52	128	128	128	128	128										
52	52	52	52	52	128	128	128	128	128										



Stride 2, Padding 0

Image						Filter					Conv Output				
128	128	128	52	52	52										
128	128	128	52	52	52										
128	128	128	52	52	52										
52	52	52	128	128	128										
52	52	52	128	128	128										
52	52	52	128	128	128										

1	1	1	1	1	x1/25
1	1	1	1	1	
1	1	1	1	1	
1	1	1	1	1	
1	1	1	1	1	

Stride 2, Padding 0

The diagram illustrates a 1D convolution operation. The input vector is of size 6, with the last three elements (52, 52, 52) highlighted in yellow. The kernel is of size 3, with the first three elements (1, 1, 1) highlighted in orange. The output vector is of size 4, with the first element (92) highlighted in purple. The operation is labeled as $\times 1/25$.

Image	Filter	Conv Output																																																																			
<table> <tr><td>128</td><td>128</td><td>128</td><td>52</td><td>52</td><td>52</td></tr> <tr><td>128</td><td>128</td><td>128</td><td>52</td><td>52</td><td>52</td></tr> <tr><td>128</td><td>128</td><td>128</td><td>52</td><td>52</td><td>52</td></tr> <tr><td>52</td><td>52</td><td>52</td><td>128</td><td>128</td><td>128</td></tr> <tr><td>52</td><td>52</td><td>52</td><td>128</td><td>128</td><td>128</td></tr> <tr><td>52</td><td>52</td><td>52</td><td>128</td><td>128</td><td>128</td></tr> </table>	128	128	128	52	52	52	128	128	128	52	52	52	128	128	128	52	52	52	52	52	52	128	128	128	52	52	52	128	128	128	52	52	52	128	128	128	<table> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </table>	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	<table> <tr><td>92</td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td></tr> </table>	92															
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92																																																																					

Filtering Examples



Original
Cameraman



Cameraman blurred by convolution
Filter impulse response

$$\frac{1}{25} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Filtering Examples - Effect of Filter Size

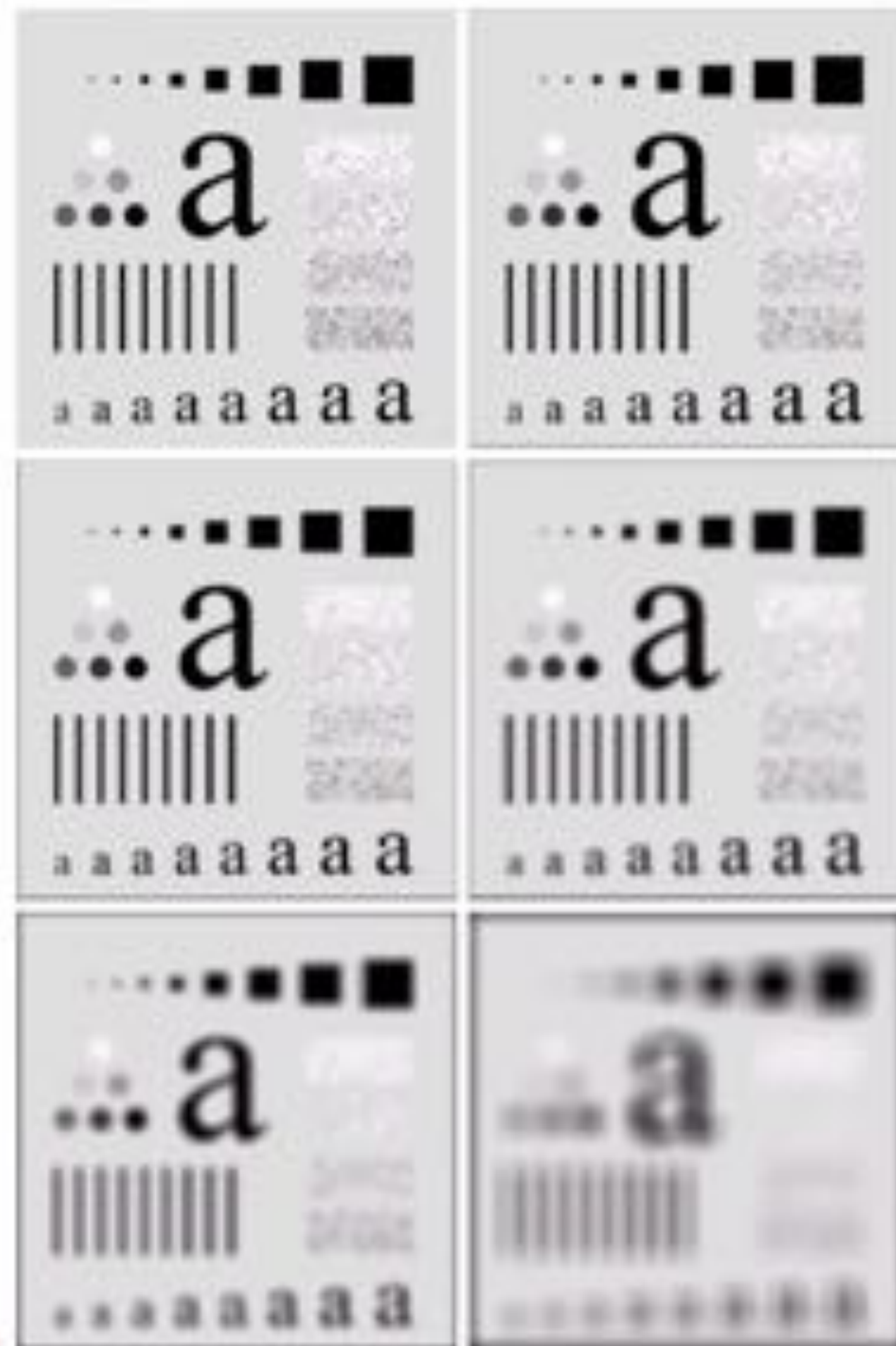


FIGURE 3.35 (a) Original image, of size 500×500 pixels. (b)-(f) Results of smoothing with square averaging filter masks of sizes $n = 3, 5, 9, 15$, and 35 , respectively. The black squares at the top are of sizes $3, 5, 9, 15, 25, 35, 45$, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their gray levels range from 0% to 100% black in increments of 20% . The background of the image is 10% black. The noisy rectangles are of size 50×120 pixels.

Final Exam Syllabus

Lecture 5 and onwards



Finding Edges - Differentiation



Finding Edges - Differentiation

► Continuous form

$$\frac{\partial f}{\partial x} = \lim_{\Delta h \rightarrow 0} \frac{f(x + \Delta h) - f(x)}{\Delta h}$$

Finding Edges - Differentiation

- ▶ Continuous form

$$\frac{\partial f}{\partial x} = \lim_{\Delta h \rightarrow 0} \frac{f(x + \Delta h) - f(x)}{\Delta h}$$

- ▶ Discrete form (put $\Delta h = 1$)

$$f_x = \frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

Discrete Derivatives in 2D

$$f(x+1) - f(x)$$

$$h_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$f(y+1) - f(y)$$

$$h_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



synthetic image



output



output

Discrete Derivatives in 2D

$$f_x = \frac{\partial f}{\partial x} = f(x+1) - f(x) \quad f_y = \frac{\partial f}{\partial y} = f(y+1) - f(y)$$

$$h_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$h_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



synthetic image



h_y output

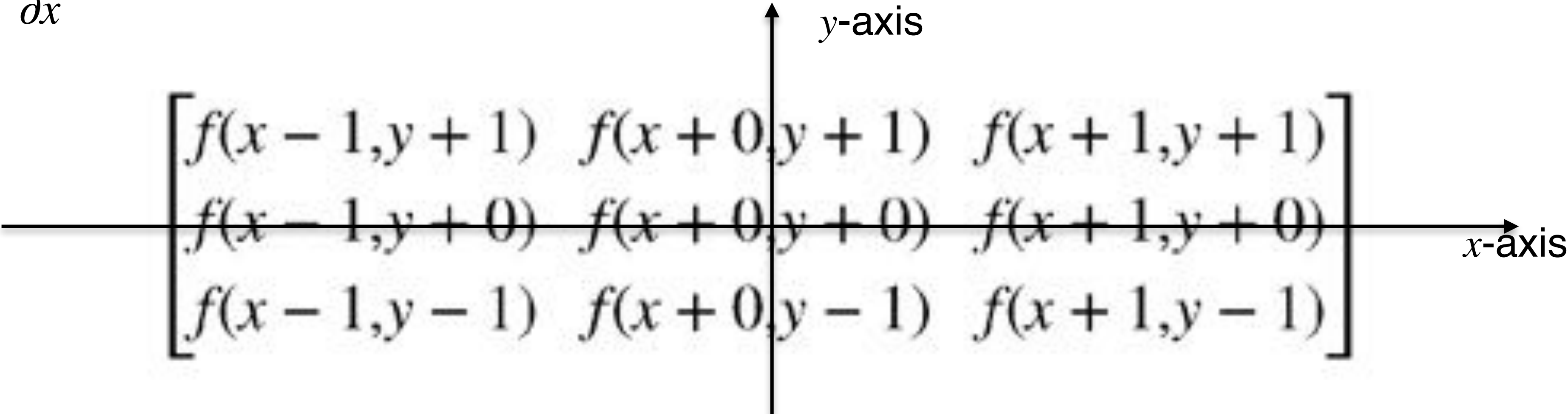


h_x output

$$f_x = \frac{\partial f}{\partial x} = f(x+1) - f(x)$$

$$f_x = \frac{\partial f}{\partial x} = f(x+1) - f(x+0)$$

$$\frac{\partial f(x,y)}{\partial x} = f(x+1,y+0) - f(x+0,y+0)$$



Indicies

h_x	$(-1,+1)$	$(0,+1)$	$(+1,+1)$
	$(-1,0)$	$(0,0)$	$(+1,0)$
	$(-1,-1)$	$(0,-1)$	$(+1,-1)$

Values

h_x	0	0	0
	0	-1	1
	0	0	0

Any Questions

Vertical Edge Detection

$$h_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$



Original Image: Opera



Vertical Edge Detection

$$h_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$



Original Image: Opera



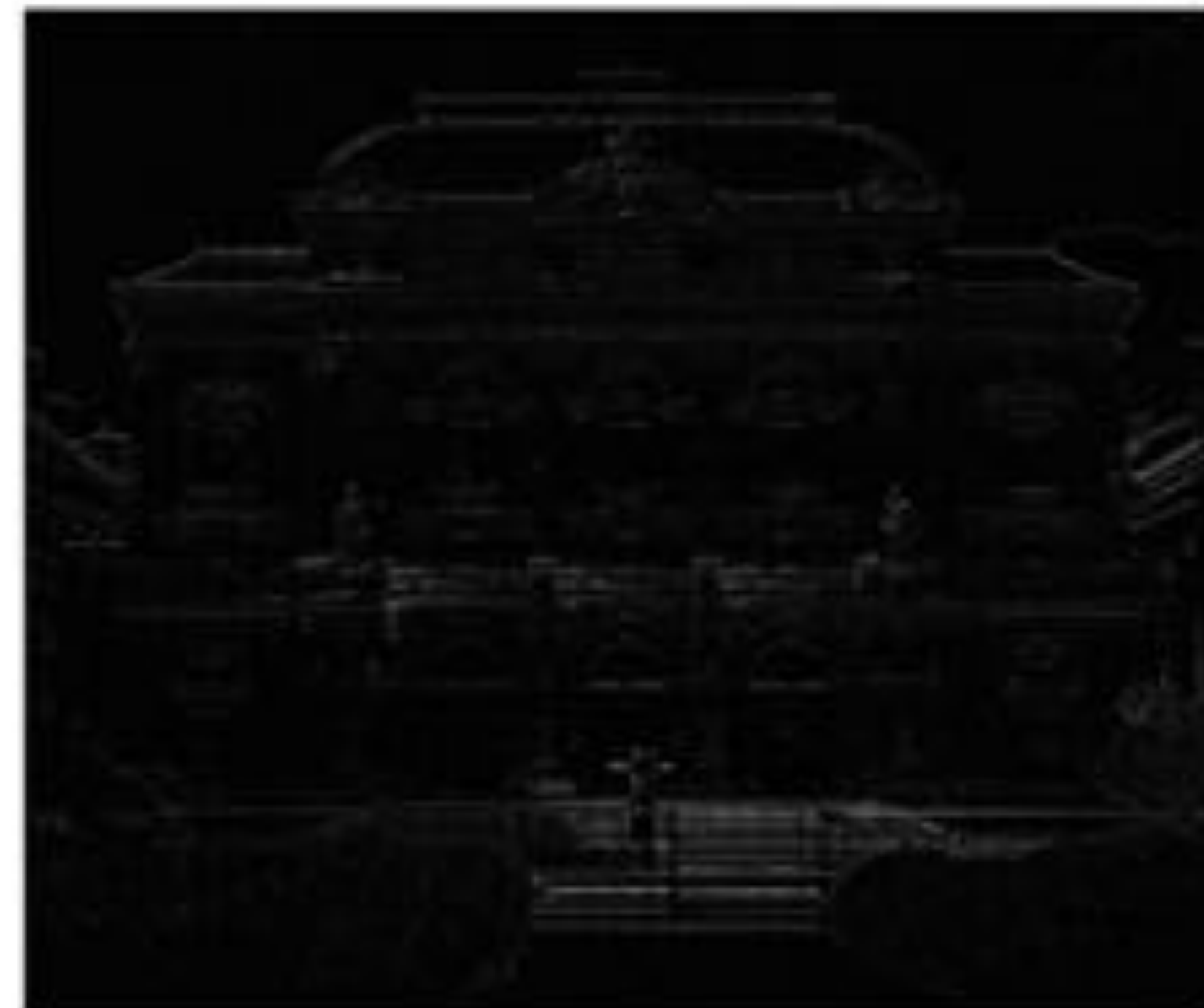
Inverted Image for Visualisation

Horizontal Edge Detection

$$h_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Original Image: Opera

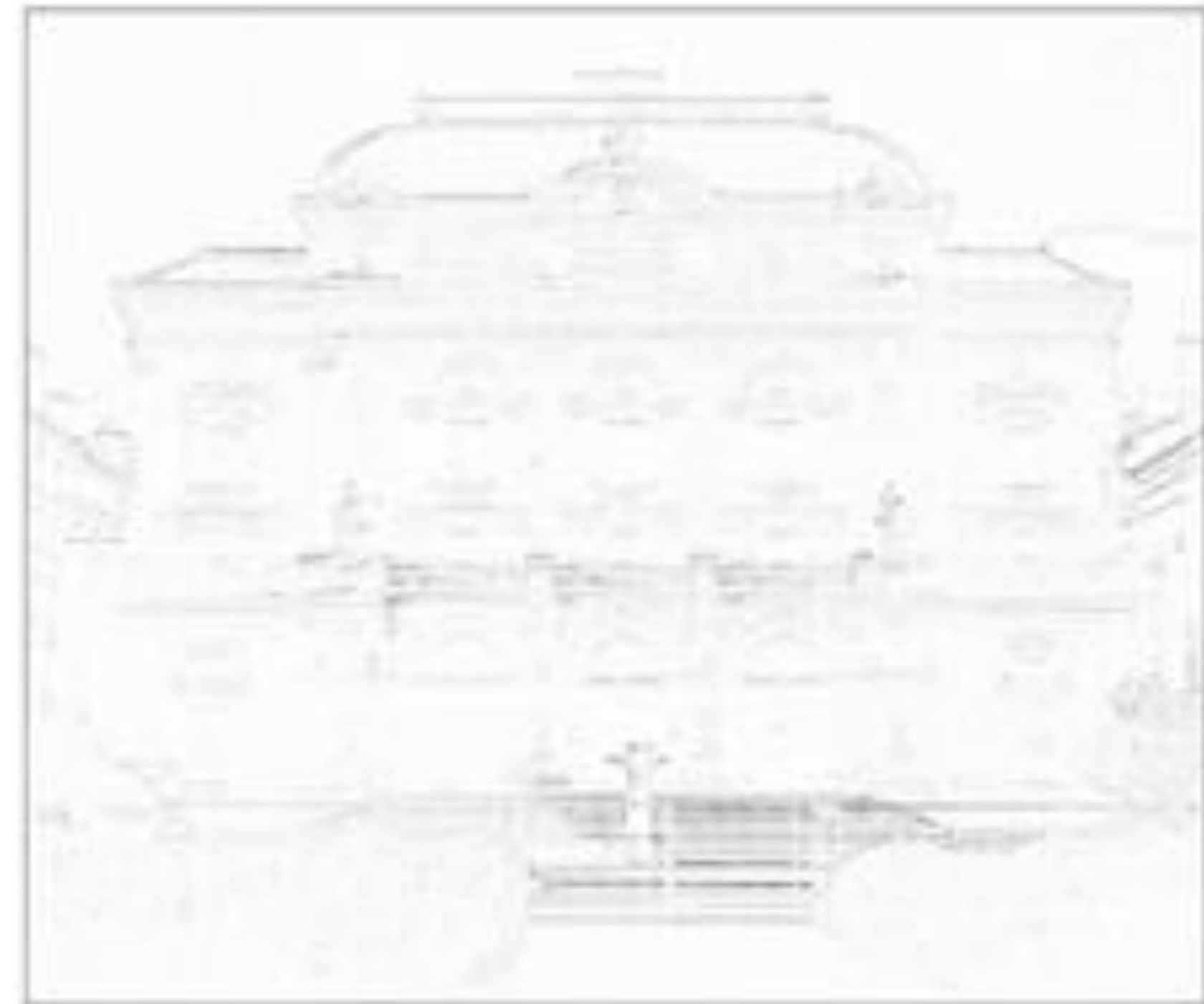


Horizontal Edge Detection

$$h_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Original Image: Opera



Inverted Image for Visualisation

Sobel, Prewit & Roberts Operators

Sobel

$$h_x = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

Prewit

$$h_x = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Roberts

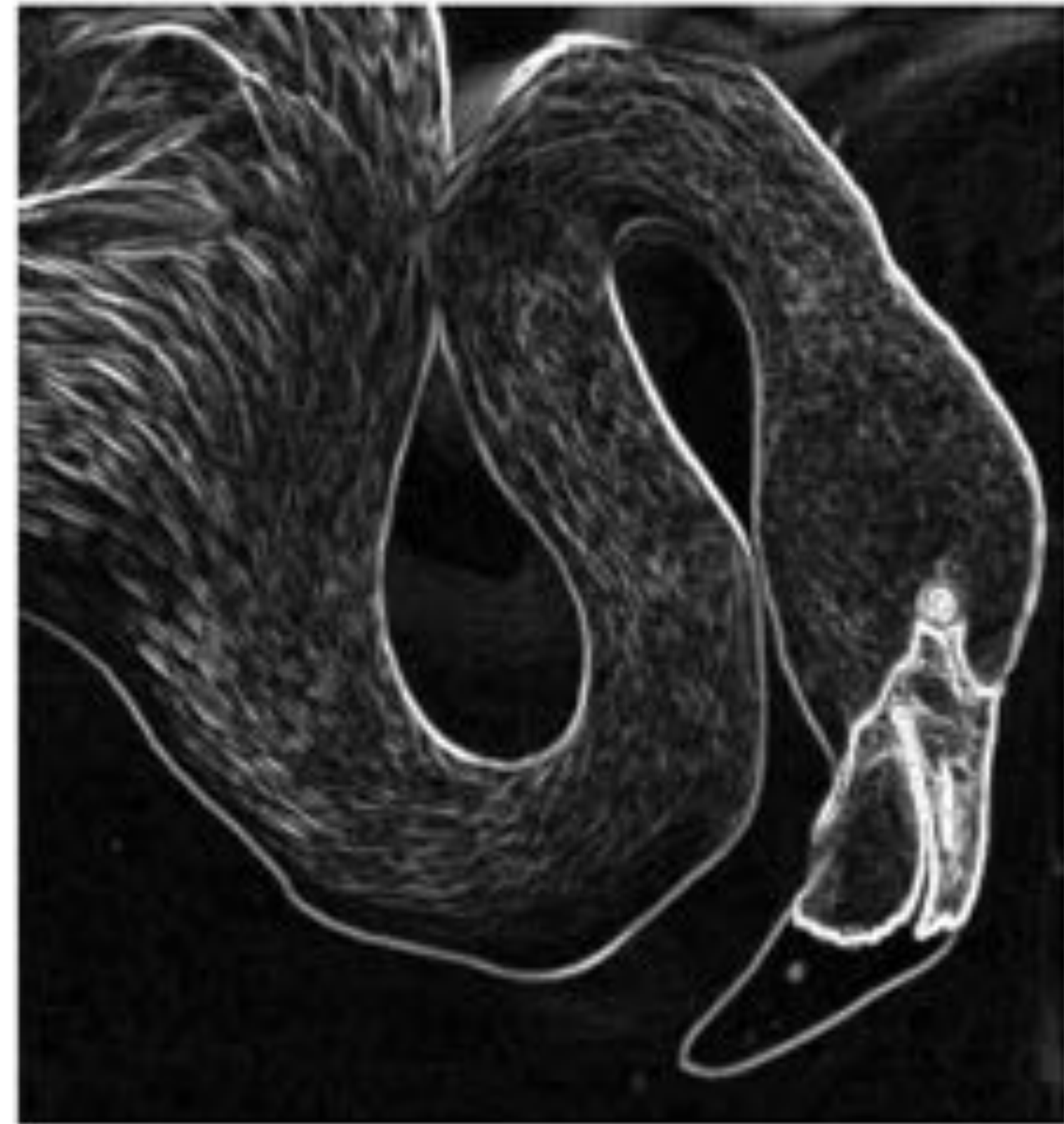
$$h_x = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$h_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$h_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h_y = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Sobel



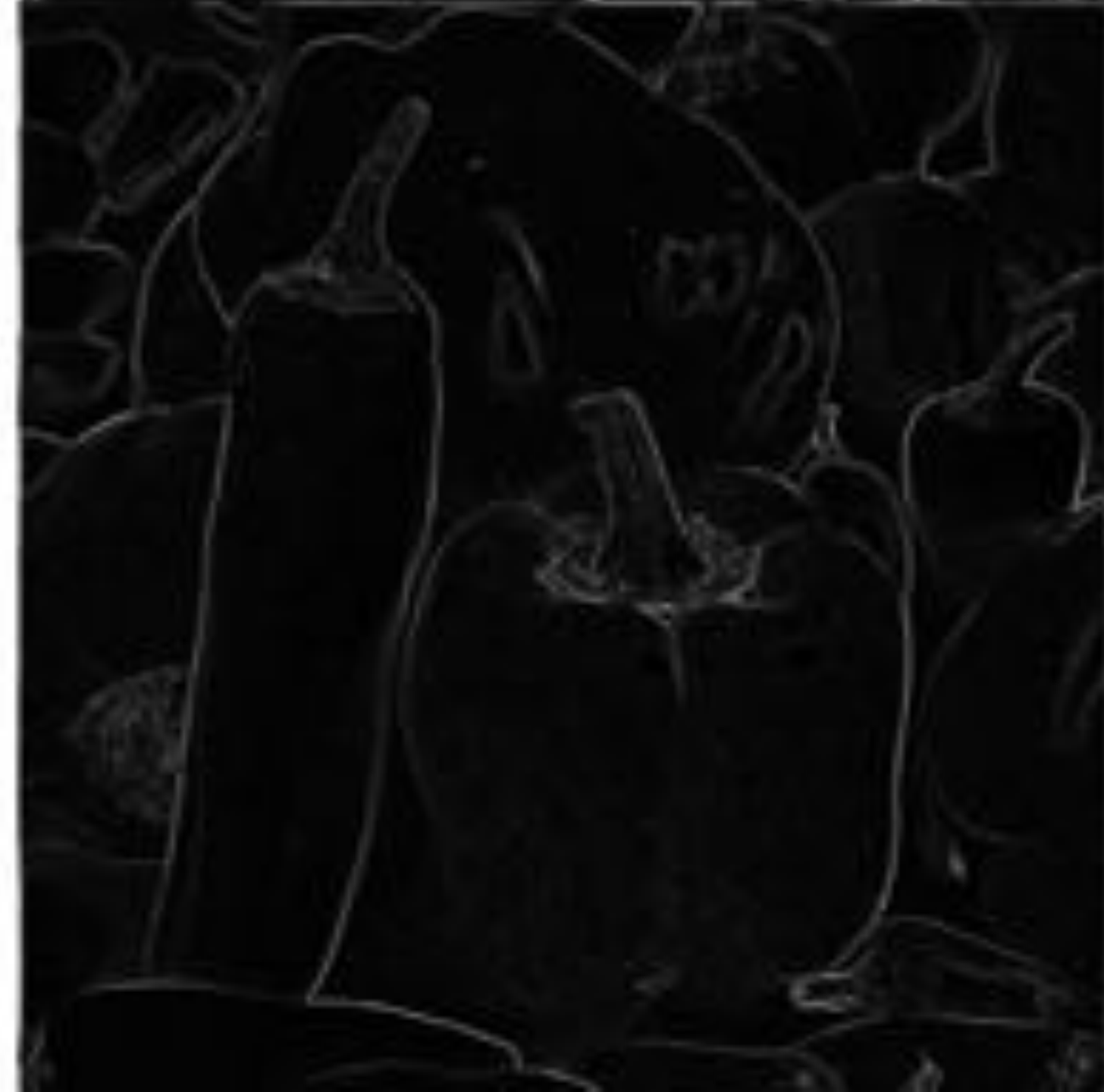
Sobel



Original



Roberts



Prewitt



Sharpening Example



Original
Cameraman

0	0	0
0	1	0
0	0	0

+

0	-1	0
-1	4	-1
0	-1	0

=

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & [5] & -1 \\ 0 & -1 & 0 \end{pmatrix}$$



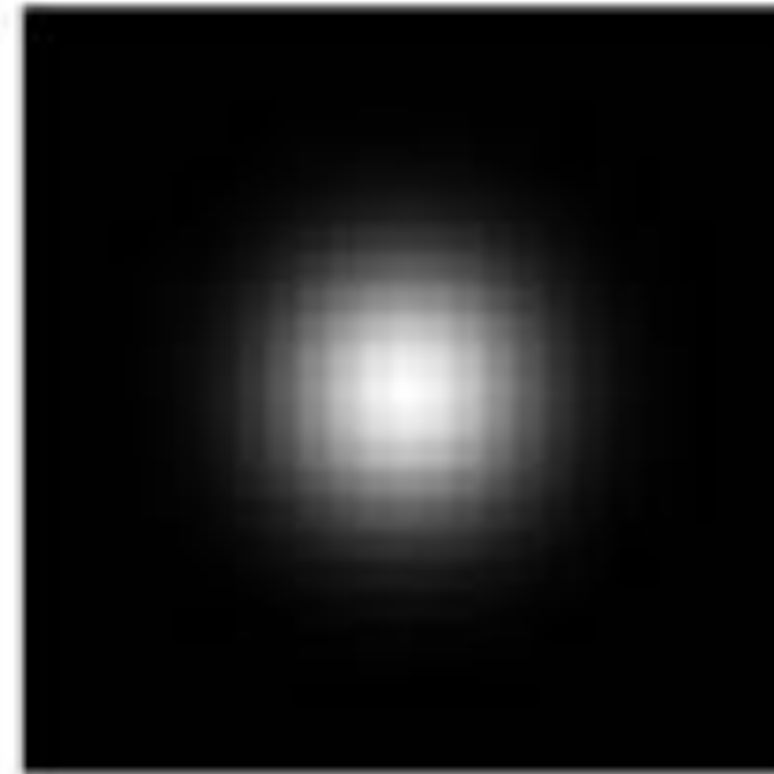
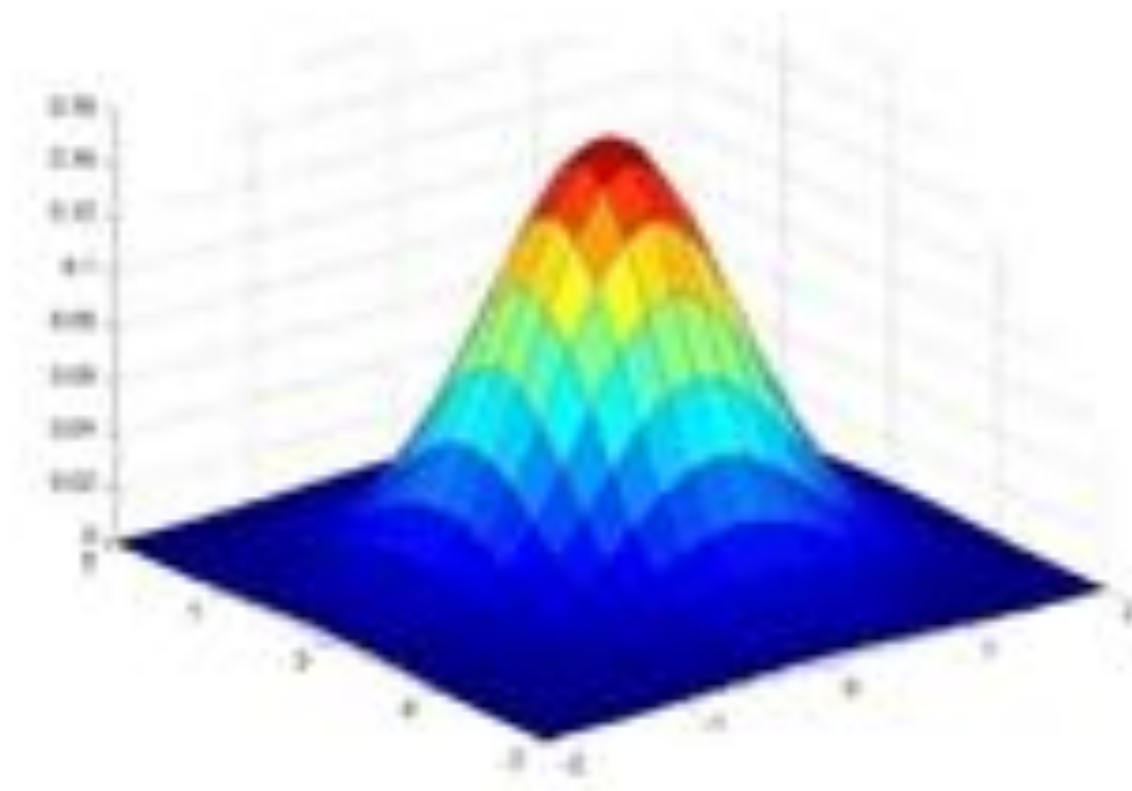
Cameraman sharpened
Filter impulse response



Cameraman sharpened
Filter impulse response

$$\frac{1}{4} \begin{pmatrix} 0 & -1 & 0 \\ -1 & [8] & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Important Filter: Gaussian

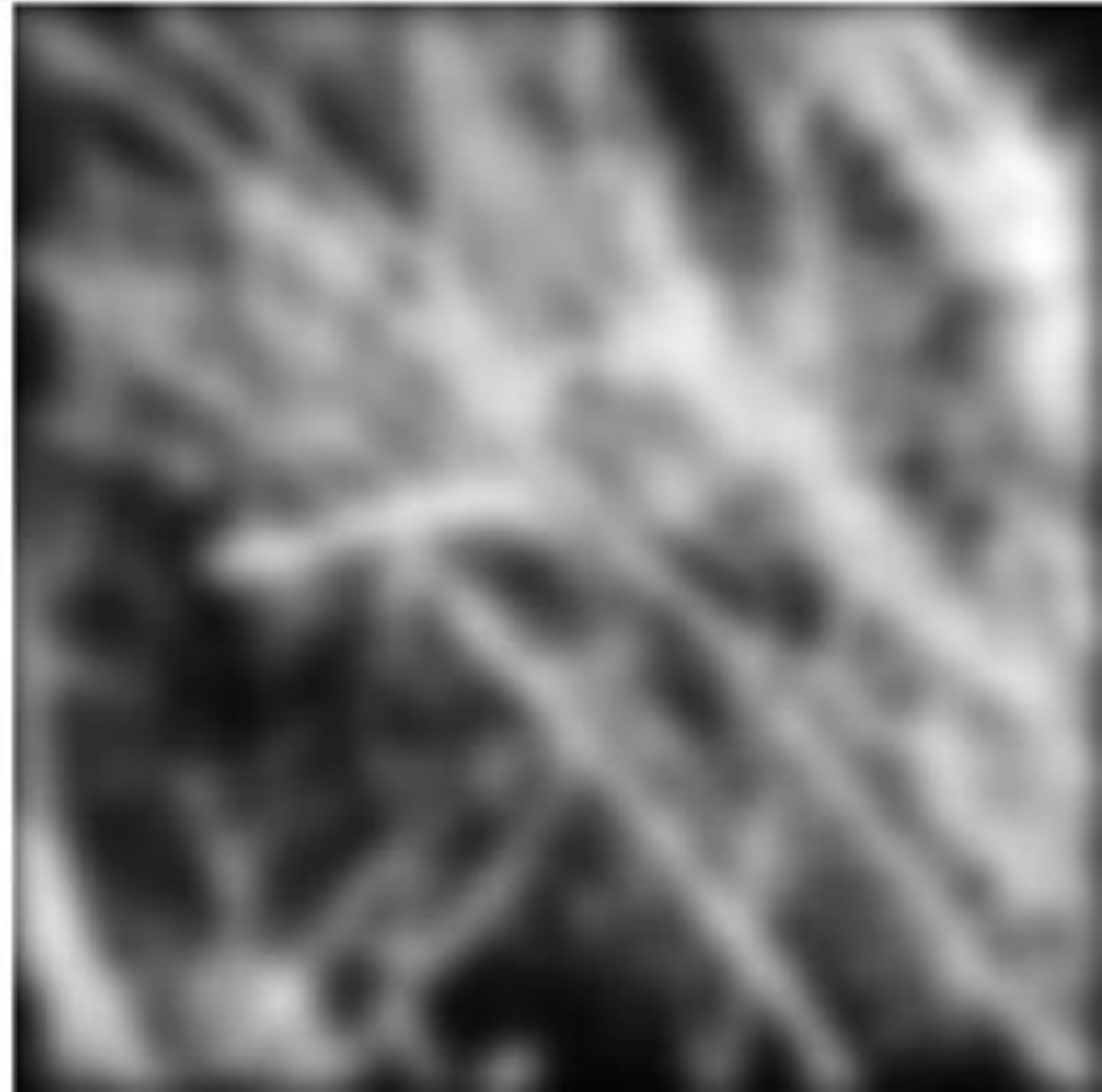


0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, $\sigma = 1$

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

Smoothing with Gaussian Filter



Practice with Linear Filters



Original

0	0	0
0	1	0
0	0	0

?



Practice with Linear Filters



Original

0	0	0
0	1	0
0	0	0



Filtered
(no change)

Practice with Linear Filters



Original

0	0	0
0	0	1
0	0	0

?



Practice with Linear Filters



Original

0	0	0
0	0	1
0	0	0



Shifted left
By 1 pixel

Practice with Linear Filters



Original

0	0	0
0	2	0
0	0	0

-

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

(Note that filter sums to 1)



Practice with Linear Filters



Original

0	0	0
0	2	0
0	0	0

-

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

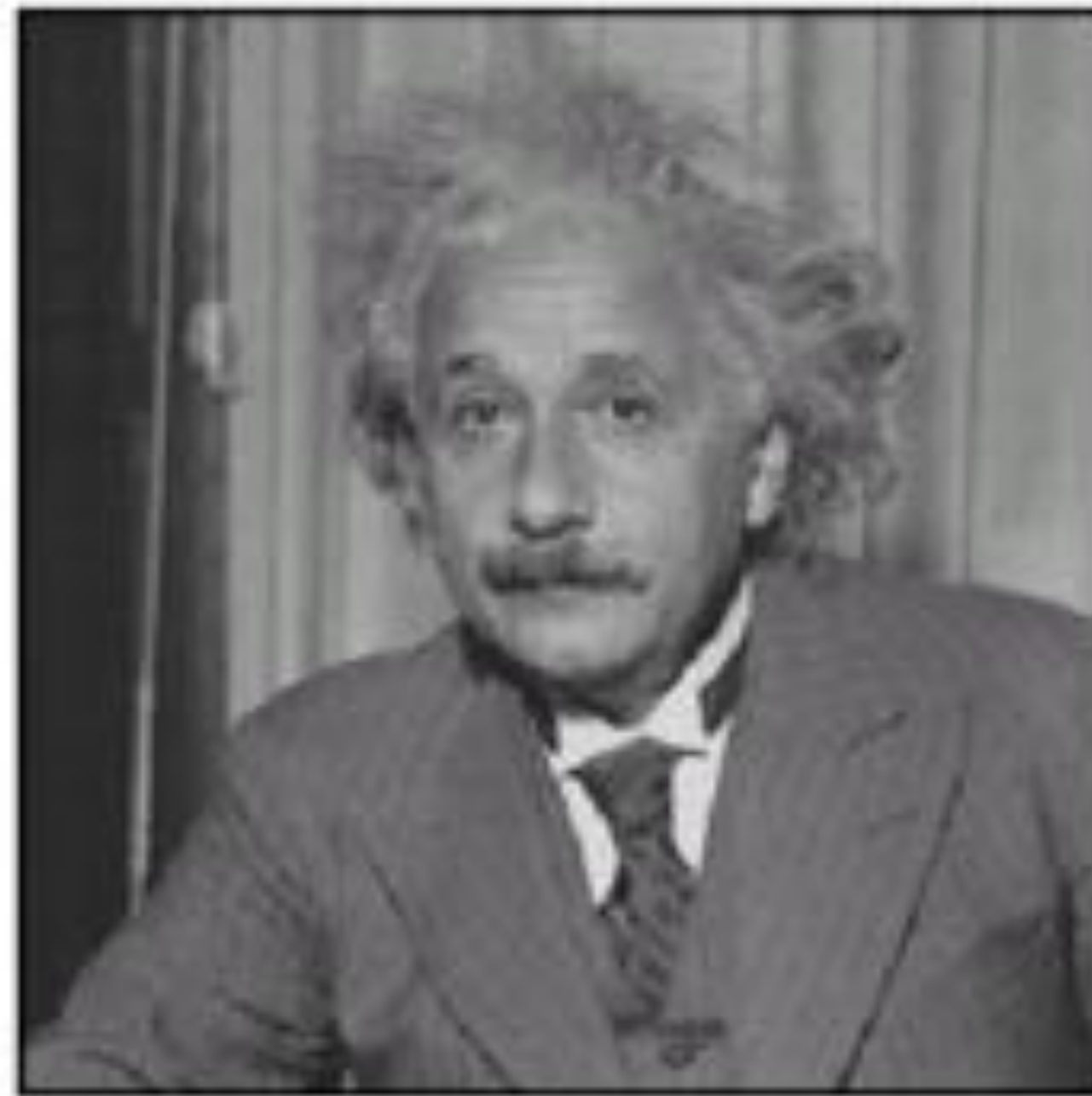


Sharpening filter

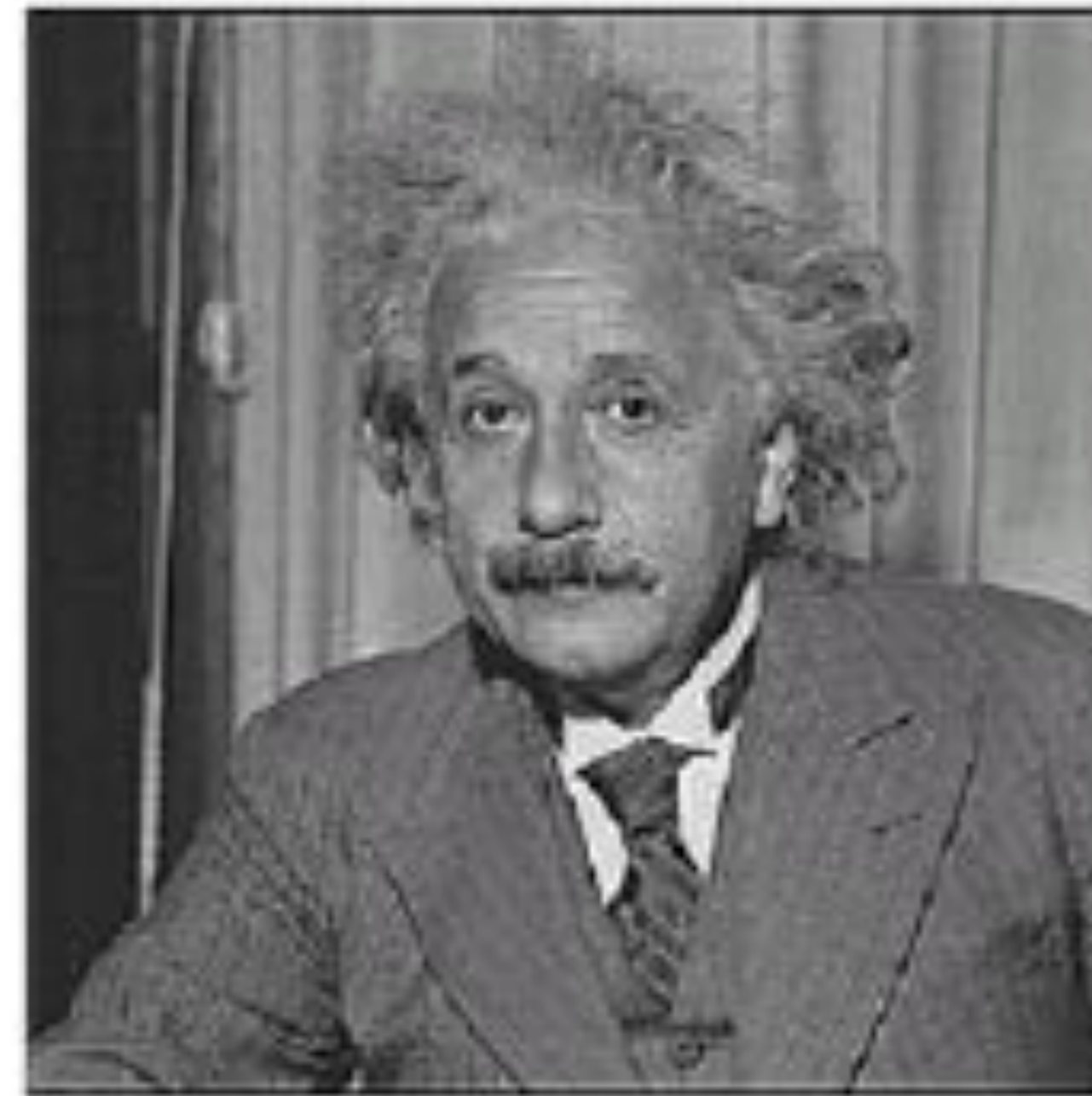
- Accentuates differences with local average



Sharpening



before

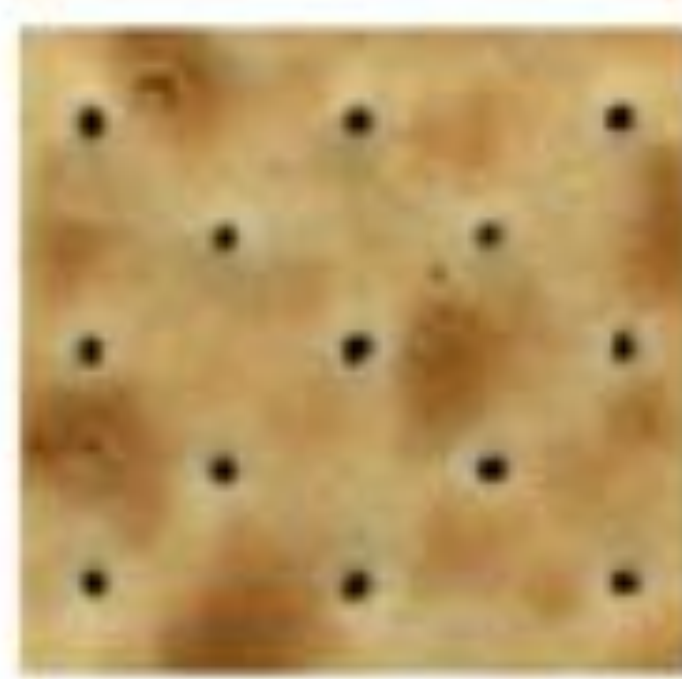


after



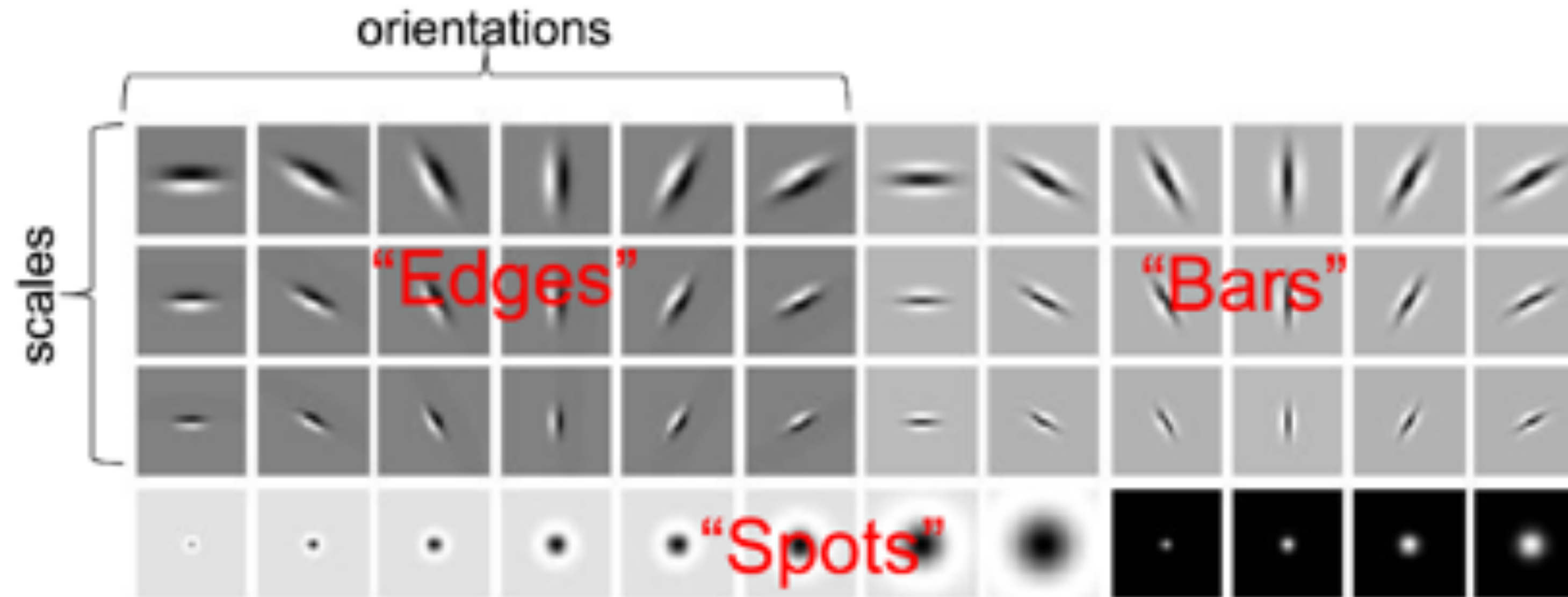
Image Classification Using Filter Bank





Filter Banks

- ▶ What filters to put in the bank?
 - ▶ Typically we want a combination of scales and orientations, different types of patterns.



Matlab code available for these examples: <http://www.robots.ox.ac.uk/~vgg/research/texclass/filters.html>

Filter Banks

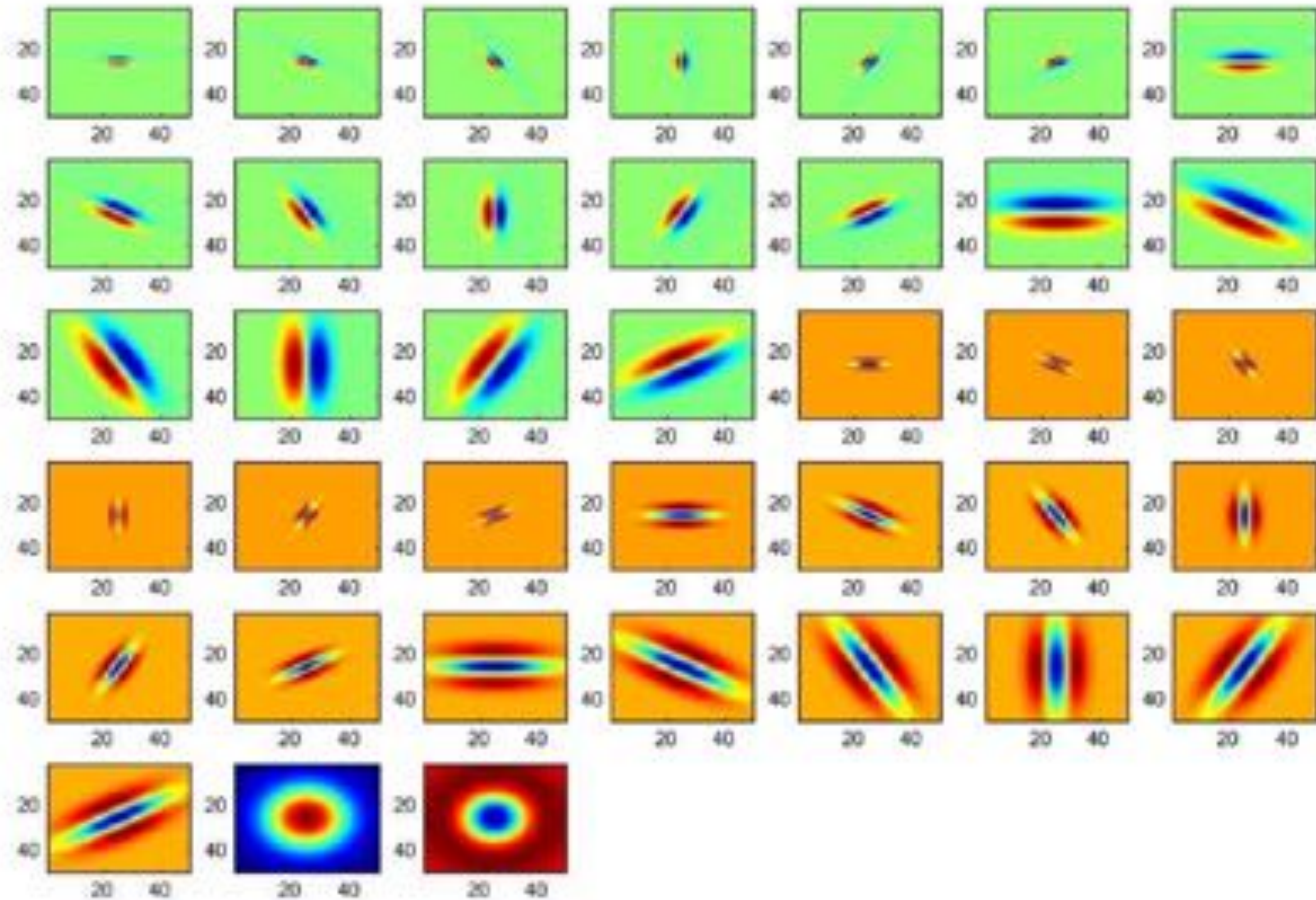
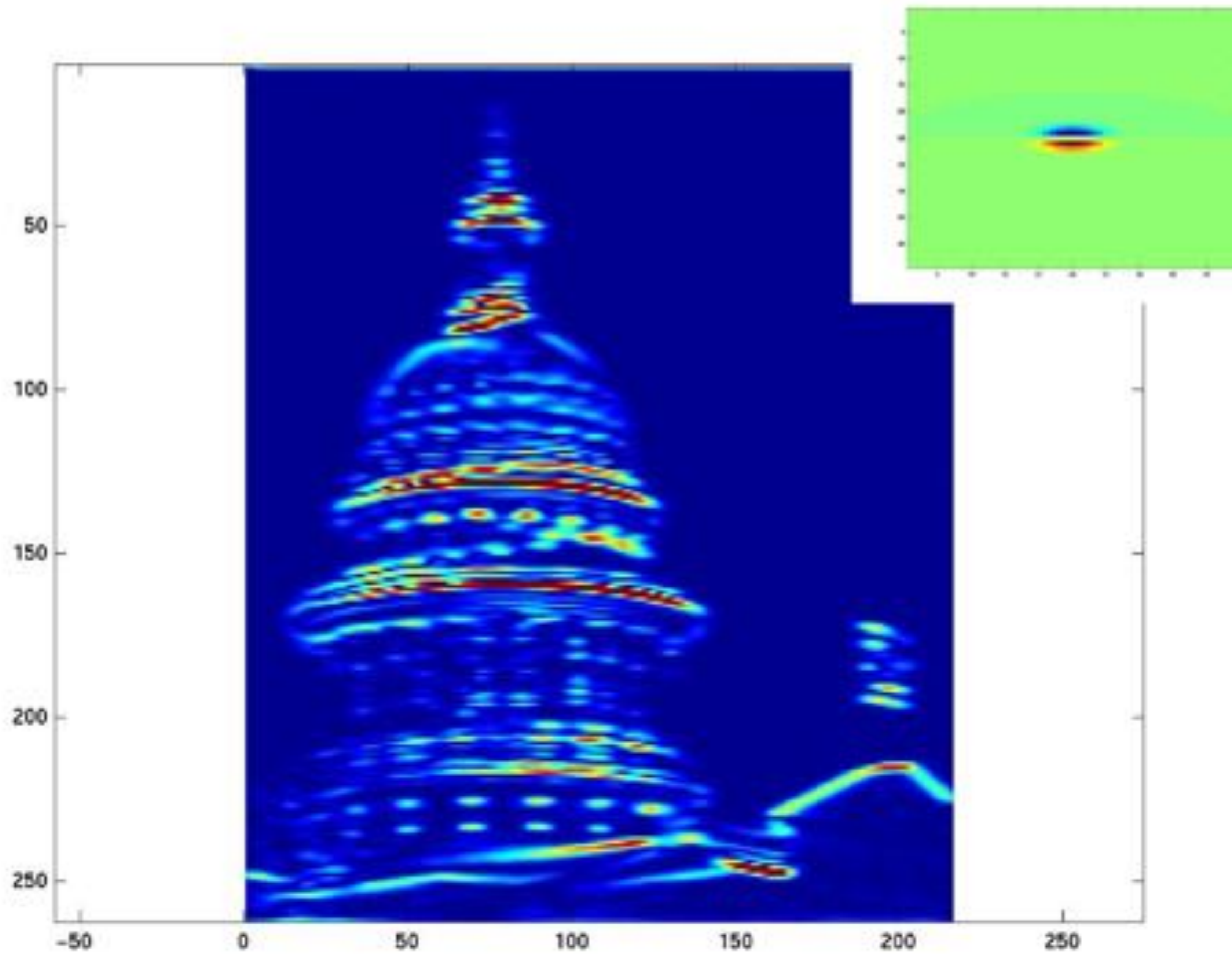
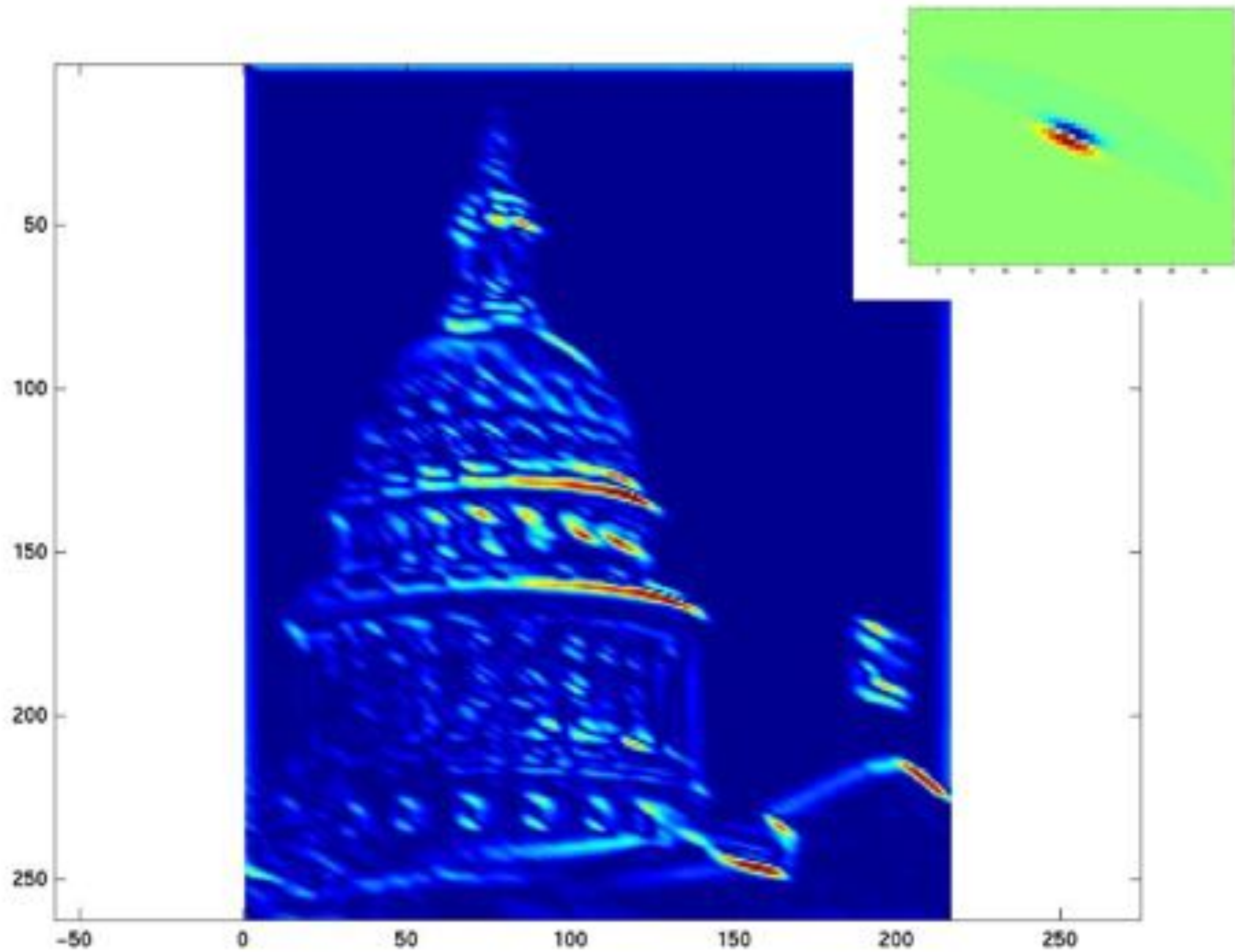


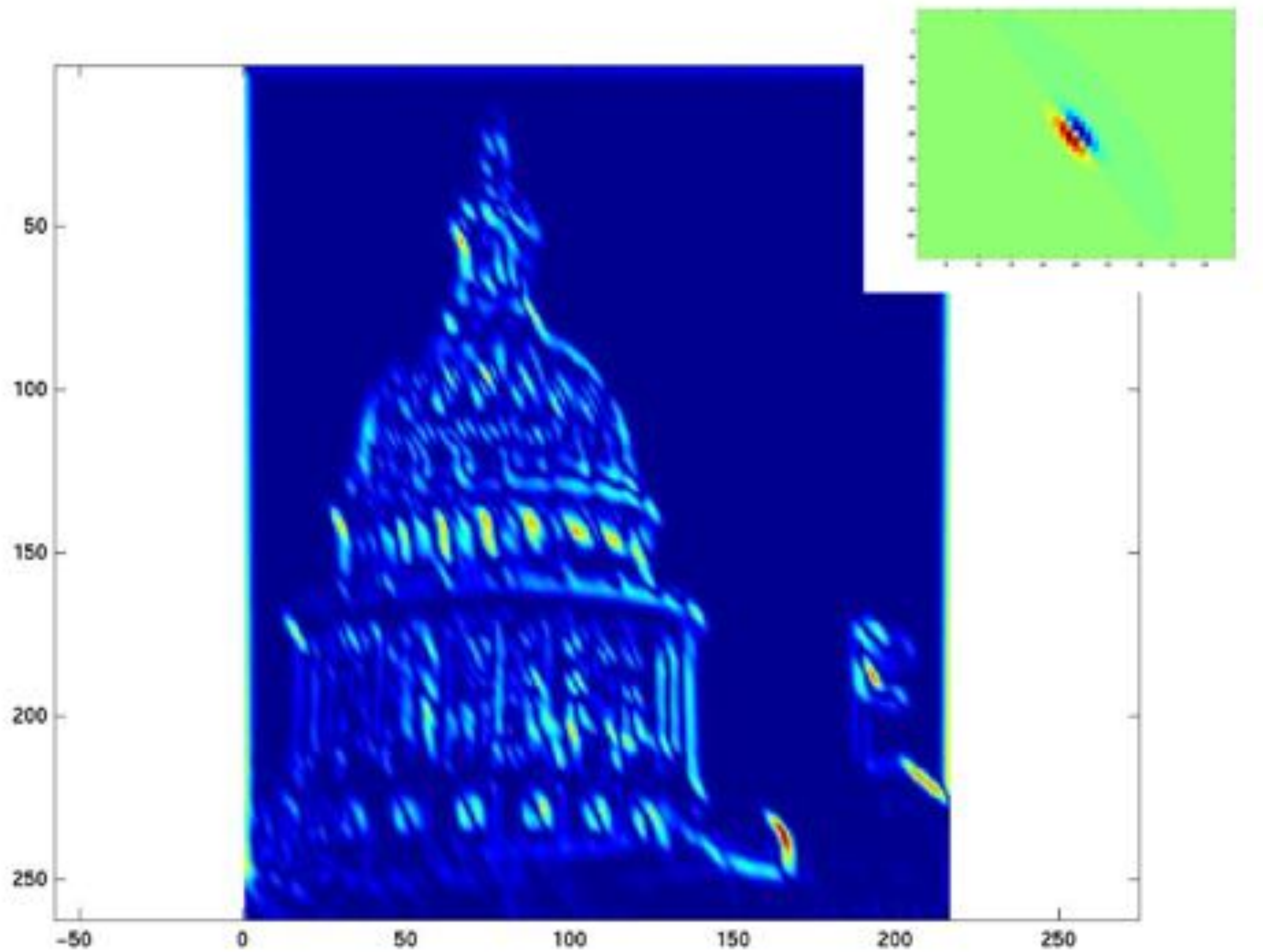
Image from <http://www.texasexplorer.com/austincap2.jpg>

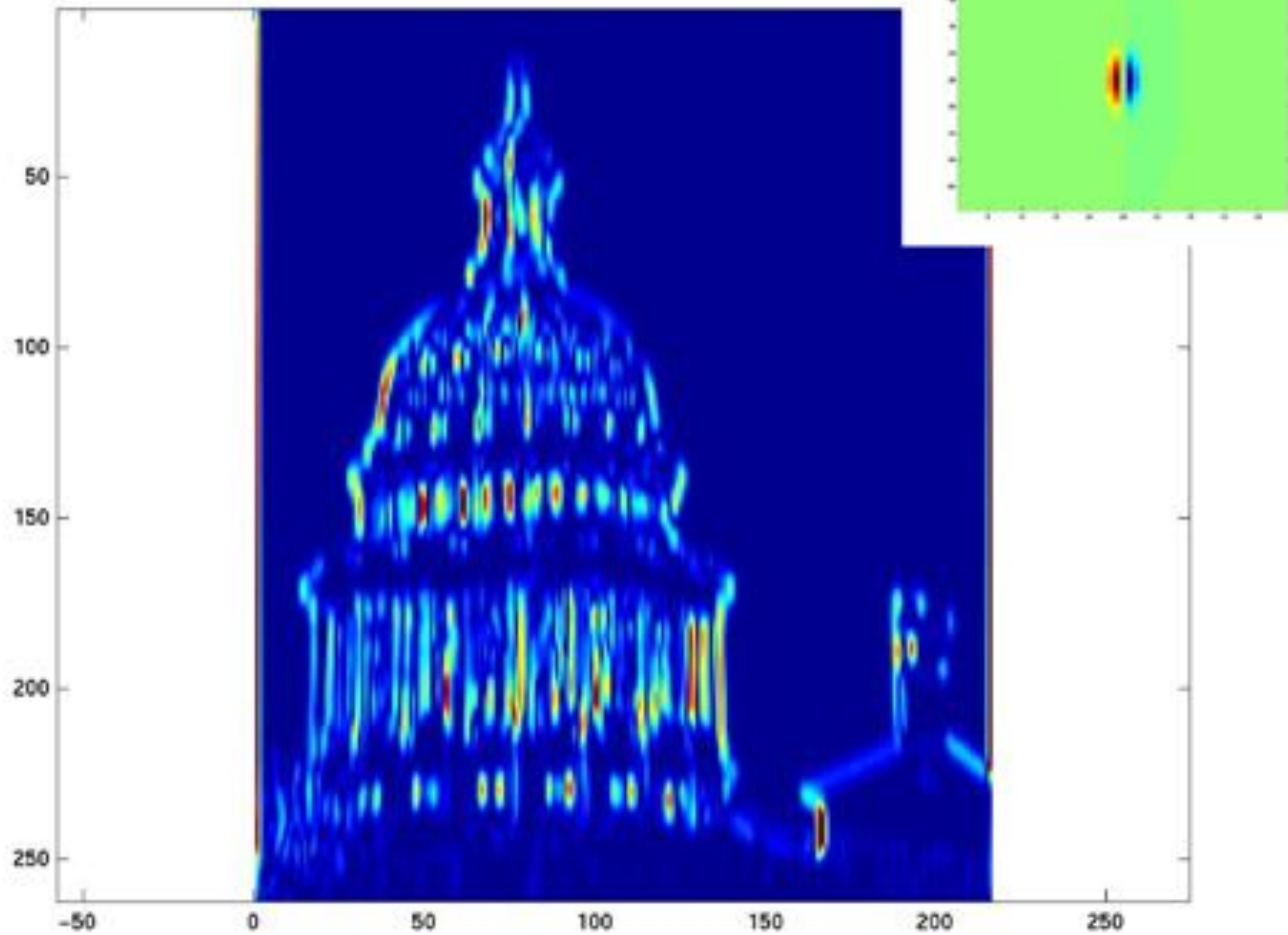
Kristen Grauman

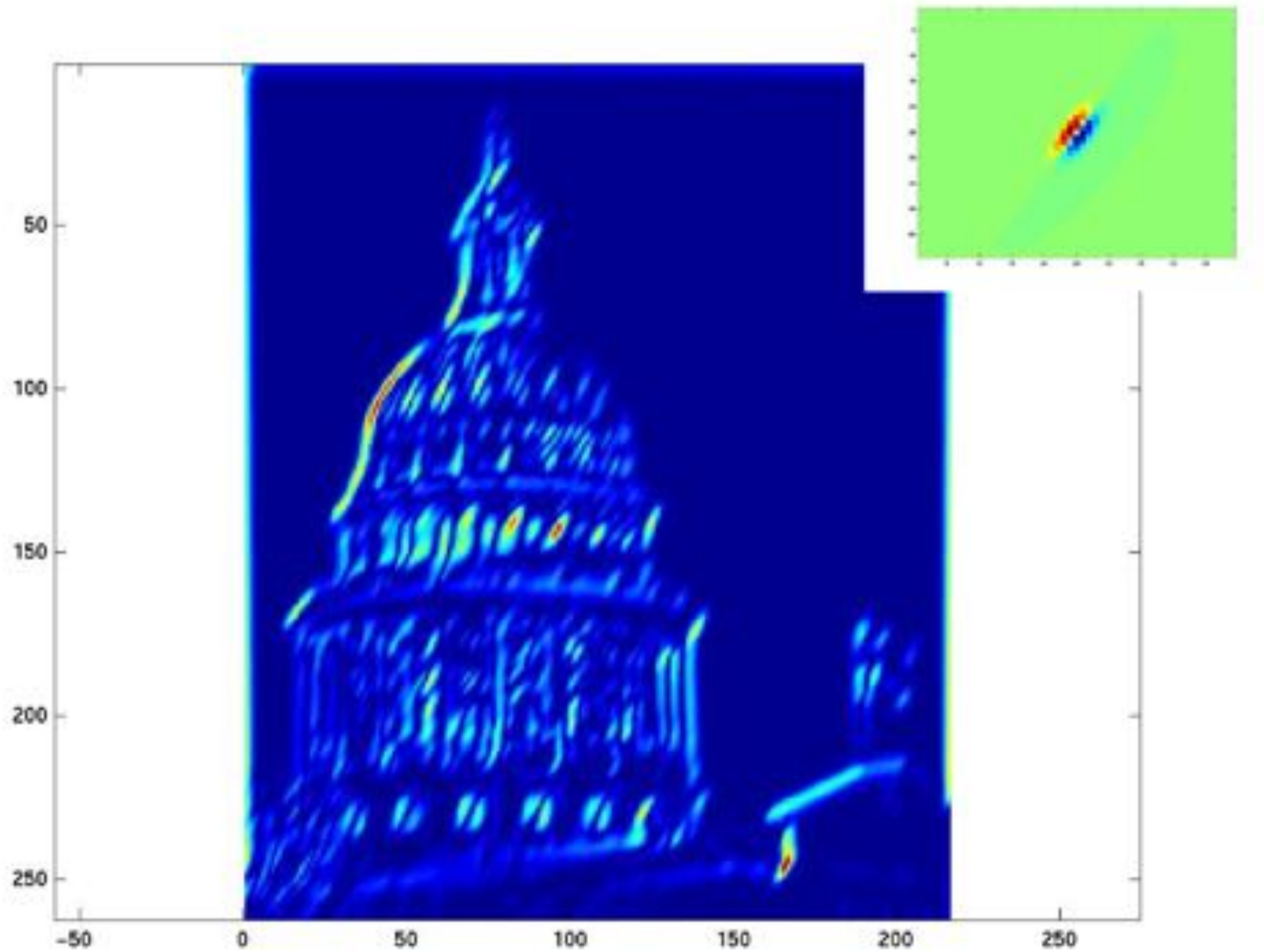


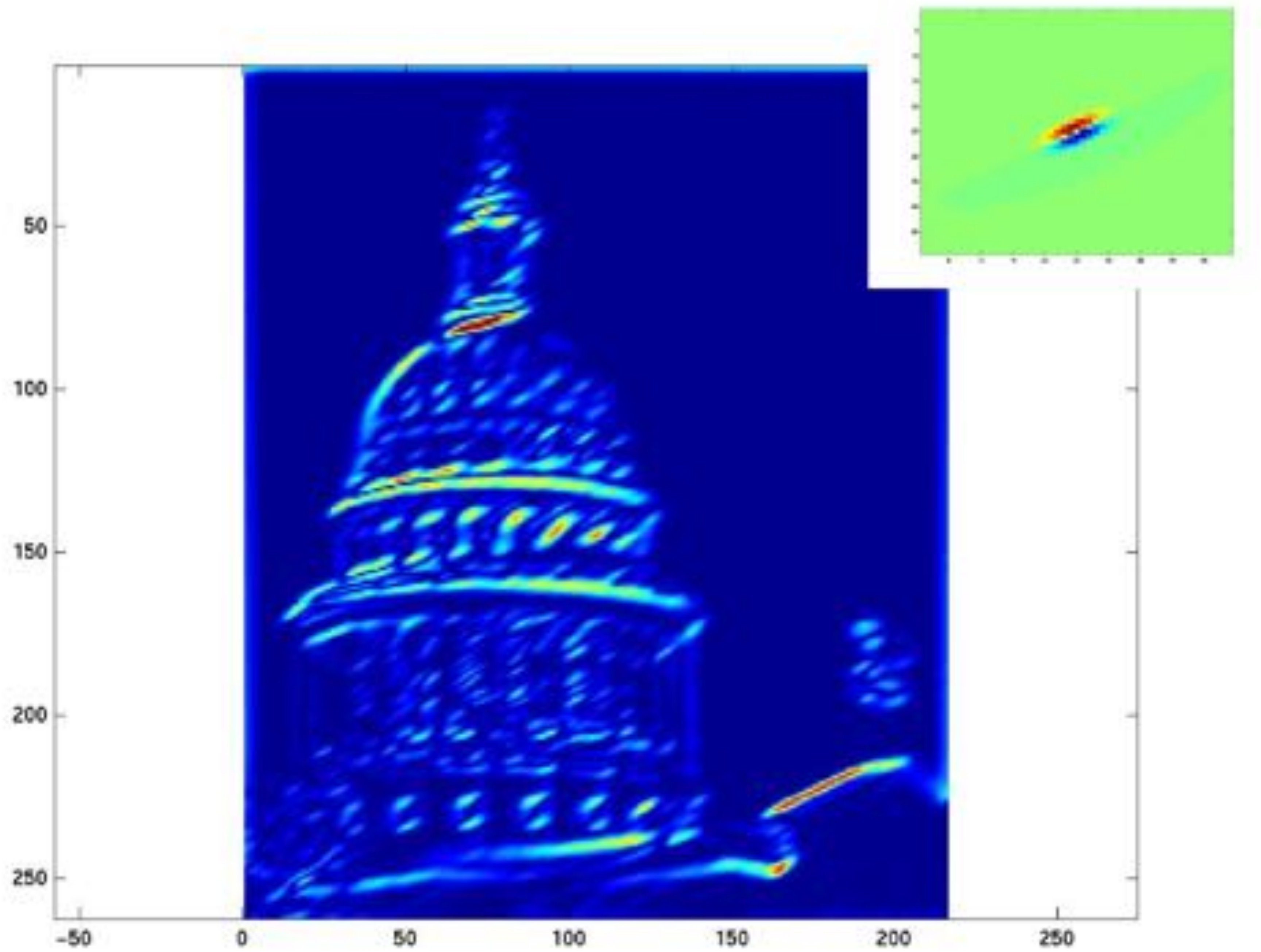


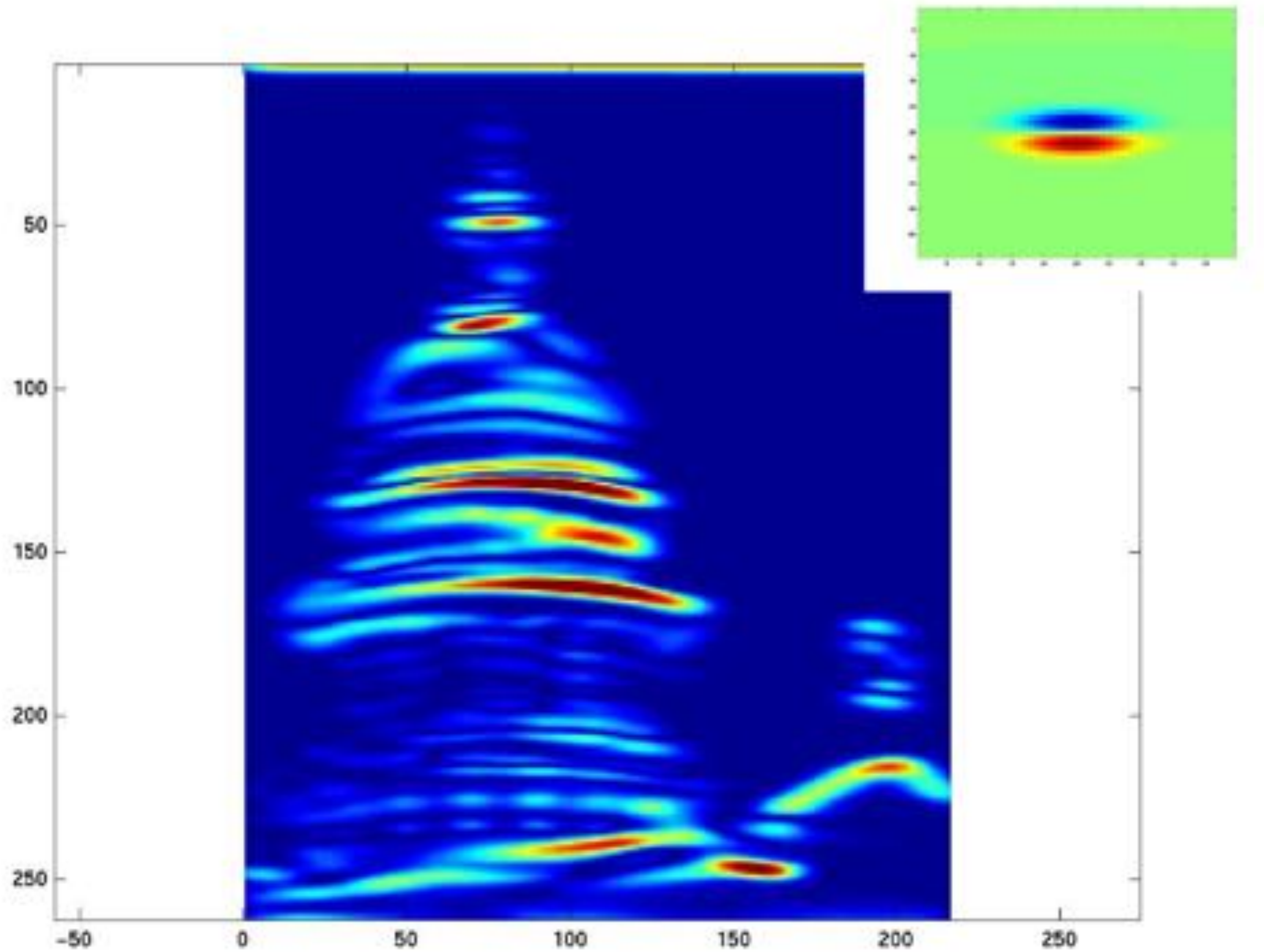


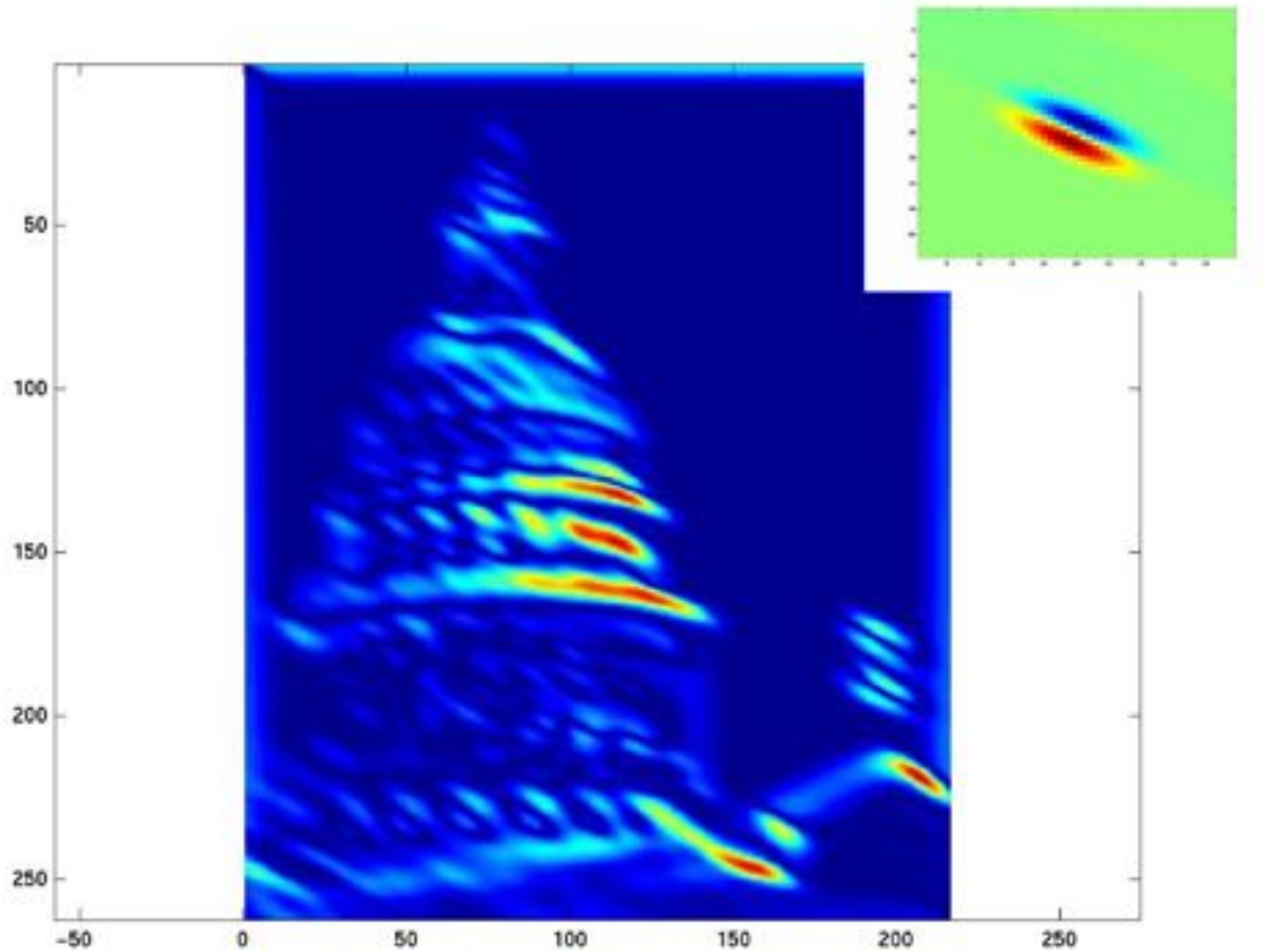


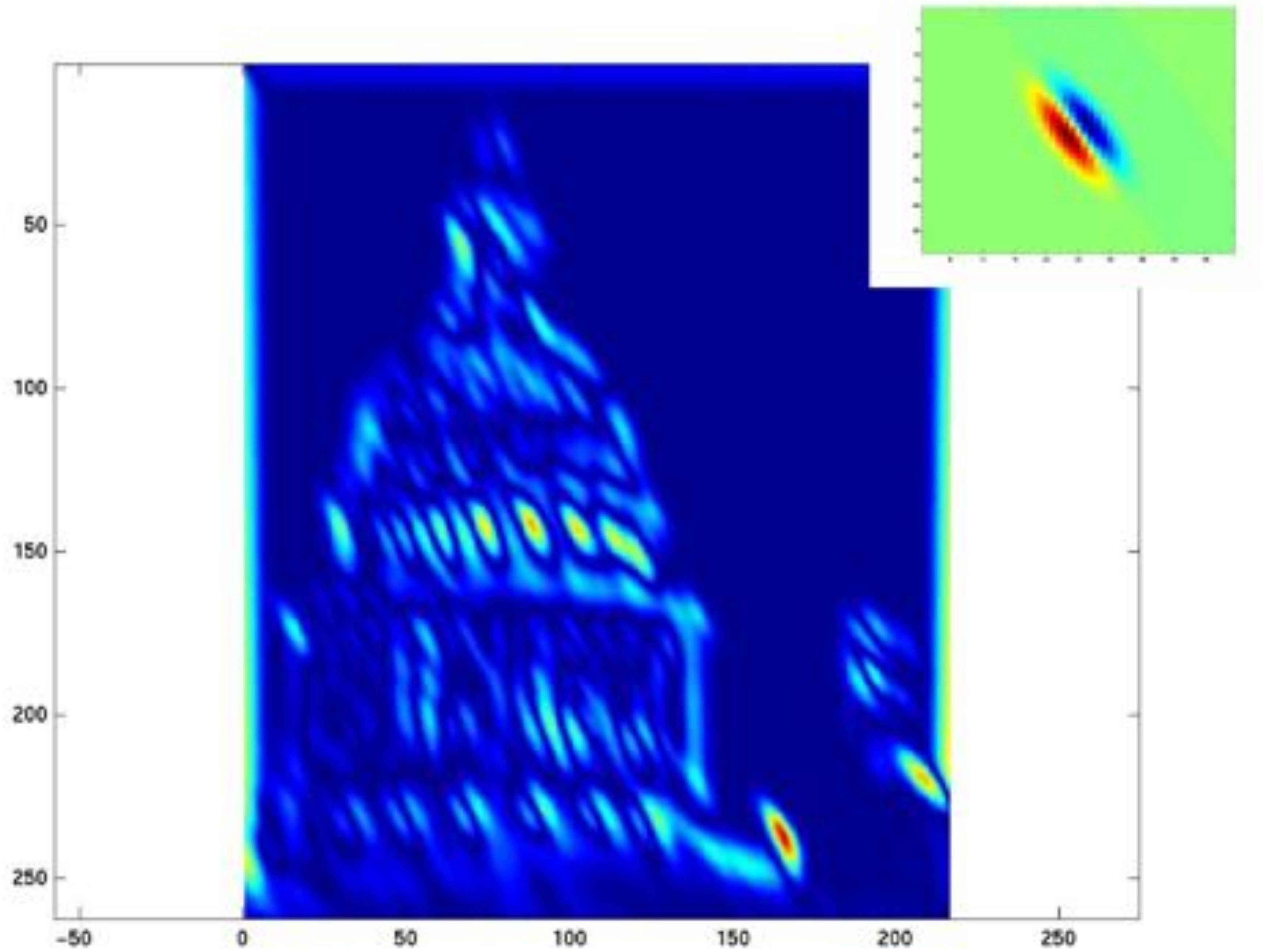


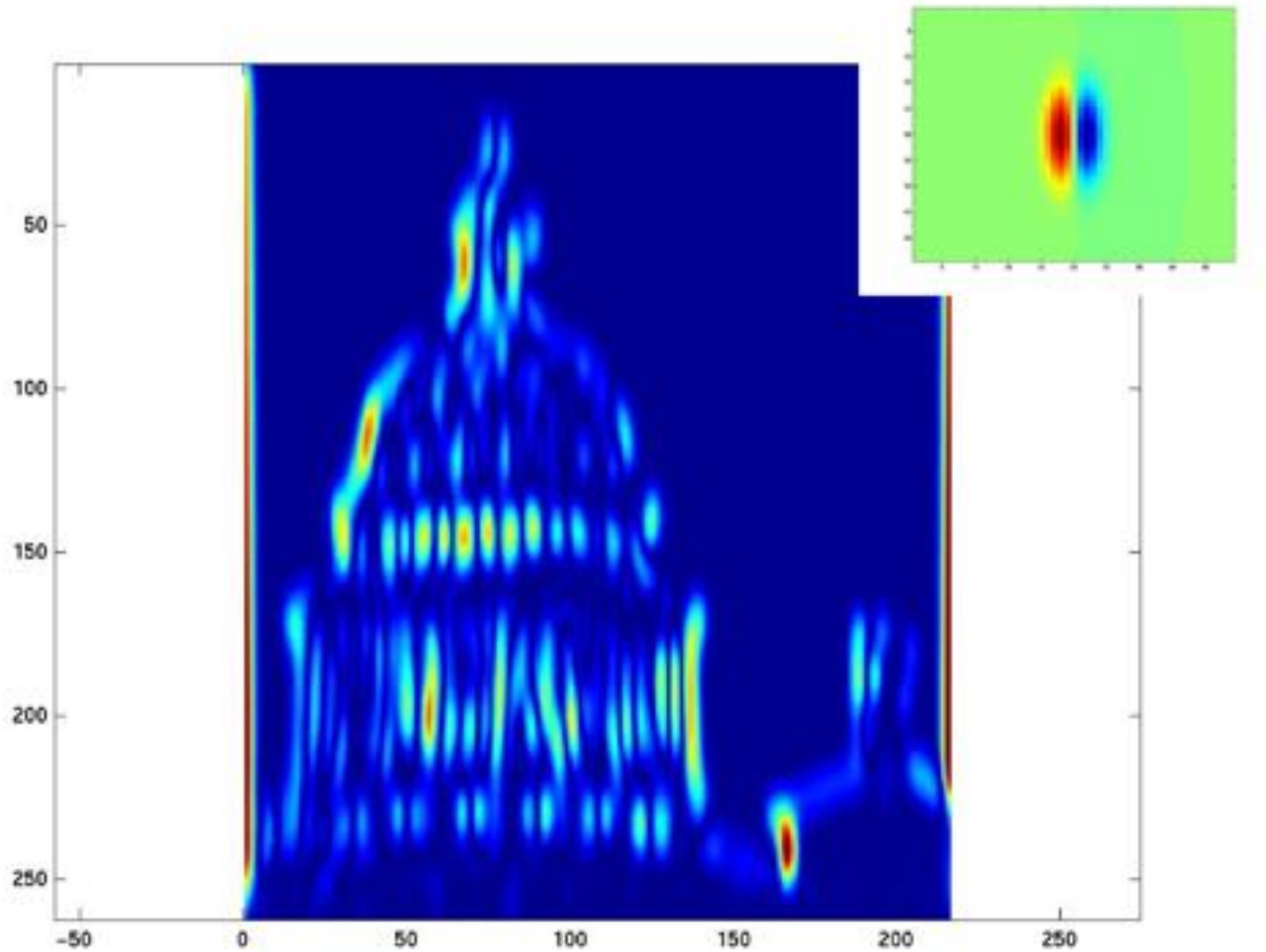


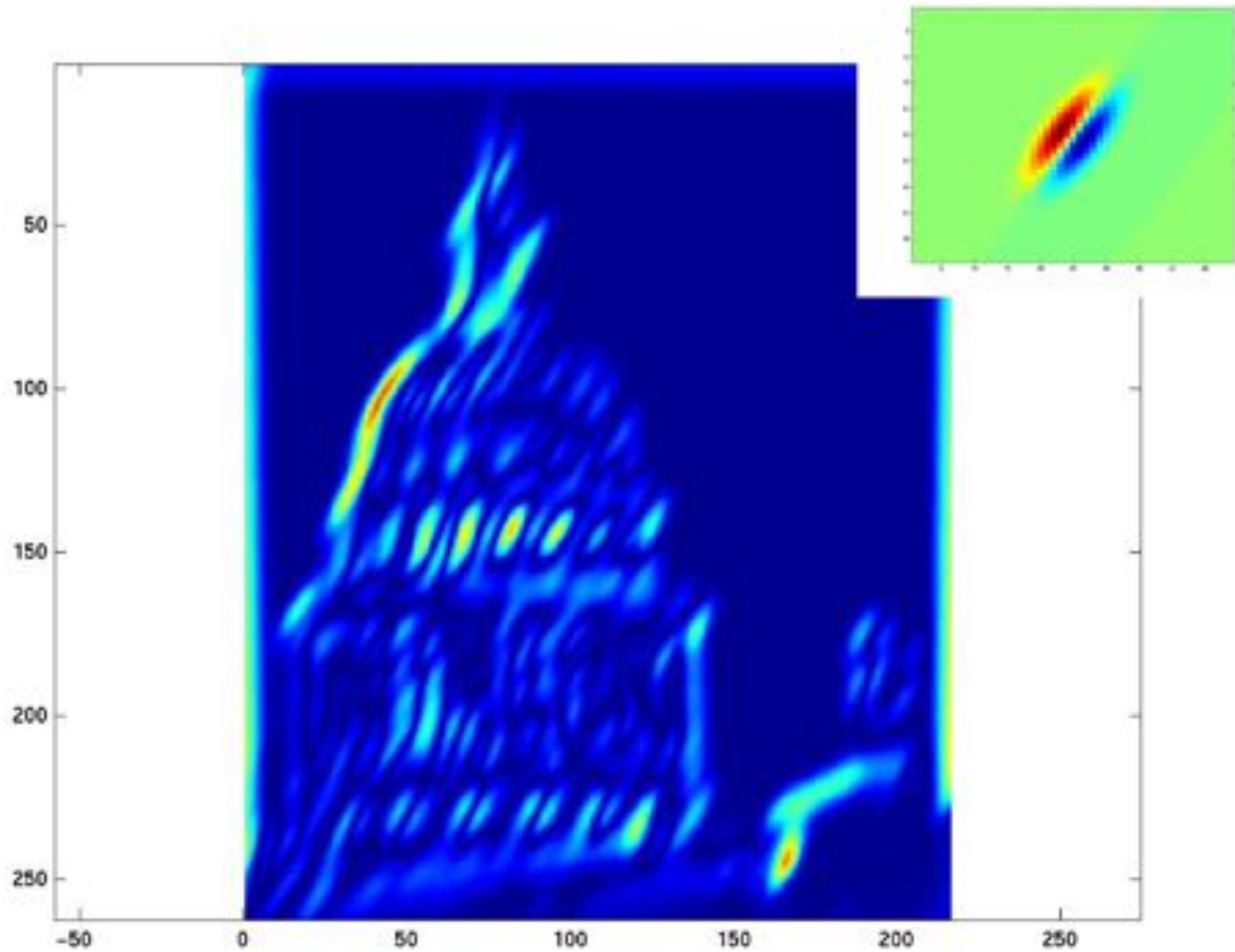


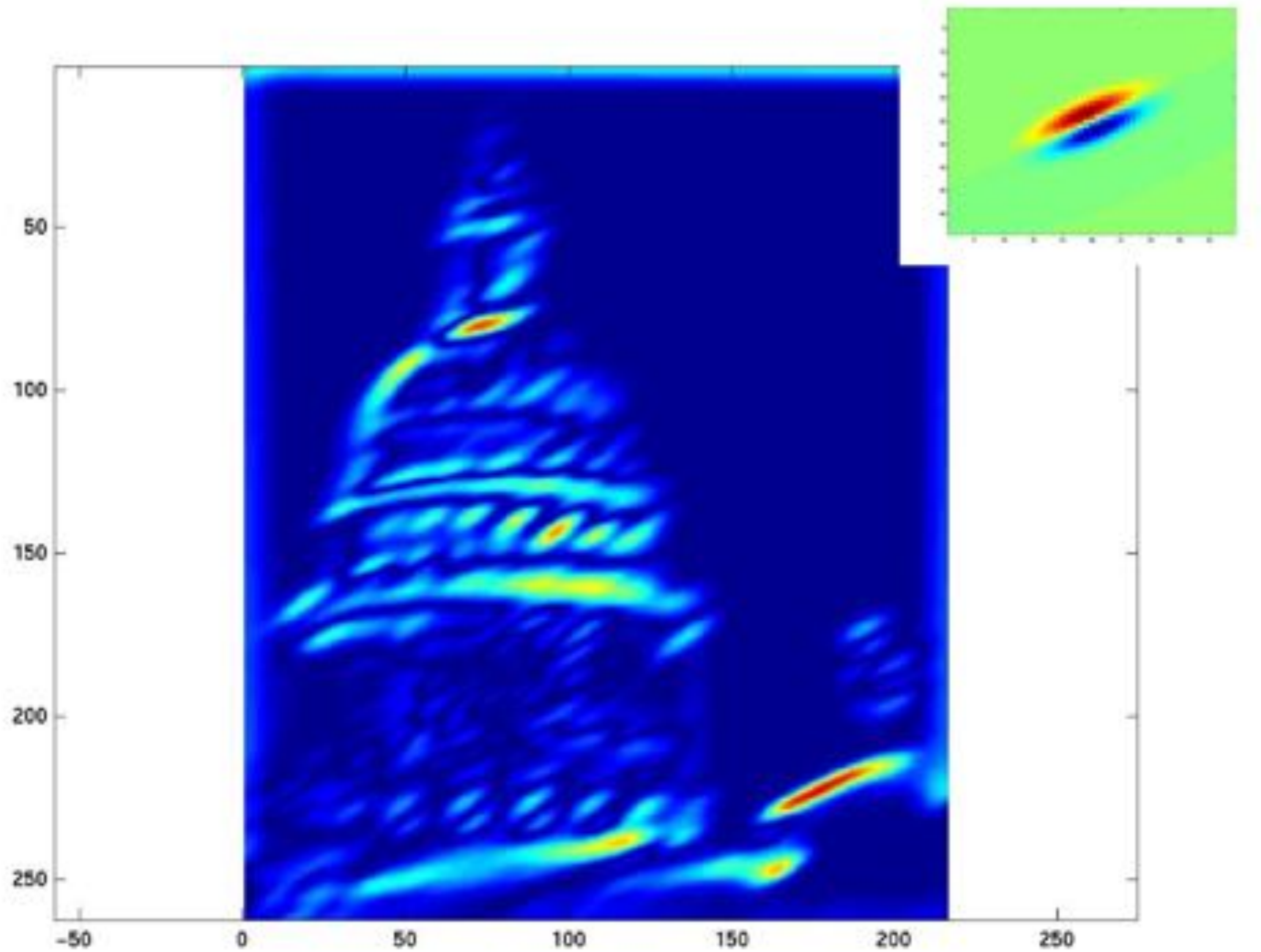


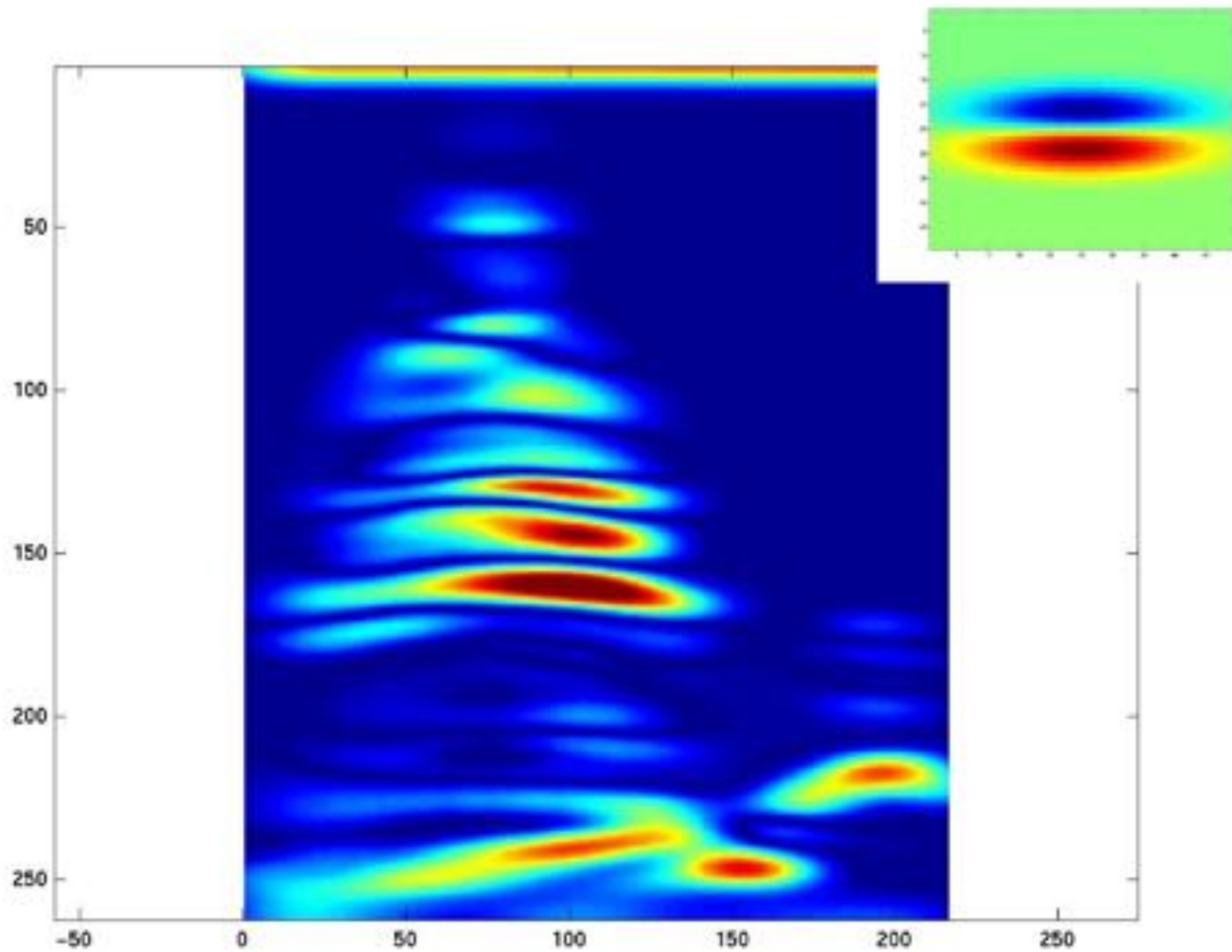


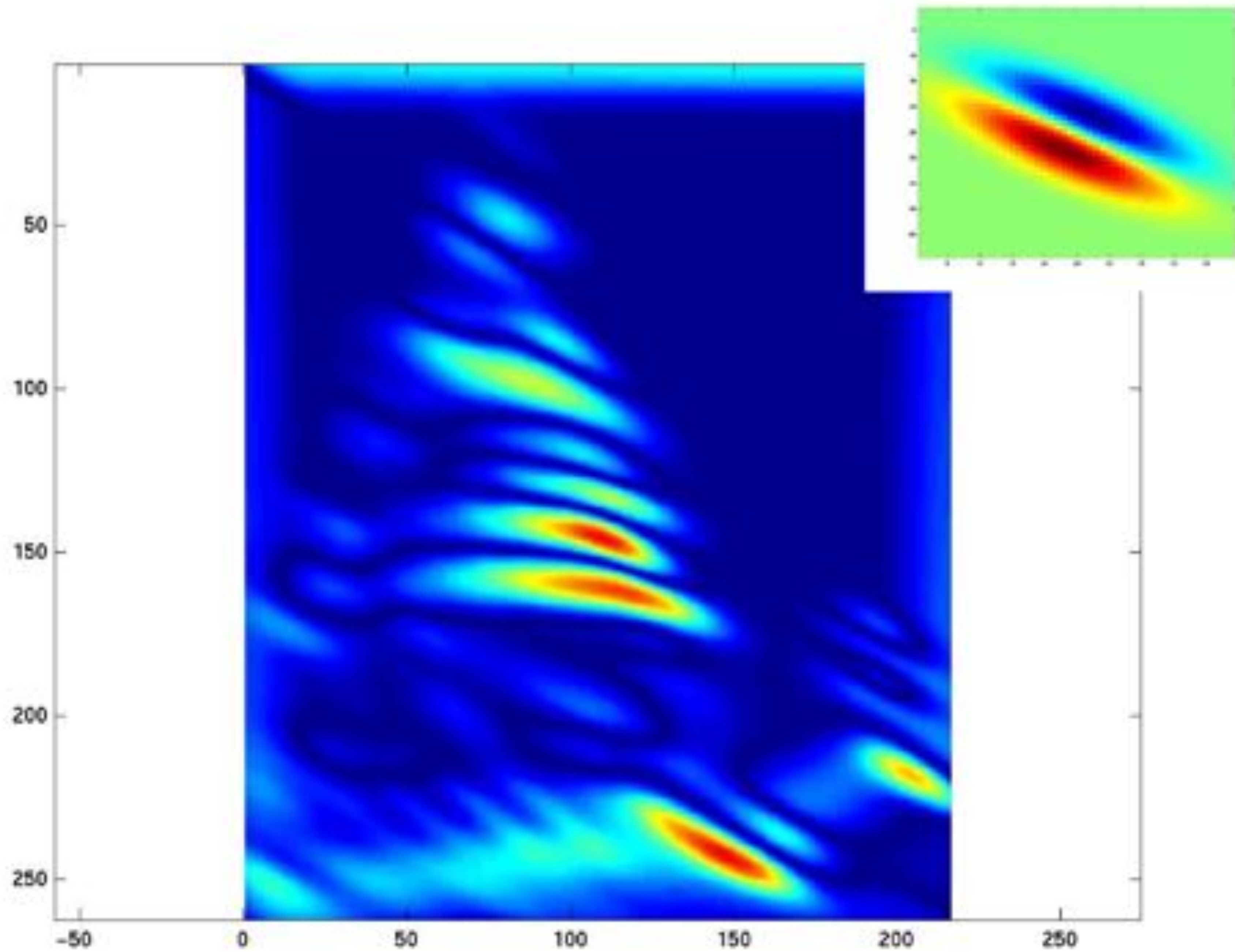


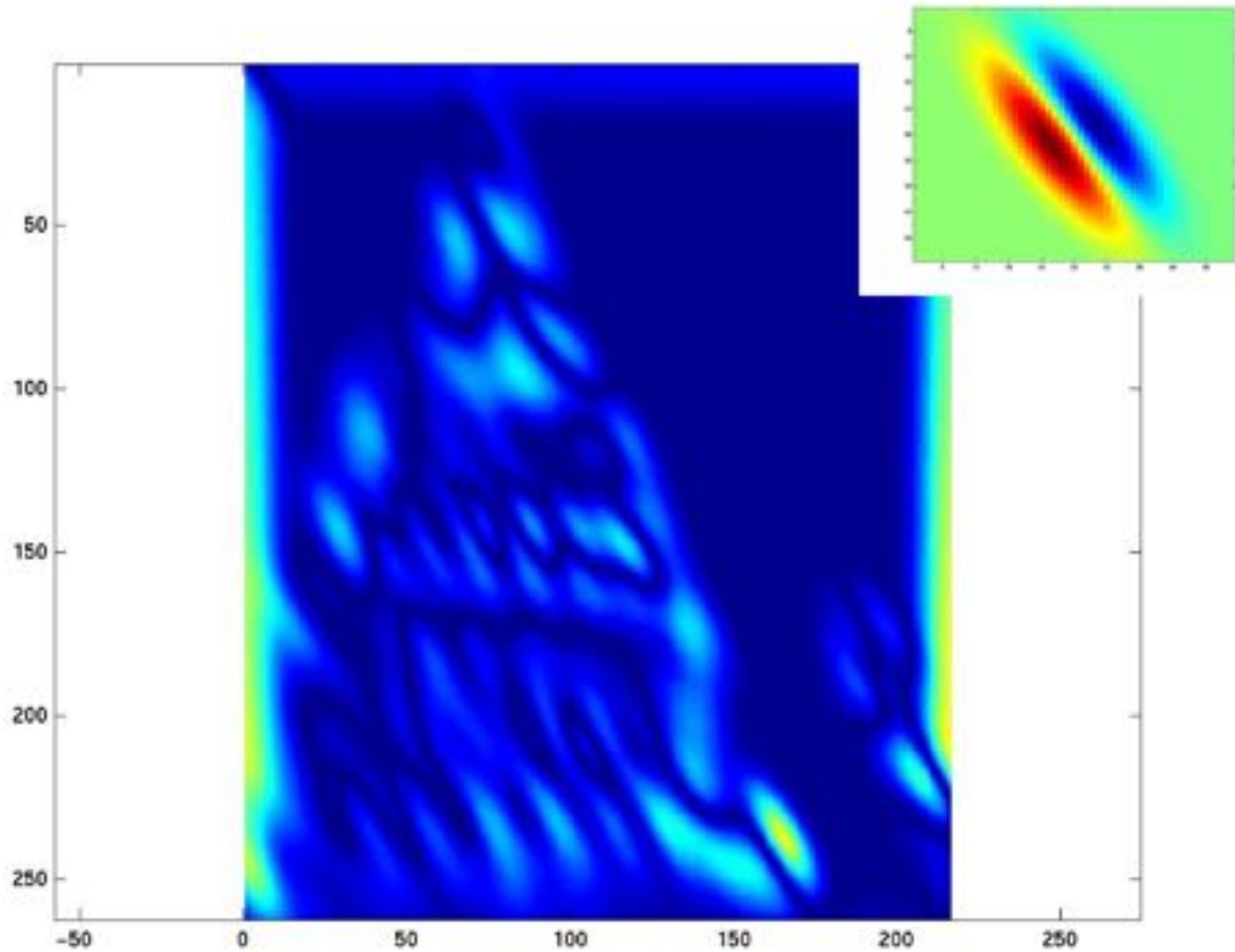


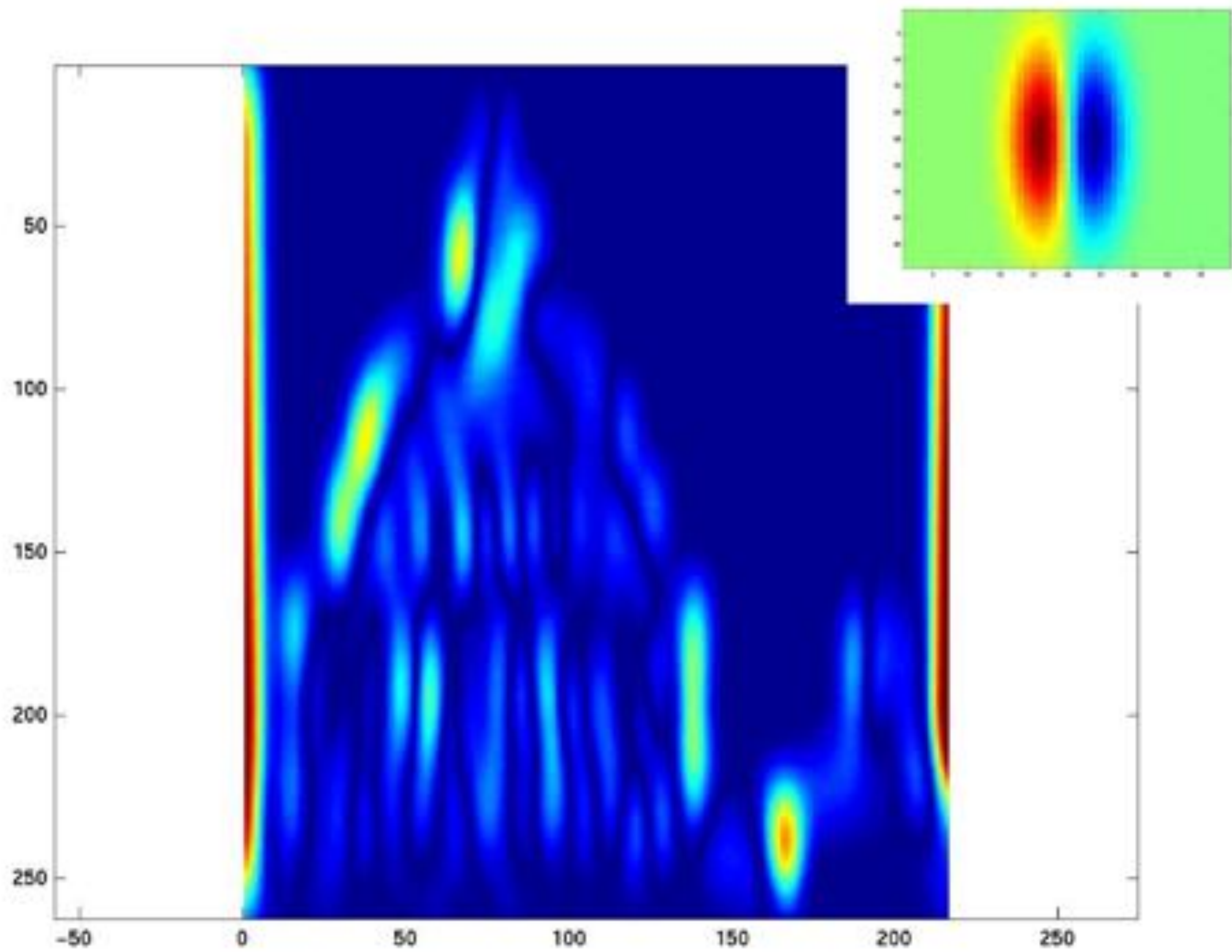


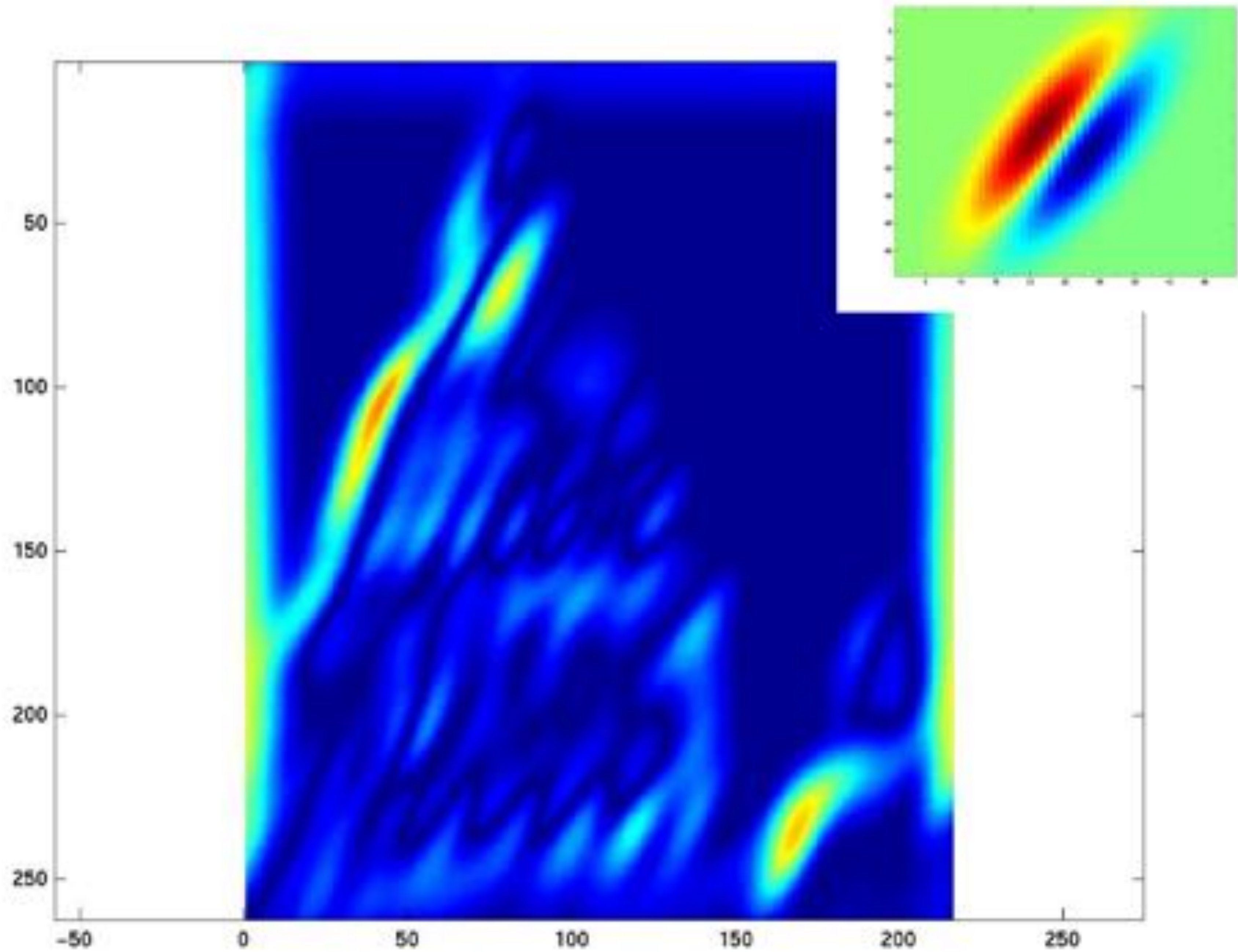


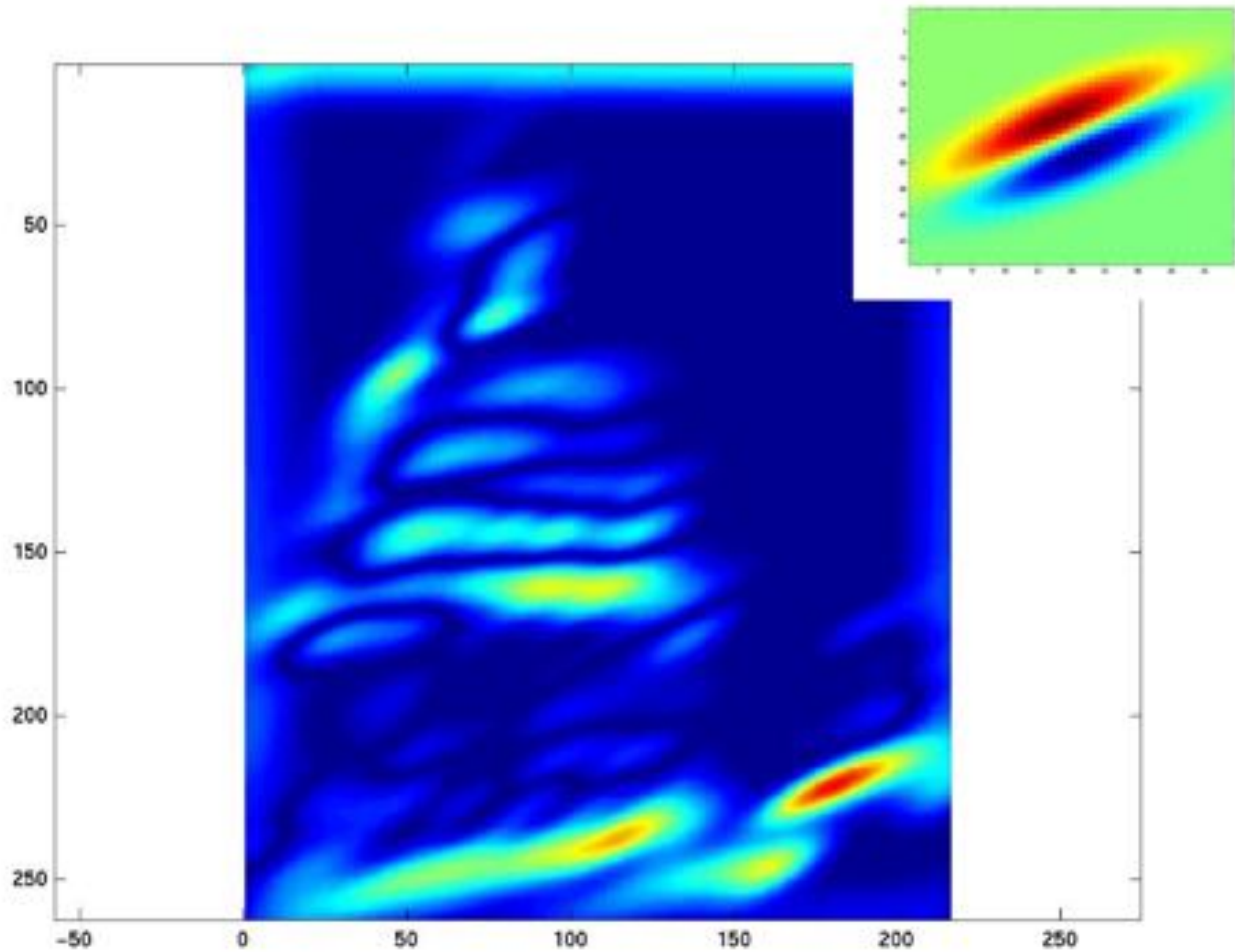


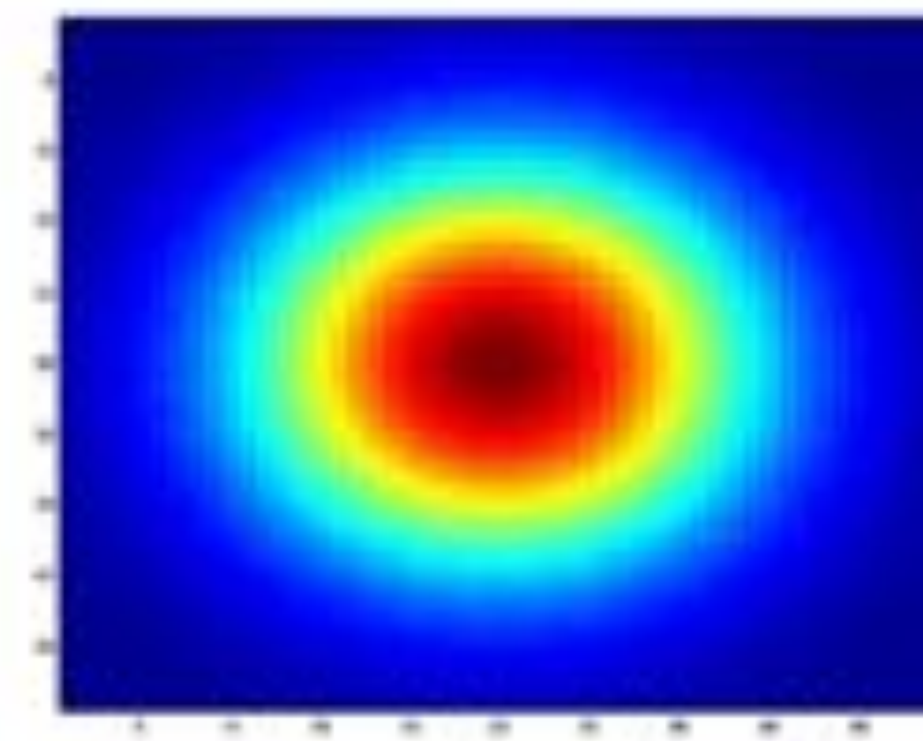
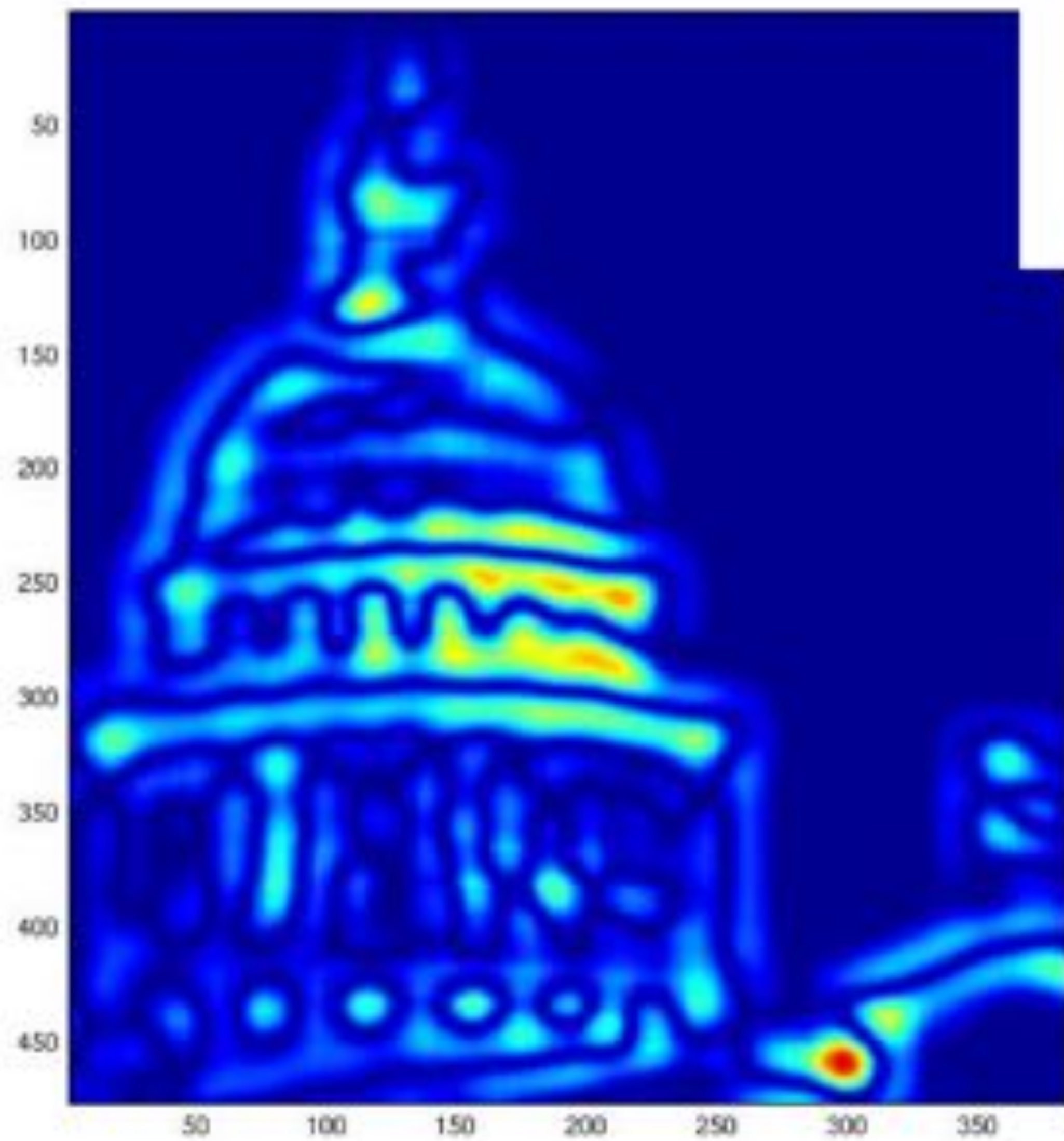


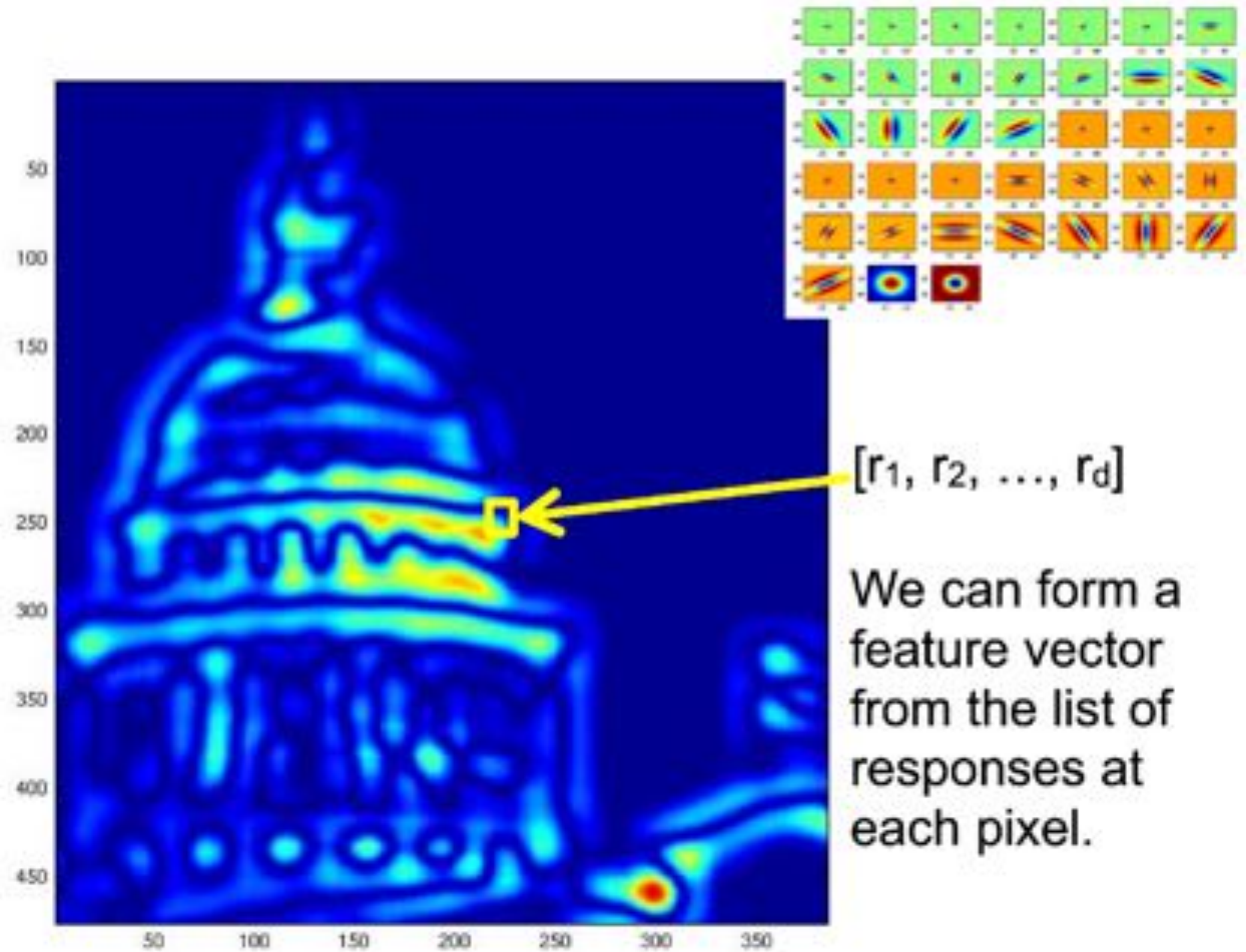


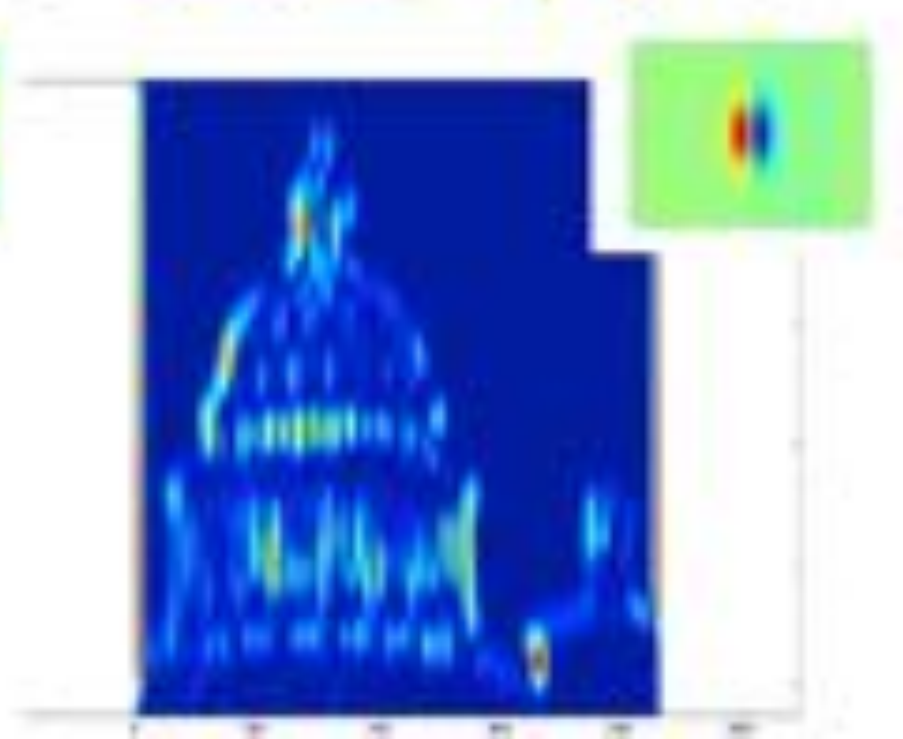
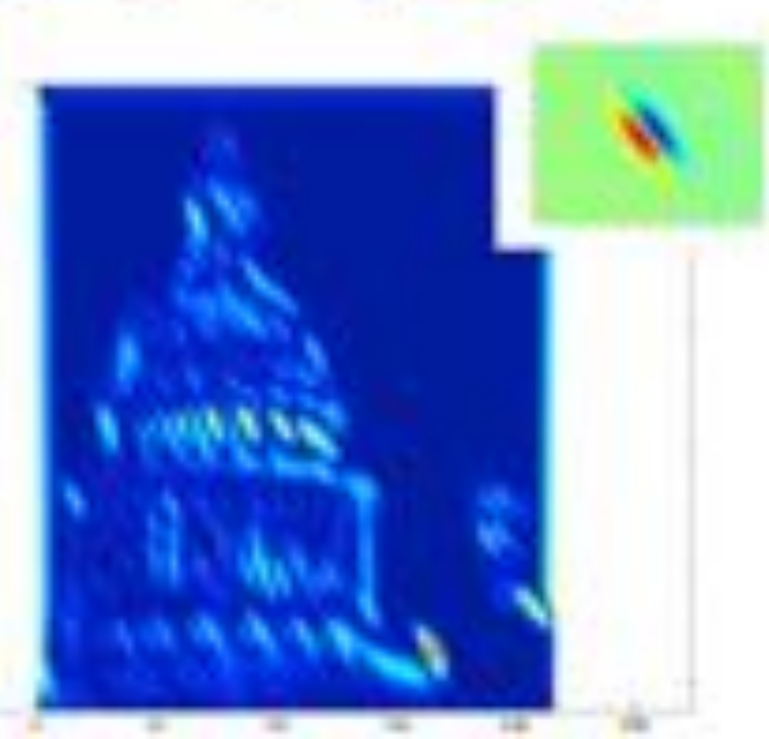
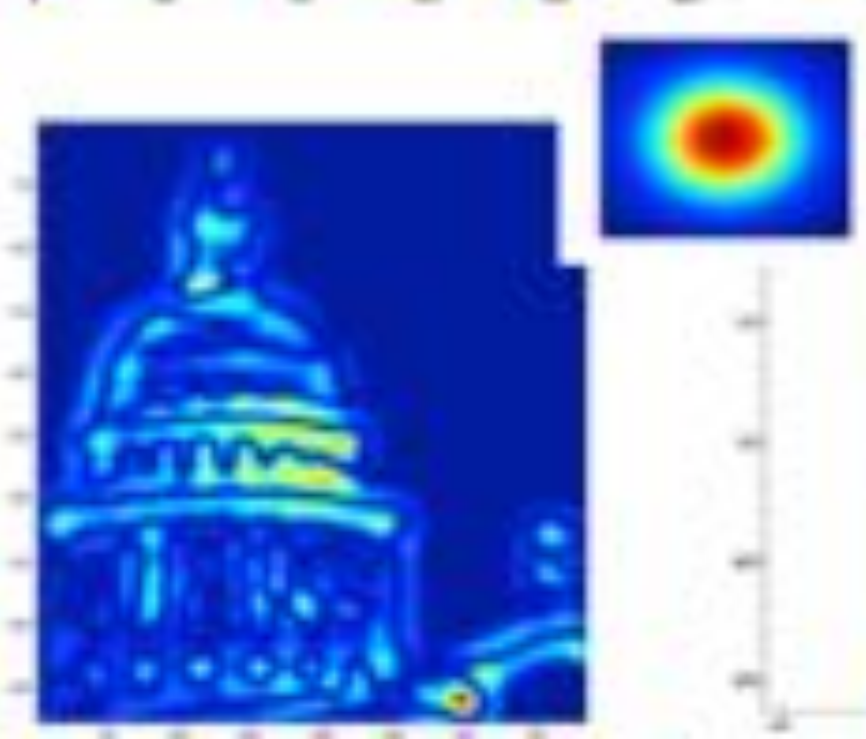
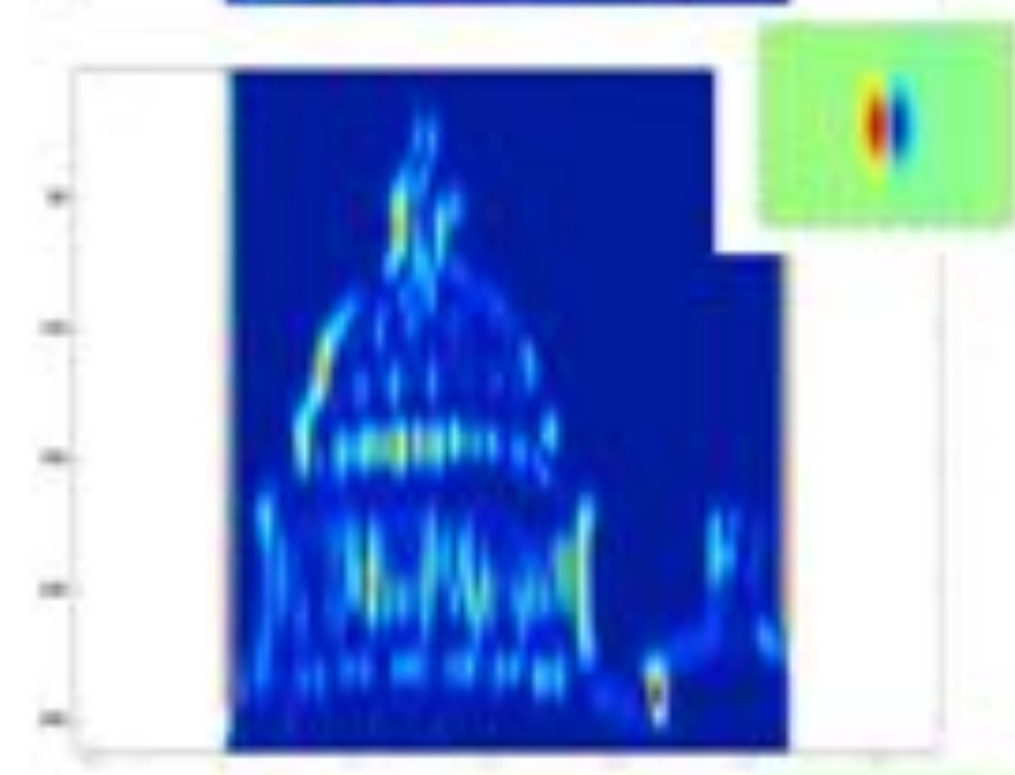
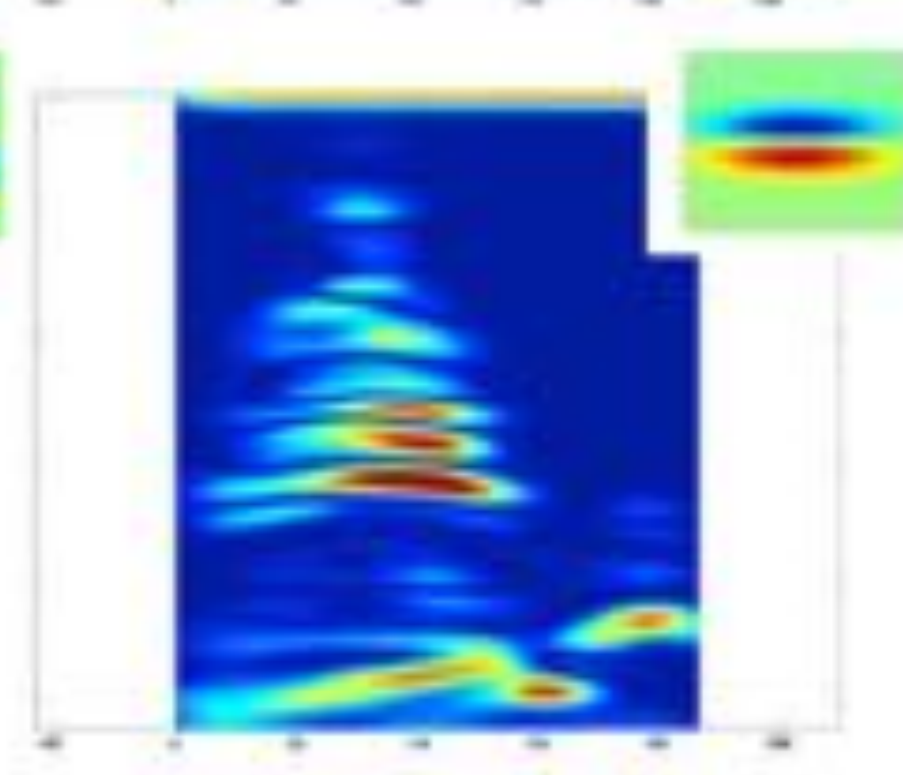
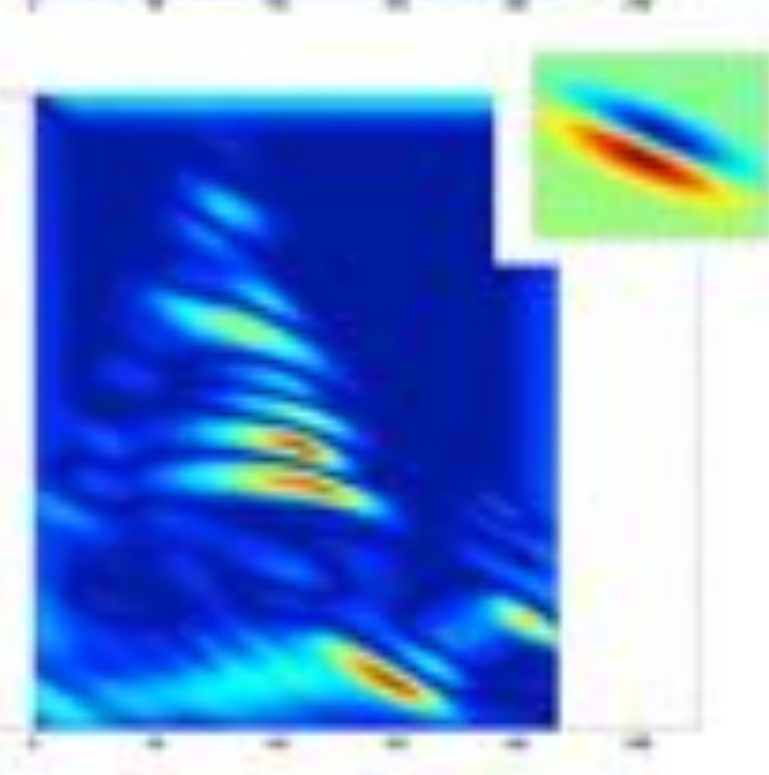
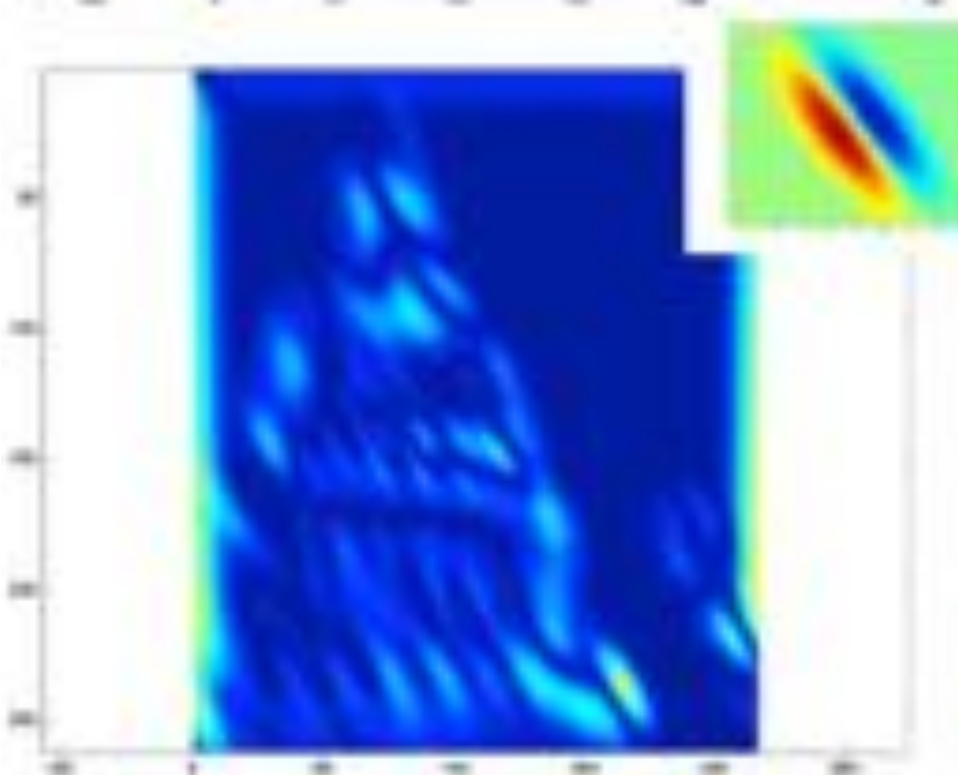
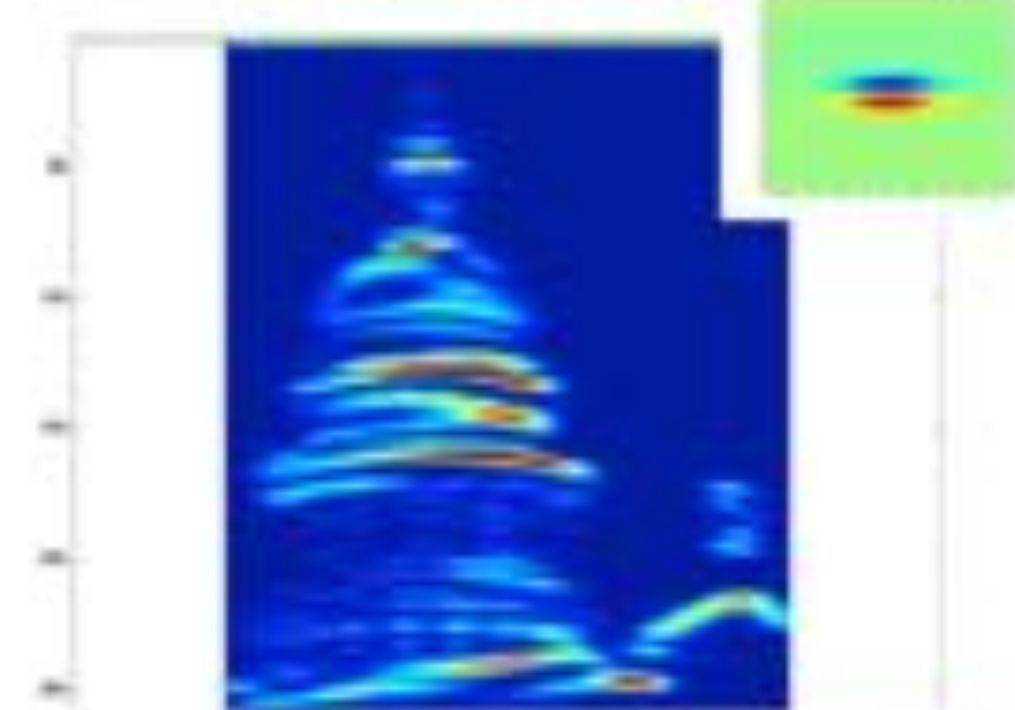
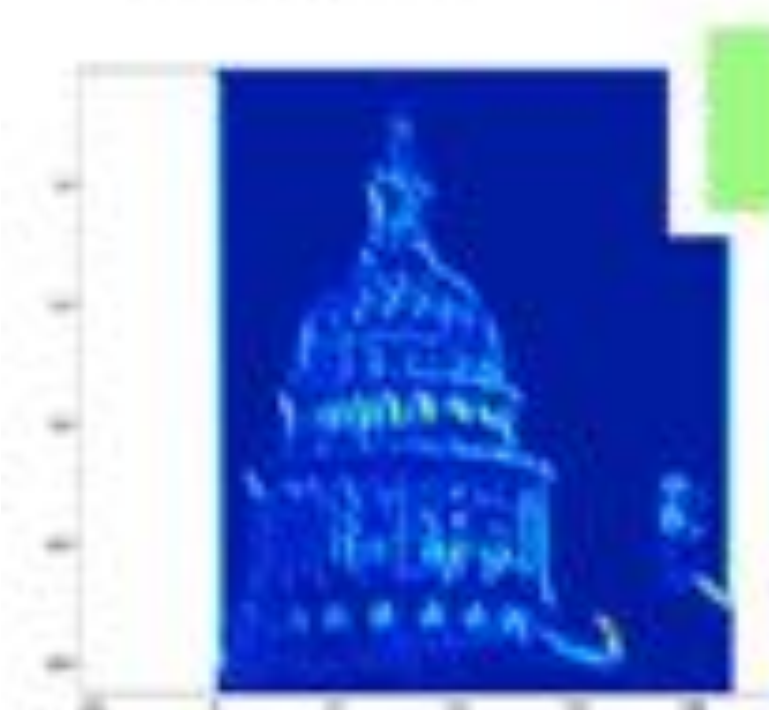
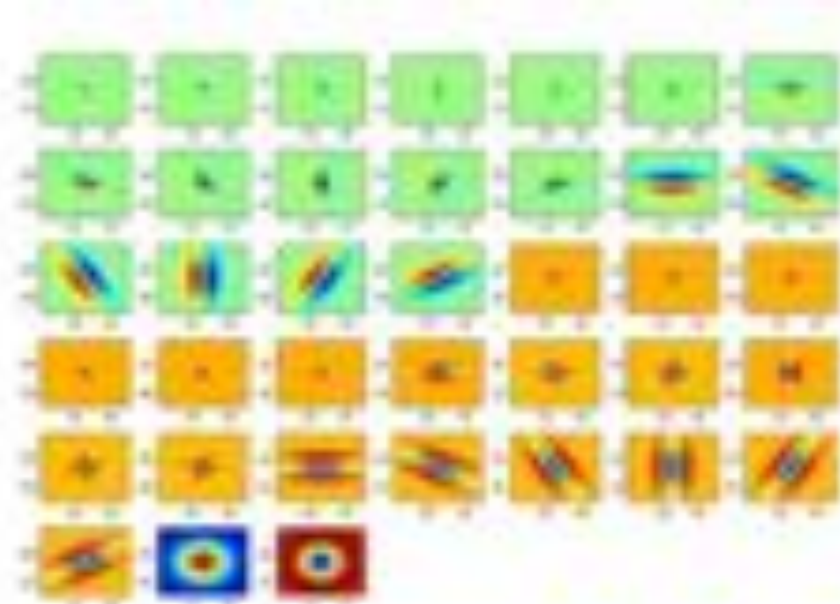




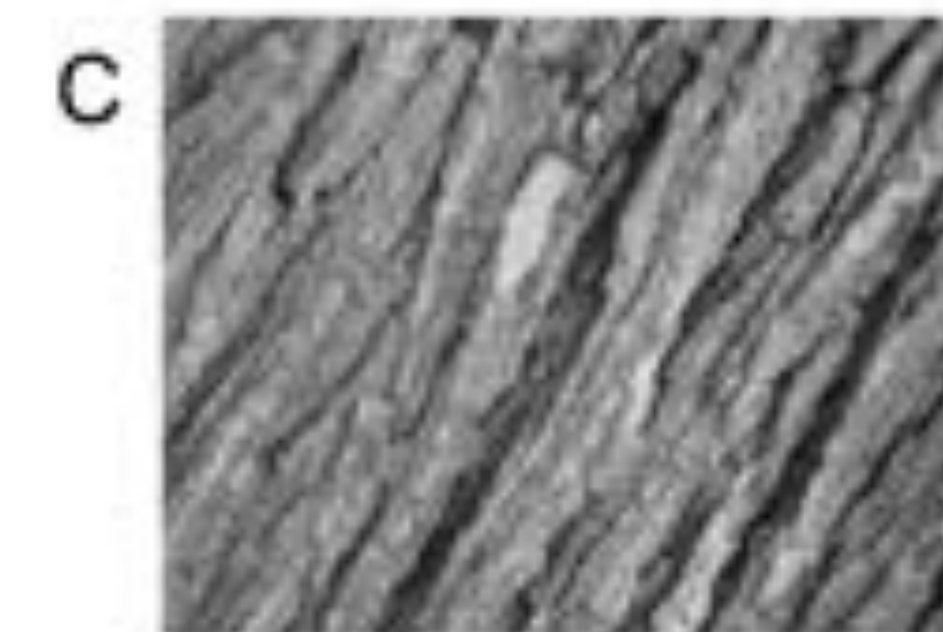
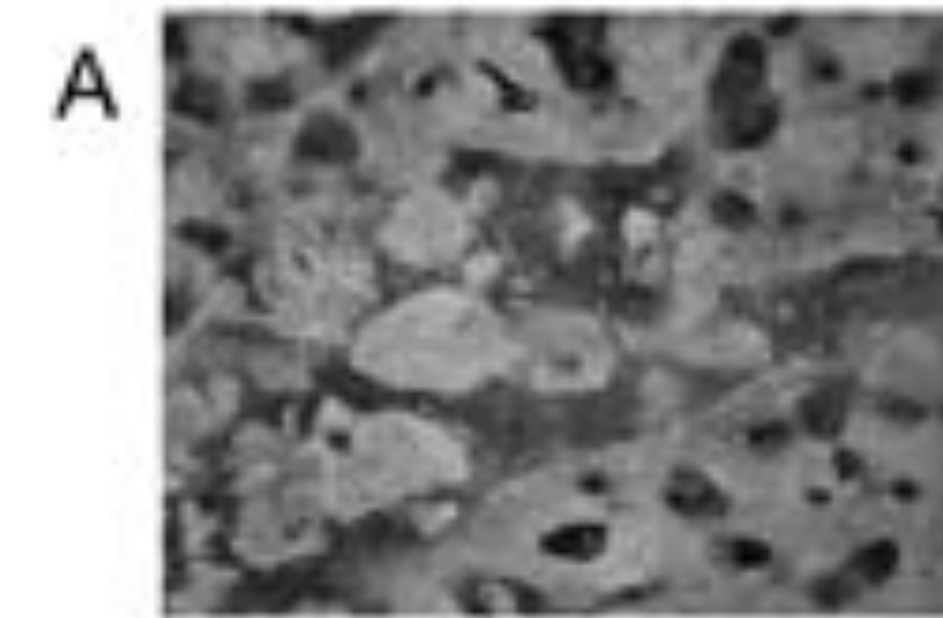
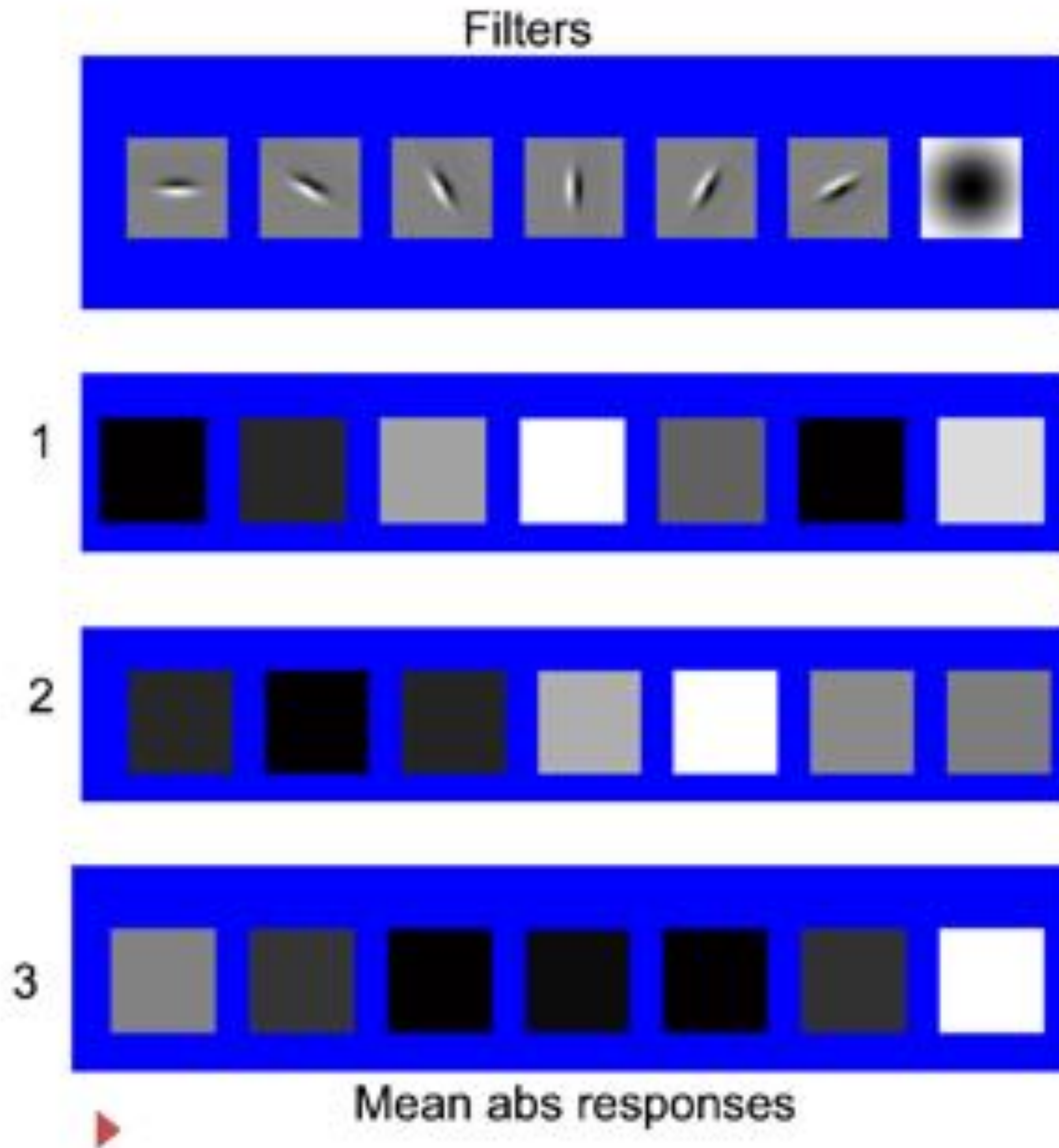




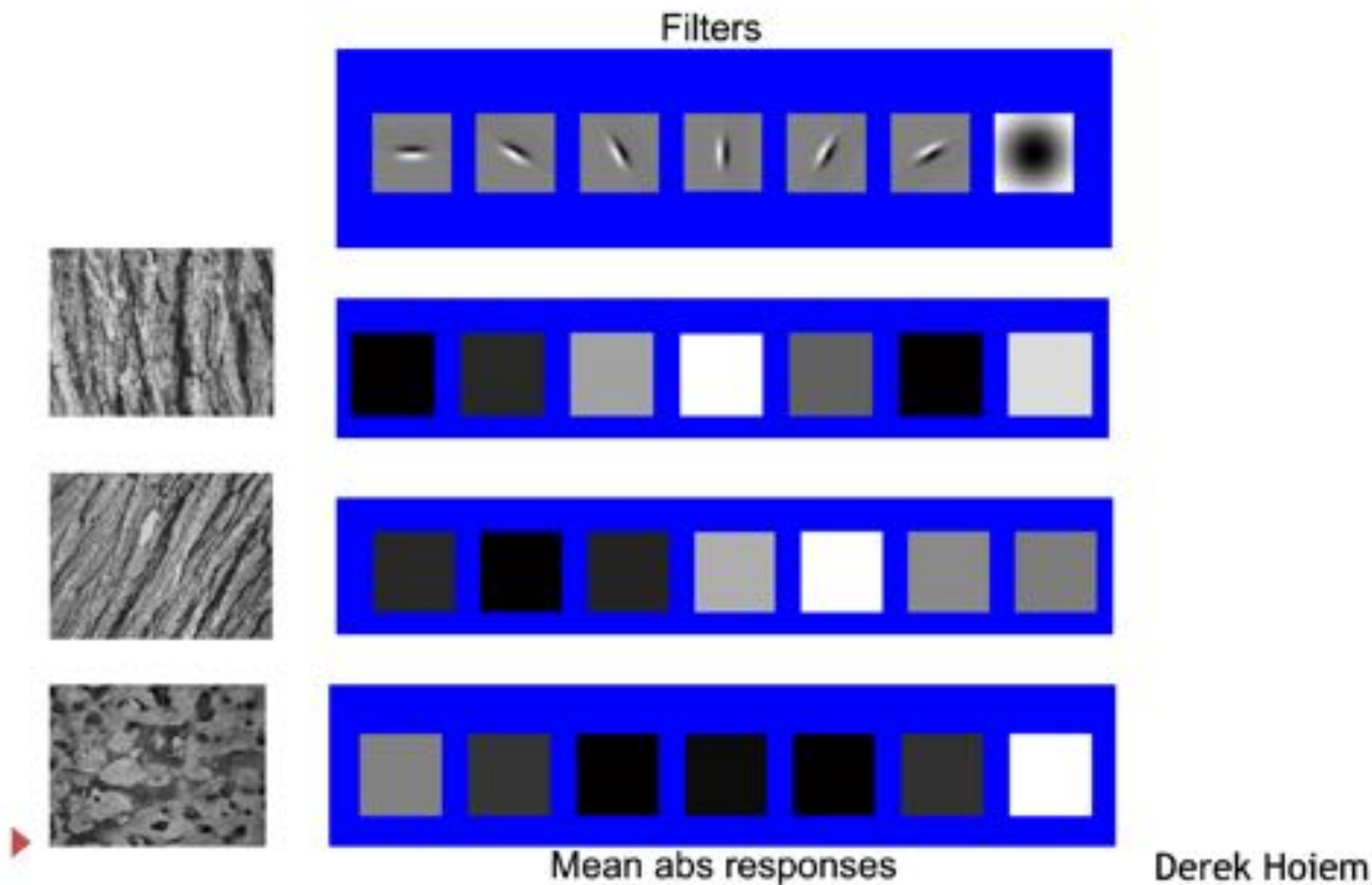




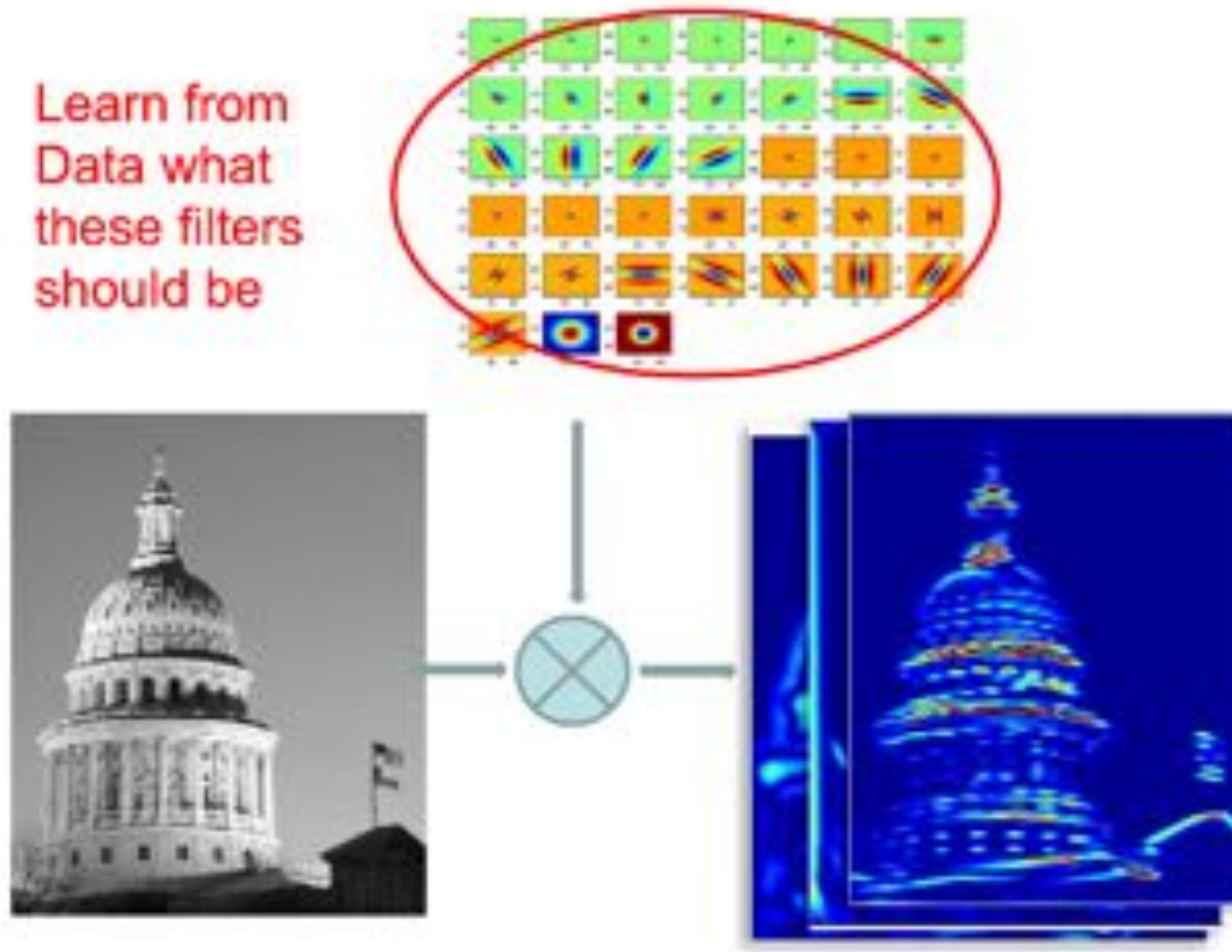
You try: Can you match the texture to the response?



Representing texture by mean abs response

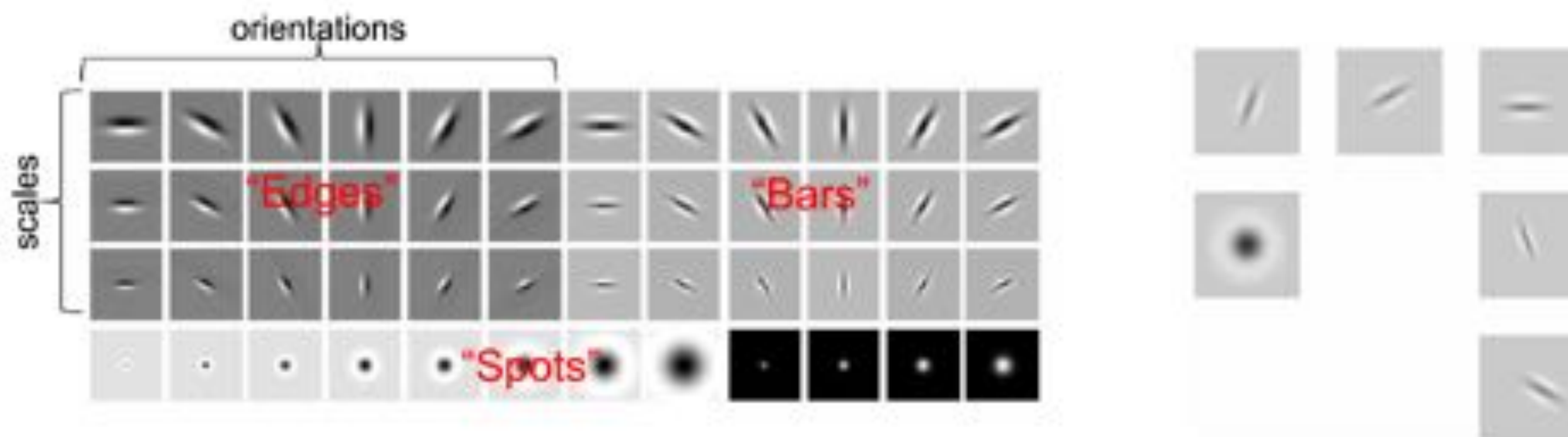


Convolutional Neural Network

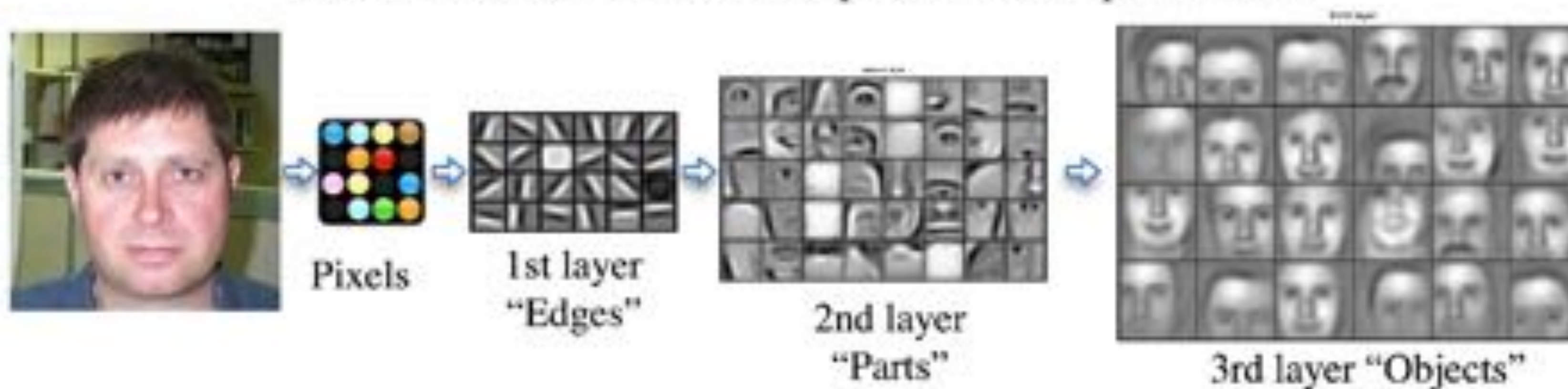


Filter Banks

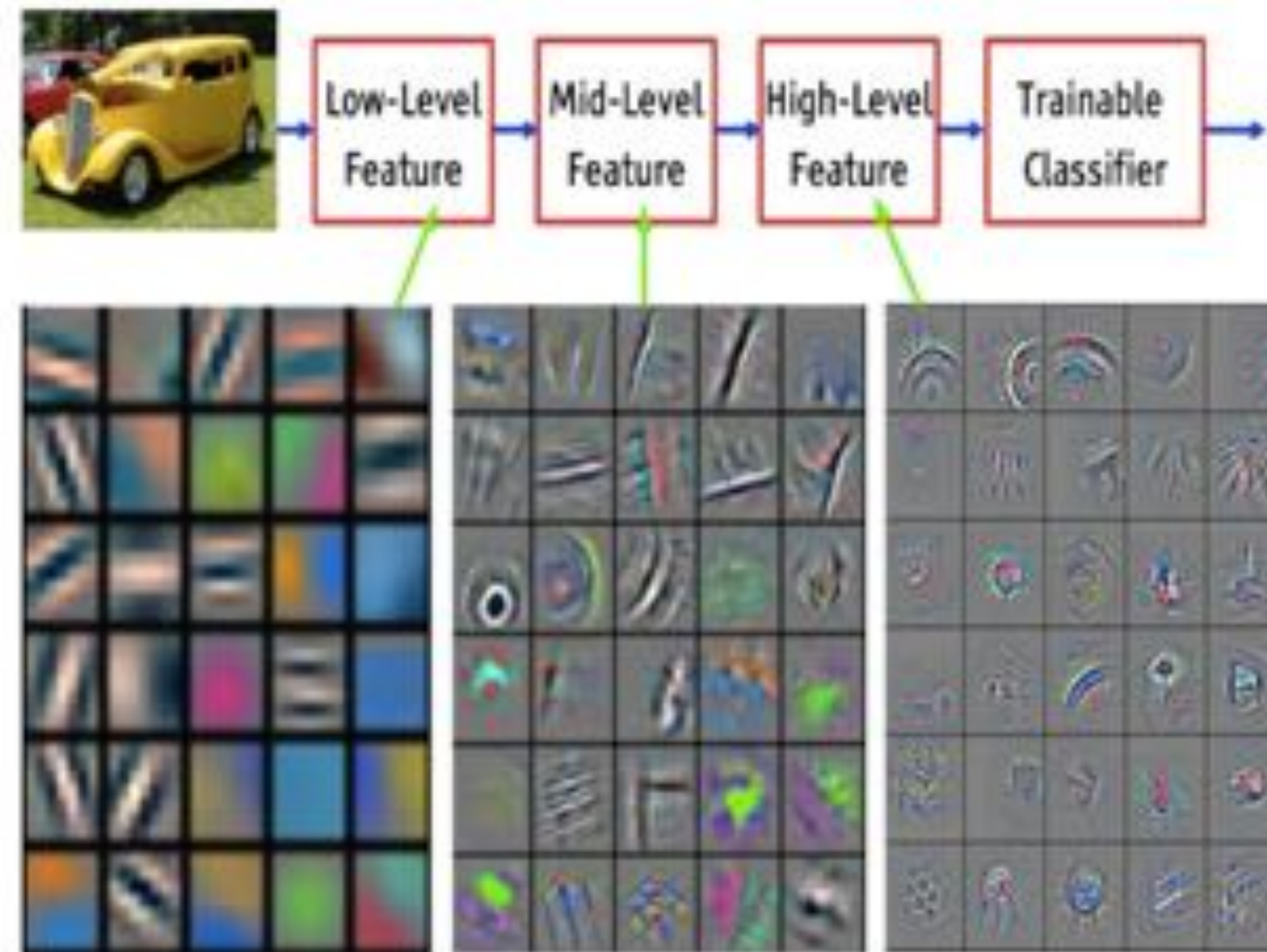
- ▶ From Low-Level to Mid & High Level Filters



Learned hierarchical feature representation: Sparse DBN



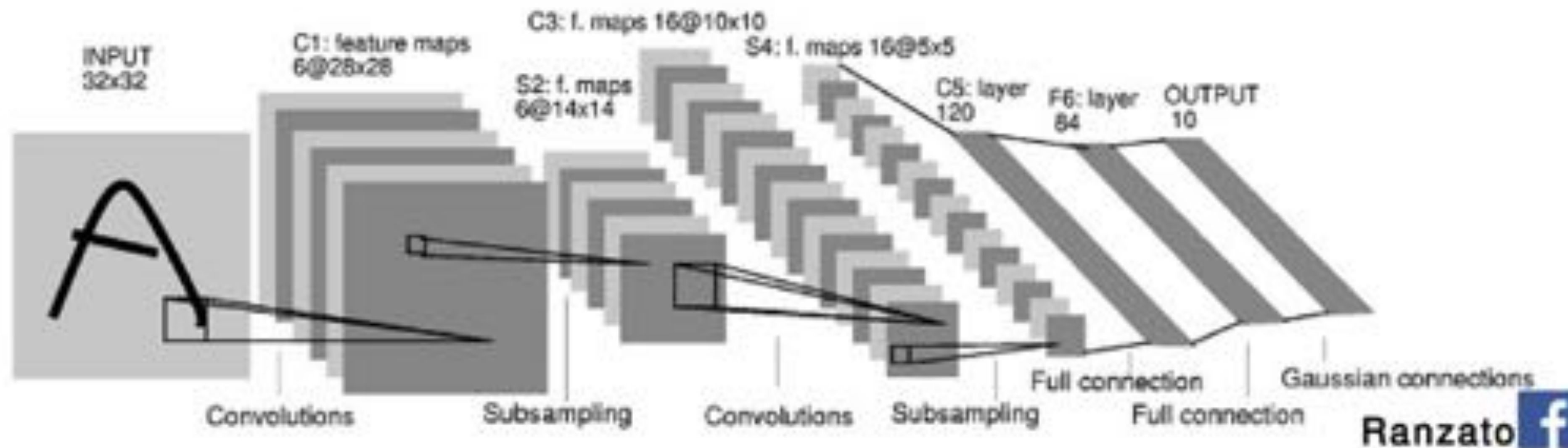
Convolutional Neural Network



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Convolutional Neural Network

► Le Cun 1998



Readings



- Deep Feedforward Nets
 - <https://www.deeplearningbook.org/contents/mlp.html>
- Regularization for DL
 - <https://www.deeplearningbook.org/contents/regularization.html>

Next: Transforming Multilayer Perceptron to Convolutional Neural Network

